Four-Color Boundary-Order Notes

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Abstract

This note records the current status of the boundary-order layer in the Lean formalization of the Goertzel four-color proof route. The key point is now formalized as an explicit regression: the present `PlaneEmbeddingWithBoundary` API only exposes unordered face-boundary incidence, and this is too weak to recover the ordered face-boundary geometry used downstream by the selected-boundary and annulus arguments. Moreover, even the current cyclic edge-list and cyclic edge/vertex-walk surrogates are still strictly weaker than an honest actual facial closed-walk source interface. The newest obstruction goes in the opposite direction: once the honest closed-walk source is used to extract a separated two-component annulus boundary split, the current all-roots-on-the-outer-boundary peel package cannot use that same split. A later diamond-with-triangle forcing-edge calibration now also shows that repaired previous-boundary witness geometry does not imply the interior-dual root-distance package.

1 Current source interfaces

The development now distinguishes ten source-facing or immediately derived boundary-order layers above `PlaneEmbeddingWithBoundary`:

1. `OrderedFaceBoundaryData`: each ambient face boundary is listed as a linear chain of distinct edges with shared endpoints between consecutive edges.
2. `OrderedFaceBoundaryVertexData`: the same linear edge order together with one explicit chosen shared primal endpoint for each adjacent pair of boundary edges.
3. `OrderedFaceBoundaryVertexSequenceData`: the same linear edge order together with one explicit ordered list of shared primal vertices, aligned with the boundary edge list itself.
4. `CyclicOrderedFaceBoundaryData`: the same linear data, plus a wrap condition saying the last and first boundary edges also share an endpoint.
5. `CyclicOrderedFaceBoundaryVertexData`: the cyclic edge order together with one explicit chosen shared primal endpoint for each adjacent pair in the closed cycle.
6. `CyclicOrderedFaceBoundaryVertexSequenceData`: the same cyclic edge order together with one explicit cyclic list of shared primal vertices, again stored in boundary order.
7. `FaceBoundaryClosedVertexWalkGeometry`: a bundled cyclic alternating edge/vertex walk, where each boundary edge is incident both to its own listed vertex and to the next listed vertex in cyclic order.
8. `FaceBoundaryClosedRunGeometry`: the same cyclic information repackaged as one explicit closed boundary run per ambient face, i.e. a pure boundary-walk layer.

9. `FaceBoundaryClosedWalkGeometry`: a stronger bundled source object recording for each face an actual nonempty closed trail in the graph whose traversed edges are exactly the face boundary.

10. FourColor specializations of these structures, together with selected boundary arc data.

These interfaces are intentional temporary surrogates for honest embedding-side face-boundary walk order. They isolate exactly what the current weak API fails to provide.

2 Negative examples

Three explicit regressions are now in Lean.

First, the disconnected two-edge example shows that unordered face-boundary incidence alone does not determine any endpoint-sharing order at all. This lives in `Mettapedia/GraphTheory/OrderedPlanarEmbeddingWithBoundaryRegression.lean` and is reused in `Mettapedia/GraphTheory/FourColor/PlanarBoundaryFaceBoundaryRunRegression.lean`.

Second, the three-edge path example shows that linear ordered data is strictly weaker than cyclic ordered data and than the equivalent pure closed-run layer. The unique face boundary can be listed as `[p01, p12, p23]` or its reverse, so linear ordered data exists. But the path endpoints do not share a primal endpoint, so no cyclic closure is possible. This proves that the cyclic source layer is a genuine strengthening rather than just a repackaging.

Third, the three-edge star example shows that the current cyclic source layers are still weaker than an honest facial closed-walk source. The unique face boundary can be listed cyclically as the three spokes, because every consecutive pair shares the hub vertex. So cyclic ordered data exists, and hence also the equivalent bundled cyclic closed-run and cyclic edge/vertex-walk surrogates. But the star graph is acyclic, so it admits no nonempty actual closed trail at all. This proves that the new `FaceBoundaryClosedWalkGeometry` target is a genuine strengthening over the current cyclic surrogates on the weak embedding API.

On the FourColor side, the same star example now also lifts to the cyclic selected-boundary shells `PlanarBoundaryCyclicOrderedFaceArcEmbeddingData` and `PlanarBoundarySelectedBoundaryArcGeometry`, because in the one-face embedding every boundary edge is selected. So even these route-facing cyclic selected-boundary interfaces are still weaker than the honest actual facial closed-walk source. In particular, no generic theorem of the form “current cyclic selected-boundary shell implies honest facial closed-walk geometry” can hold over the present weak embedding API.

3 Architectural consequence

The current proof-route architecture is now sharper:

- unordered face-boundary incidence is too weak;
- linear ordered face-boundary data is enough for the present run-geometry and bundled selected-arc shells;
- the new proof-relevant linear and cyclic boundary-step vertex layers are equivalent theorem surfaces over the current ordered and cyclic edge-order layers, but they remember the actual shared primal vertices between successive boundary edges;

- the new ordered and cyclic boundary-step vertex sequence layers go one step further by storing those chosen shared vertices in the same order as the boundary walk itself, making them a closer source proxy to an honest embedding side facial walk than the pair-indexed chooser interface;

- the new bundled cyclic edge/vertex walk layer is equivalent to the cyclic vertex-sequence interface, but packages the same information in the explicit alternating walk form that future honest embedding-side derivations are most likely to target directly;

- the new honest face-boundary closed-walk layer is strictly stronger than the cyclic ordered / cyclic closed-run / cyclic edge/vertex-walk surrogate layers over the current weak API;

- cyclic ordered data is a closer proxy to genuine planar face-boundary order and is strictly stronger;

- the new pure closed-run layer gives an equivalent "boundary walk" presentation of the cyclic source data, while the new edge/vertex walk layer refines that presentation by keeping the shared primal vertices explicitly in cyclic order;

- the annulus boundary-connectivity and construction shells now also accept the split same-embedding interface "cyclic alternating edge/vertex face-boundary walk + selected-boundary arc on each induced linear run" directly, so downstream theorem endpoints no longer have to manually unwrap the cyclic walk layer before using the current selected-arc reductions;

- the stronger honest source surface "facial closed walk + selected-boundary arc on each induced linear run" now also lowers directly to the current cyclic ordered face-arc shell and to the same annulus boundary-connectivity / boundary-support / positive-frontier / construction endpoints, so future work can target actual facial walks without first repackaging them by hand as cyclic surrogate data;

- the honest facial closed-walk lowering now constructs the cyclic edge/vertex boundary-walk layer directly from the actual walk darts: the boundary edges are the dart edges in order, and the shared vertices are the dart terminal vertices. This removes an avoidable detour through the cyclic edge-order surrogate and makes the proof-relevant vertex data traceable to the source walk itself;

- the current same-embedding annulus theorem surfaces "ordered face-arc data", "cyclic ordered face-arc data", "cyclic edge/vertex walk + selected-boundary arc", and "honest facial closed walk + selected-boundary arc", each combined with the live interior-dual adjacency-distance peel data and the current positive-frontier hypotheses, now also lower directly to the pure annulus decomposition endpoint, removing one more manual construction-to-decomposition composition step from the theorem-4.9 route;

- on the construction side, the same four honest boundary-order source surfaces now also lower directly to the weaker "FacePartitionData" shell, replacing the stronger per-face non-peeled selected-boundary-contact hypothesis by the more honest finite-set face-cover obligation;

- on the live boundary-root adjacency-distance source route, that finite face-cover obligation is now discharged internally: the interior-dual root package already proves that every non-peeled face is a boundary-root face and hence meets selected boundary support. The generic construction bridge now lowers boundary-component reachability plus this interior-dual package directly to the boundary-support, face-partition, and positive-frontier shells, leaving the positive current-frontier facts as the explicit remaining frontier-side inputs;

- the positive-frontier package now has a checked finite-descent consequence: whenever one annulus stage is strictly later than another, the surviving current-face set has strictly smaller cardinality. Adjacent stages therefore form a genuine finite decreasing chain, so the current-frontier assumptions are not merely interface markers but yield a measurable progress invariant. Since the final current-face set is nonempty and stage zero covers all ambient faces under the face-partition hypothesis, the package also proves the global finite bound that the number of collars is at most the number of ambient faces;

- the shifted boundary-root adjacency-distance construction now also has an exact obstruction theorem for parent-witness orientation: failure of strict parent orientation is equivalent to the existence of a peeled face with shifted annulus distance zero. This records precisely why the first collar must be excluded before the parent-oriented well-founded witness surface can be used;

- the same boundary-root adjacency-distance source now also has a checked constructive repair: a root-distance annulus construction that keeps the unshifted interior-dual distance. On peeled faces, the shifted collar-indexing distance plus one is exactly this root-distance height, and the root-distance construction reaches the parent-oriented well-founded witness surface without any first-collar exclusion. Thus the formal route separates the collar-indexing construction from the parent-orientation construction instead of trying to make one shifted distance serve both purposes;

- on the outer-boundary-root side, the formal interface now also proves that every chosen root face already lies in the ambient outer-boundary face set. So the unresolved face-cover obstruction is genuinely about non-root faces, not about localizing the root set itself;

- the honest closed-walk source now also exposes a structural obstruction to the current all-outer-root route: if `PlanarBoundaryOuterBoundaryRootAdjDistancePeelData` uses the same two-component boundary split extracted from `PlanarBoundaryClosedWalkAnnulusBoundarySource`, then it is inconsistent. The theorem `false_of_closedWalkAnnulusBoundarySource_and_outerBoundaryRootAdjDistancePeelData_sameBoundaryData` proves the fixed-embedding contradiction, and `not_exists_embedding_closedWalkAnnulusBoundarySource_and_outerBoundaryRootAdjDistancePeelData_sameBoundaryData` packages it at graph level. The same file now also proves the stronger parent-level versions `false_of_closedWalkAnnulusBoundarySource_and_outerBoundaryRootParentPeelData_sameBoundaryData` and `not_exists_embedding_closedWalkAnnulusBoundarySource_and_outerBoundaryRootParentPeelData_sameBoundaryData`. So the obstruction is not peculiar to the weaker adjacency-distance interface: even the stronger all-roots-on-the-outer-boundary parent package is already incompatible with the honest same-split closed-walk source. This means the surviving root-distance route needs a two-sided/root-per-boundary formulation or a different boundary split before it can be honestly sourced from closed facial walks. The same file now also proves `false_of_boundaryReachabilityData_and_interiorDualBoundaryRootParentPeelData_of_rootsOuterAmbientBoundary_and_boundarySelectedBoundaryArcOnFace`, so once a boundary-order shell already carries facewise selected-boundary arcs, the stronger outer-root parent package is impossible there even before one asks for an honest closed-walk annulus source. The same file now also proves the same-boundary local nonexistence theorems `not_exists_planarBoundaryOuterBoundaryRootAdjDistancePeelData_sameBoundaryData_of_boundaryReachabilityData_and_boundarySelectedBoundaryArcOnFace` and `not_exists_planarBoundaryOuterBoundaryRootParentPeelData_sameBoundaryData_of_boundaryReachabilityData_and_boundarySelectedBoundaryArcOnFace`, so the live selected-arc boundary-order shell already rules out both outer-root packages as soon as they try to reuse that shell's own annulus boundary split;

- the two-sided replacement formulation is now a Lean interface, not just an instruction for future work. The file `Theorem49PlanarBoundaryAnnulusRootInteriorDual.lean` introduces `PlanarBoundaryAnnulusRootAdjDistancePeelData`, whose roots may touch either extracted boundary component. The constructor from `PlanarBoundaryAnnulusBoundaryData` plus generic `InteriorDualBoundaryRootAdjDistancePeelData` and the raw cover/separated-root constructor over `outerAmbientBoundary` union `innerAmbientBoundary` both lower to the existing boundary-root pipeline. The same file also has direct closed-walk constructors from `PlanarBoundaryClosedWalkAnnulusBoundarySource` plus generic interior-dual data into the two-sided target, so this is the current positive target for closed-walk sourcing;

- the theorem-4.9 synthesis layer now factors through that same generic interior-dual route. The file `Theorem49Synthesis.lean` defines the core `Theorem49BoundaryRootSynthesis` package directly from generic `PlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData`, and then re-exports it as both `Theorem49OuterBoundaryRootSynthesis` and `Theorem49AnnulusRootSynthesis`. So once boundary-order work supplies the generic interior-dual route, the purified theorem-4.9 endpoint is already available on the two-sided annulus-root surface without any extra synthesis repackaging;

- the route file `PlanarBoundaryClosedWalkSourceRoute.lean` now closes that positive interface explicitly for the honest source objects already used in this note: theorems `exists_theorem49AnnulusRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct` and `exists_theorem49AnnulusRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_direct` show that explicit closed-walk or successor-cycle boundary-order data plus generic interior-dual data already reaches the two-sided theorem-4.9 synthesis endpoint directly. The same route file now also exposes the more direct certificate-side corollaries `admitsPlanarBoundaryAttachedRootedFacePeelCertificate_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`, and the corresponding successor-cycle versions, so once the generic boundary-root interior-dual package is in hand the honest source shells can land directly on the attached-certificate and boundary-root synthesis surfaces without going back through annulus-construction parent orientation. The same successor-cycle shell now also reaches the non-vacuous two-sided annulus-root surface: `theorem49AnnulusRootNonemptySynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_` selectedBoundarySupport and exists_theorem49AnnulusRootNonemptySynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_with_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport show that once the canonical annulus construction on that embedding has peeled-face endpoint no-touch and the raw interior-edge endpoint carrier is nonempty, the live boundary-order shell already reaches Theorem49AnnulusRootNonemptySynthesis before any source-fixed parent package is added;

- the same live successor-cycle shell now also reaches the generic source-fixed IDBRAD surface itself: interiorDualBoundaryRootAdjDistancePeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover is the same-embedding constructor, and admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePe
of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover is the graph-level route theorem. So once the successor-cycle boundary-order shell supplies the source boundary-face root cover/separation, boundary-free peeled faces, rooted shared-edge coverage, and strict root-distance child-membership fields, no additional source-to-IDBRAD repackaging step remains. The same concrete rooted-cover package on the honest closed-walk source now also closes directly at full Theorem 4.9 through exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_direct, so even on that source-facing shell there is no further theorem-4.9-side repackaging below the named rooted shared-edge face-membership burden. The same rooted-cover package now also closes directly at the theorem-4.9 endpoint through exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_direct, so no further theorem-4.9-side repackaging remains below that concrete source-facing surface;

- the annulus-root surface now also proves the missing two-sided geometric forcing fact: any ambient face meeting either distinguished annulus boundary component is automatically non-peeled and hence a root face for the generic interior-dual adjacency-distance package. In particular, exists_root_mem_outerBoundaryFaces_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData and exists_root_mem_innerBoundaryFaces_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData show that honest closed-walk annulus source data plus generic interior-dual data already forces one chosen root face on each extracted annulus boundary component;

- the same source file now also exposes the upstream parent-orientation surface explicitly. The same-embedding theorem parentWitnessOrientation_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_rootDistance and the graph-level theorems admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct and admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation_ `of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_direct` show that honest closed-walk and successor-cycle boundary-order data already yield `PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation` through the root-distance annulus construction, before any annihilator or synthesis packaging is applied;

- the synthesis layer is now also exposed one step earlier, at the raw annulus-split interface itself: `exists_theorem49AnnulusRootSynthesis_of_exists_boundaryData_and_interiorDualBoundaryRootAdjDistancePeelData` and its nonempty counterpart show that once a single embedding already carries `PlanarBoundaryAnnulusBoundaryData` and generic interior-dual rooted peel data, the theorem-4.9 endpoint is available without first repackaging the same data through closed-walk source fields;

- the concrete annulus-construction route now also reaches the algebraic theorem-4.9 bridge directly. The new file `Theorem49PlanarBoundaryAnnulusConstructionSpanning.lean` proves `boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryAnnulusConstruction_of_parentWitness`: construction data plus the explicit `ParentWitnessOrientation` hypothesis already implies the fixed-embedding annihilator endpoint "boundary-zero plus Definition 4.8 orthogonality implies zero". From that generic bridge hypothesis it derives the corrected span identity on the projected generator subspace, the corresponding theorem-4.9 target equality and inclusion, pointwise separation on the purified endpoint carrier, kernel-triviality of the pairing map, the resulting finrank inequality, and a raw Kempe-span preimage theorem. So once boundary-order work delivers honest parent-witness orientation on the annulus construction, the route can already enter the algebraic theorem-4.9 bridge without rebuilding the older interior-dual boundary-root package;

- the same algebraic bridge has now been pushed down to the boundary-aware witness-cover surfaces themselves. `Theorem49SpanningGap.lean` proves `boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryHeightOrderedFacePeelWitnessData` and `boundaryZeroAnnihilatorTrivialForEmbedding_of_planarBoundaryCollarLayerFacePeelWitnessData` `Theorem49SpanningGap.lean` also exposes graph-level height-ordered and collar-layer wrappers for the ordinary and relative submodule-duality interfaces; the chain-dot versions build the duality package directly from the same-embedding target/source subspaces and their identification lemmas. `Theorem49KempeGeneratorSpan.lean` fixes the canonical Definition 4.8 generator side for both surfaces. Its graph-level wrappers `exists_theorem49_kempeGeneratorSubspace_spanning_of_admitsPlanarBoundaryHeightOrderedFacePeelWitnessData` and `exists_theorem49_kempeGeneratorSubspace_spanning_of_admitsPlanarBoundaryCollarLayerFacePeelWitnessData` instantiate the concrete submodule-duality package using the actual Kempe-closure generator subspace, removing the separate orthogonality-to-annihilator obligation from this canonical source path. Then `Theorem49BoundaryProjection.lean` proves the corresponding projected-source equalities `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_planarBoundaryHeightOrderedFacePeelWitnessData` and `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_planarBoundaryCollarLayerFacePeelWitnessData`, with matching boundary-zero Kirchhoff target equality and inclusion theorems. Thus height-ordered or finite collar-layer witness-cover data is enough to enter the corrected projected-generator theorem-4.9 source statement directly. The same projection file now also provides finite projected-generator representations, raw Definition 4.8 generator preimages before boundary erasure, and graph-level wrappers from `AdmitsPlanarBoundaryHeightOrderedFacePeelWitnessData` and `AdmitsPlanarBoundaryCollarLayerFacePeelWitnessData`. Those wrappers now cover the explicit-family target identity, projected and explicit-family inclusion forms, raw Definition 4.8 generator-span preimages, and finite raw-generator projection witnesses for every boundary-zero Kirchhoff target chain. They now also include the chain-dot separation theorem

$$W_0(H) \cap (\text{projectedKempeClosureGeneratorSubspace})^{\perp} = 0,$$

plus detector and pointwise orthogonality iff forms, for height-ordered and finite collar-layer data. Thus the downstream route can use those witness surfaces without rebuilding the interior-dual rooted package or assuming the extra raw-source quotient disjointness hypotheses. The weakest explicit local-remainder collar surface now has the same kind of direct annihilator access: `Theorem49PlanarBoundaryCollarPeeling.lean` proves `zero_on_allEdges_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData_of_annihilatesKempeClosureGeneratorFamily` and `zero_on_allEdges_of_exists_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData_of_annihilatesKempeClosureGeneratorFamily`. So local-remainder data can consume Definition 4.8 generator-family orthogonality directly, before being forgotten to ordinary collar-layer data. `Theorem49SpanningGap.lean` and `Theorem49BoundaryProjection.lean` now keep that local-remainder surface visible through the algebraic projection step. In particular `span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` identifies the projected Definition 4.8 span with the selected-boundary-zero subspace, and `exists_projectedKempeClosureGeneratorFamily_finset_sum_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_admitsPlanarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` packages the graph-level finite-combination theorem for the concrete boundary-zero Kirchhoff target. The same file now records the raw Definition 4.8 source projection too: `exists_kempeClosureGeneratorSubspace_preimage_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` gives a raw generator-subspace preimage, while `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` gives a finite unprojected generator-family sum whose boundary-erased projection is the target chain. It now also has the matching Kirchhoff-retaining raw source theorem `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_and_mem_kirchhoffSubmodule_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData`: under the selected-boundary avoidance hypothesis, the same kind of raw finite Definition 4.8 sum can be chosen inside the Kirchhoff submodule at the chosen target vertices. The graph-level admissibility theorem exposes this endpoint without converting the local-remainder witness into interior-dual root-distance data. The local-remainder source has also been pushed to the raw quotient/error form. `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_eq_boundaryZeroProjection_of_mem_kirchhoffSubmodule_` `of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData` represents every raw Kirchhoff chain, after selected-boundary erasure, by a finite raw Definition 4.8 generator-family sum that is itself Kirchhoff. The boundary-supported and kernel-decomposition forms then isolate the remaining error as a Kirchhoff chain supported on the selected boundary, equivalently in the kernel of the boundary-erasure map restricted to the Kirchhoff submodule. The same raw representative plus quotient/error package is now available at the positive projected synthesis endpoint itself via `Theorem49BoundaryRootNonemptyProjectedSynthesis.rawKirchhoffRepresentative_and_boundaryKernelDecomposition`. This theorem simply exposes the two relevant fields already inside the nested `Theorem49BoundaryRootSynthesis`, so downstream boundary-order source arguments can consume them through the same-embedding projected endpoint rather than importing another raw-projection theorem directly. The package is now named `Theorem49BoundaryRawQuotientErrorPackage`, and the current replacement positive geometry surfaces consume it directly. The fixed-embedding theorems `theorem49BoundaryRawQuotientErrorPackage_of_heightOrderedPositiveProjectedGeometryOn` and `theorem49BoundaryRawQuotientErrorPackage_of_collarLayerPositiveProjectedGeometryOn` thread the endpoint through height-ordered and finite collar-layer witness-cover data, and the graph-level methods on `Theorem49HeightOrderedPositiveProjectedGeometry` and `Theorem49CollarLayerPositiveProjectedGeometry` expose the same conclusion on the packaged embedding. This is the current non-vacuous replacement route's raw quotient/error consumption point. At the core synthesis surface this has been strengthened to a submodule equality: `Theorem49BoundaryRootSynthesis.kirchhoffSubmodule_eq_rawSource_sup_boundaryKernel`. It identifies the purified Kirchhoff submodule with the supremum of the Kirchhoff-valid raw Definition 4.8 Kempe-source submodule and the selected-boundary projection kernel error submodule. Thus the quotient/error reading is no longer only a family of finite witnesses; Lean now records the global raw-source-plus-kernel decomposition forced by any completed `Theorem49BoundaryRootSynthesis`. The same synthesis layer now also records the quotient form `Theorem49BoundaryRootSynthesis.rawSource_mkQ_eq_top`: inside the purified Kirchhoff submodule, after quotienting by the kernel of the selected-boundary projection restricted to that submodule, the Kirchhoff-valid raw Definition 4.8 source maps onto the whole quotient. This is the exact surjectivity statement needed by the quotient/error interpretation, and it avoids replacing the purified Kirchhoff quotient by a stronger ambient-space quotient. The theorem `Theorem49BoundaryRootSynthesis.finrank_quotient_le_rawSource` records the corresponding finite-dimensional consequence: the purified Kirchhoff quotient by the restricted selected-boundary kernel has finrank no larger than the raw Definition 4.8 source submodule that surjects onto it. The target-side bound `Theorem49BoundaryRootSynthesis.finrank_target_le_rawSource` also lives at the same surface: the concrete boundary-zero Kirchhoff target has finrank no larger than the Kirchhoff-valid raw Definition 4.8 source whose projection is that target. The actual quotient-target identification is now available as `Theorem49BoundaryRootSynthesis.boundaryProjectionQuotientEquivTarget`, with `Theorem49BoundaryRootSynthesis.boundaryProjectionQuotientEquivTarget_apply_mk` recording that a quotient class represented by a purified Kirchhoff chain maps to its selected-boundary-erased chain in the concrete target. The raw-source quotient version `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivTarget` now identifies the Kirchhoff-valid raw Definition 4.8 source quotient, modulo the selected-boundary projection kernel on that source, with the same concrete target; its `...rawSourceQuotientEquivTarget_apply_mk` theorem gives the corresponding representative computation rule. The raw and purified quotient models are now connected by `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivKirchhoffQuotient`. This linearly identifies the raw-source quotient with the purified Kirchhoff quotient through their common concrete theorem-4.9 target, and `Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivKirchhoffQuotient_apply_mk` records the representative computation: a raw Kirchhoff-valid source class maps to the same edge-chain class in the purified Kirchhoff quotient. This is a genuine comparison map between the quotient/error surfaces, not another package threading variant. The same synthesis layer now also records the finite representative form `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_for_kirchhoffQuotientClass`: each purified Kirchhoff quotient class admits a representative lying in the raw Definition 4.8 source and given as a finite linear combination of raw generator-family terms. The conclusion already includes equality in the purified Kirchhoff quotient, so the selected-boundary projection kernel conversion is discharged once at the synthesis surface. The quotient/error surface now carries the separation test as well: `Theorem49BoundaryRootSynthesis.kirchhoffQuotientClass_eq_zero_iff_projectedGenerator_orthogonal` identifies zero purified Kirchhoff quotient classes with classes whose selected-boundary-erased representatives are orthogonal to all projected Definition 4.8 generators. Its detector companion `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_chainDot_ne_zero_of_ne_zero_kirchhoffQuotientClass` extracts a projected generator with nonzero chain-dot pairing from any nonzero purified quotient class. This makes projected-generator separation usable directly modulo selected-boundary kernel error. The combined theorem `Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_kirchhoffQuotientClass` now gives, for every nonzero purified quotient class, a finite raw Definition 4.8 generator-sum representative together with a projected generator that detects that representative after selected-boundary erasure. This is the usable synthesis-surface form for nonzero quotient classes. The quotient surface also now has a named linear-map form: `Theorem49BoundaryRootSynthesis.kirchhoffQuotientProjectedGeneratorPairingMap` pairs purified Kirchhoff quotient classes with projected Definition 4.8 generators after selected-boundary erasure. Its `..._apply_mk` theorem computes the pairing on representatives, and `Theorem49BoundaryRootSynthesis.ker_kirchhoffQuotientProjectedGeneratorPairingMap_eq_bot` records the quotient-level trivial-kernel statement. `Theorem49BoundaryRootSynthesis.kirchhoffQuotientProjectedGeneratorPairingMap_eq_zero_iff` and `Theorem49BoundaryRootSynthesis.exists_projectedGenerator_pairingMap_ne_zero_of_ne_zero_kirchhoffQuotientClass` turn that kernel statement into the map-level detector for arbitrary purified quotient classes. `Theorem49BoundaryRootSynthesis.finrank_kirchhoffQuotient_le_projectedGeneratorSubspace` then extracts the corresponding finite-dimensional bound for the purified Kirchhoff quotient from that named pairing map, rather than appealing only to the older target-side finrank field, and `Theorem49BoundaryRootSynthesis.finrank_range_kirchhoffQuotientProjectedGeneratorPairingMap` identifies the pairing-map image dimension with the purified quotient dimension. The new equivalence `Theorem49BoundaryRootSynthesis.kirchhoffQuotientEquivPairingMapRange` then uses this same injective map to model purified quotient classes as functionals in the pairing-map image, with Theorem49BoundaryRootSynthesis.kirchhoffQuotientEquivPairingMapRange_apply_mk retaining the representative-level chain-dot computation. Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivPairingMapRange now gives the parallel image model for the raw Definition 4.8 source quotient itself, and Theorem49BoundaryRootSynthesis.rawSourceQuotientEquivPairingMapRange_apply_mk computes raw-source quotient representatives by selected-boundary erasure and chain-dot pairing. The raw-source quotient also has its own projected-generator separation test: Theorem49BoundaryRootSynthesis.rawSourceQuotientClass_eq_zero_iff_projectedGenerator_orthogonal and Theorem49BoundaryRootSynthesis.exists_projectedGenerator_chainDot_ne_zero_of_ne_zero_rawSourceQuotientClass characterize zero raw-source classes and detect nonzero ones without leaving the quotient/error surface. The finite representative theorem Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_rawSourceQuotientClass keeps the same raw-source quotient class while choosing an explicit finite raw Definition 4.8 generator-family sum and a projected generator that detects it. The concrete target-level theorem Theorem49BoundaryRootSynthesis.exists_rawGeneratorSum_and_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace now combines the raw projection representation and pointwise separation: every nonzero boundary-zero Kirchhoff target has a finite raw generator-family preimage and a projected generator that detects that same target. The sharpened theorem Theorem49BoundaryRootSynthesis.exists_nonempty_rawGeneratorSum_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace records the non-vacuity content of the same conclusion: the finite raw generator-family representative is not the empty sum, and the projected detector is itself nonzero. The strengthened support-sensitive theorem Theorem49BoundaryRootSynthesis.exists_nonzero_rawGeneratorSum_and_nonempty_coeffSupport_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace adds that this raw representative is a nonzero chain and that the coefficient support is nonempty, not only the ambient term finset. The term-level witness theorem Theorem49BoundaryRootSynthesis.exists_rawGenerator_with_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace extracts a concrete raw Definition 4.8 generator-family chain from that support with nonzero coefficient, while keeping the same nonzero raw representative and projected detector data. Theorem49BoundaryRootSynthesis.exists_nonzero_rawGenerator_with_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace removes the last degeneracy in this witness: the extracted support chain can be chosen nonzero, so the target-level detector is now tied to an actual nonzero raw Definition 4.8 generator-family chain. Theorem49BoundaryRootSynthesis.exists_rawGenerator_with_nonzero_boundaryZeroProjection_and_nonzero_coeff_and_nonzero_projectedGenerator_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace adds the boundary-facing non-vacuity condition: the extracted support chain can be chosen with nonzero selected-boundary erasure, so a nonzero theorem-4.9 target is not represented only by support terms killed by boundaryZeroProjection. The raw-source quotient now also has the rank consequences of that image model: Theorem49BoundaryRootSynthesis.finrank_rawSourceQuotient_eq_pairingMapRange identifies its dimension with the pairing-map image dimension, and Theorem49BoundaryRootSynthesis.finrank_ `rawSourceQuotient_le_projectedGeneratorSubspace` bounds it by the projected Definition 4.8 generator subspace. This separation layer has now been lowered through `PlanarBoundaryClosedWalkSourceProjection.lean` to the actual source shells: canonical witness choice, the local at-most-one criterion, and same-boundary one-collar geometry, for both closed-walk sources and successor-cycle boundary-order sources, give the same chain-dot separation endpoint through height-ordered and finite collar-layer witness data. The source file now lowers the matching pointwise detector and orthogonality iff forms too, so these same source shells can test each nonzero boundary-zero Kirchhoff chain by a projected Definition 4.8 generator; it now also lowers the raw preimage and finite raw-generator sum forms, so each such chain has a raw Definition 4.8 generator-span representative and a finite raw-generator combination projecting to it. The positive non-vacuous lane now has a single same-embedding package, `Theorem49BoundaryRootNonemptyProjectedSynthesis`, combining `Theorem49BoundaryRootNonemptySynthesis` with the corrected projected subspace and projected-family span equalities. Height-ordered and finite collar-layer witness data construct the package under nonempty purified carrier, and the closed-walk and successor-cycle source wrappers expose it for canonical witness choice and same-boundary one-collar geometry. The at-most-one/nonempty-carrier branch is intentionally absent from this positive package because the current purified carrier makes that branch inconsistent;

- `Theorem49PositiveGeometricSource.lean` names the candidate geometric source packages that were isolated for testing: `Theorem49ClosedWalkCanonicalPositiveGeometry`, `Theorem49ClosedWalkOneCollarPositiveGeometry`, `Theorem49SuccessorCycleCanonicalPositiveGeome` and `Theorem49SuccessorCycleOneCollarPositiveGeometry`. These packages keep the source data, canonical witness choice or source-preserving one-collar geometry, and nonempty purified carrier on the same embedding, then lower to `Theorem49BoundaryRootNonemptyProjectedSynthesis` for any Tait coloring. The fixed-embedding predicate forms now lower directly to the same endpoint on the same embedding;

- `Theorem49PositiveGeometricSourceRegression.lean` supplies that focused obstruction for the wheel-with-inner-triangle benchmark. The fixed embedding retains `wheelWithInnerTriangleTaitEdgeColoring_isTait` and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_wheelWithInnerTriangle`, but Lean proves the four package-level failures `not_theorem49ClosedWalkCanonicalPositiveGeometryOn_wheelWithInnerTriangle`, `not_theorem49ClosedWalkOneCollarPositiveGeometryOn_wheelWithInnerTriangle`, `not_theorem49SuccessorCycleCanonicalPositiveGeometryOn_wheelWithInnerTriangle`, and `not_theorem49SuccessorCycleOneCollarPositiveGeometryOn_wheelWithInnerTriangle`. Thus the wheel benchmark is no longer just an informal separator between carrier and geometry: it is a fixed-embedding regression test for the current positive package layer;

- `Theorem49PositiveGeometricSourceImpossibility.lean` upgrades that benchmark calibration to a structural route correction. Canonical witness choice implies facewise at-most-one interior edge, and honest closed-walk data plus facewise at-most-one empties the purified carrier. Same-boundary one-collar geometry induces canonical witness choice, so `not_nonempty_currentTheorem49PositiveGeometricSourcePackages` refutes all four current graph-level positive package structures. The boundary-order next step is therefore no longer to instantiate these packages, but to replace the source surface or carrier condition so that the nonempty projected endpoint is not forced empty. The same obstruction now has direct live-endpoint forms: `not_hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice` and `not_hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySource_and_oneCollarGeometry_with_sourceBoundaryData` say that canonical choice and source-preserving one-collar geometry rule out the local endpoint witness itself. Equivalently, `not_nonempty_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint` blocks canonical choice from an honest source plus `HasUnblockedInteriorEndpoint`, and `not_exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint` blocks same-boundary one-collar geometry;

- `Theorem49PositiveGeometricSourceReplacement.lean` now records the direct replacement surface. The fixed-embedding predicates `Theorem49HeightOrderedPositiveProjectedGeometryOn` and `Theorem49CollarLayerPositiveProjectedGeometryOn` pair the corresponding boundary-aware witness-cover data with nonempty purified carrier on the same embedding and lower directly to `Theorem49BoundaryRootNonemptyProjectedSynthesis`. The graph-level packages `Theorem49HeightOrderedPositiveProjectedGeometry` and `Theorem49CollarLayerPositiveProjectedGeometry` expose the same endpoint, and the theorem `heightOrderedPositiveProjectedGeometryOn_iff_collarLayerPositiveProjectedGeometryOn` identifies the two fixed-embedding forms using the existing height/collar conversions. The construction task has therefore been sharpened to building one of these direct witness-data packages from boundary-order geometry, while the file's compatibility lemmas prevent silently reintroducing canonical-choice or source-preserving one-collar fields on a nonempty carrier embedding. The same file now exposes the graph-level annulus bridge explicitly: `theorem49CollarLayerPositiveProjectedGeometry_of_annulusCollarGeometry` and `theorem49HeightOrderedPositiveProjectedGeometry_of_annulusCollarGeometry` turn annulus collar geometry plus nonempty purified carrier into the replacement packages, with existential versions and repaired previous-boundary-witness specializations. The same minimal hypotheses now also reach the nonempty projected synthesis endpoint directly via `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_annulusCollarGeometry_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct`. The repaired previous-boundary-witness lane now has the endpoint-witness version as well: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_annulusPreviousBoundaryWitnessGeometry_and_hasUnblockedInteriorEndpoint` uses `HasUnblockedInteriorEndpoint` directly, and the route file adds the closed-walk source-facing bridge `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_annulusPreviousBoundaryWitnessGeometry_and_hasUnblockedInteriorEndpoint`. The acyclic witness-cover lane now has the same direct endpoint: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint` uses well-founded parent witness-cover data plus `HasUnblockedInteriorEndpoint` to reach the projected synthesis surface, while the route-facing full theorem-4.9 theorem `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_wellFoundedFacePeelWitnessData_and_hasUnblockedInteriorEndpoint` keeps the honest closed-walk source visible while making the acyclic data the actual geometric input. The same file now also gives `Theorem49ClosedWalkAnnulusWellFoundedPositiveProjectedGeometry.boundaryRootSynthesis` and its successor-cycle analogue, so the named well-founded route packages themselves are no longer merely projected-endpoint surfaces. This direct acyclic lane now also consumes endpoint-support separation and raw endpoint-carrier nonemptiness without first manufacturing the named endpoint witness. The generic theorem `theorem49BoundaryRootNonemptyProjectedSynthesis_of_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` has height-ordered, finite-collar, graph-level, and existential variants, while the source-facing wrappers `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_wellFoundedFacePeelWitnessData_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` show that honest closed-walk and successor-cycle source shells can use the same endpoint-support bridge once explicit well-founded witness-cover data is available and immediately reach full theorem-4.9. This is a packaging advance, not a derivation of the acyclic witness-cover datum itself. The root-distance interior-dual route now produces that acyclic input on the same embedding: `planarBoundaryWellFoundedFacePeelWitnessData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` turns honest source data plus generic boundary-root interior-dual adjacency-distance data into explicit well-founded witness-cover data, and `planarBoundaryHeightOrderedFacePeelWitnessData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData` turns the same root-distance parent order into the height-ordered witness-cover surface directly. The graph-level source theorem is `admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_direct`. The obstruction side has also been factored at this same witness-cover interface. In `Theorem49PlanarBoundaryPeeling.lean`, a non-boundary remainder edge whose value is another peeled face's witness edge forces a parent edge in the well-founded package, or a strict height inequality in the height-ordered package, provided the witness choice is injective on peeled faces. The resulting two- and three-cycle lemmas rule out directed cycles of these non-boundary remainder-witness dependencies before any wheel-specific argument is used. This has now been generalized to arbitrary nonempty finite closed dependency chains through `false_of_isChain_height_lt_and_getLast_lt_head` and the well-founded and height-ordered `false_of_witnessRemainderDependency_cycle_of_injective_witness` lemmas. The wheel-with-inner-triangle regression now exercises that list-level lemma directly: the two possible orientations of the height-ordered spoke chase are closed as explicit three-face finite cycles, not as separate ad hoc irreflexivity arguments. The well-founded parent side has also been separated from this height presentation. The certificate builder proves `false_of_isChain_parentRel_and_getLast_parent_head`, saying that a well-founded parent map has no nonempty finite closed `ParentRel` chain, and the witness-cover finite-cycle blocker now uses that parent-chain theorem after extracting the actual parent edges. Thus the minimal upstream lemma for deriving acyclic witness-cover data from raw closed-walk or successor-cycle boundary-order source hypotheses is now precise: construct `InteriorDualBoundaryRootAdjDistancePeelData` on the same embedding, or construct an equivalent cycle-breaking well-founded parent witness. The source-side primitive constructor `interiorDualBoundaryRootAdjDistancePeelDataOfClosedWalkAnnulusBoundarySourceCoverSeparated` now spells out the first option: starting from a `PlanarBoundaryClosedWalkAnnulusBoundarySource`, it is enough to supply cover/separated annulus-boundary roots, boundary-free peeled faces, canonical rooted shared-edge coverage of every interior edge, child-face membership with strict root-distance increase, and the ambient two-incidence bound. The graph-level theorem `admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData_of_exists_closedWalkAnnulusBoundarySource_and_coverSeparatedAnnulusBoundaryRootsCanonicalRootedShare` records the same source-side obligation. The refined constructor `interiorDualBoundaryRootAdjDistancePeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRo` then fixes the roots to the source's outer-or-inner boundary faces, fixes the fallback edge to the honest closed-walk fallback, and uses the embedding's two-incidence theorem. Consequently the live source-side obligation is cover/separation of the source boundary-face roots, boundary-free peeled faces, rooted shared-edge coverage of the interior support, and strict root-distance child membership. The parent-oriented sufficiency option is now formalized: `PlanarBoundaryClosedWalkAnnulusBoundarySource.wellFounded_parentRel_boundaryFaceRoots_parentTowardsRoot` proves the acyclic orientation part of the sufficiency direction: once the source boundary-face roots cover and separate the interior-dual components, the canonical shortest-path parent map toward those roots is well founded. Lean also packages the parent-oriented fixed-root route as `interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsCa` and its graph-level theorem `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover`. Thus the source-fixed sufficiency target can now be attacked as IDBRAD from the raw rooted shared-edge cover; canonical parent child-membership is derived internally by `canonicalParentChildMembership_of_canonicalParentCover`. The older `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` theorem remains a compatibility endpoint for callers already carrying the derived child-membership field. The actual successor-cycle boundary-order shell now has the matching lowering package `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover`, `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover`, and the direct graph-level endpoints `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_direct`. All four lower through `PlanarBoundaryClosedWalkAnnulusBoundarySource.ofDartSuccessorCycleFields`, so once the live successor-cycle shell carries this concrete source-fixed canonical-parent cover package, no further projection step remains before full Theorem 4.9 synthesis. The same layer now has a proved empty-interior-edge base case:

interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsNo and admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges construct the source-fixed parent package with peelFaces=\emptyset whenever the source boundary-face roots cover and separate and interiorEdgeSupport=\emptyset. Hence the nondegenerate burden is now precisely the pairwise-unique shared-edge selector input, root cover/separation, boundary-free peeled faces, and rooted shared-edge coverage of actual interior edges. The empty-support case obtains the required shared-edge uniqueness internally from pairwiseUniqueSharedInteriorEdges_of_interiorEdgeSupport_eq_empty, so it does not assume pairwise uniqueness as an independent source hypothesis. The nonempty-support branch now has a constructor-side witness theorem instead of only selector-level consequences. The generic lemma exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_canonicalParentCover_of_mem_interiorEdgeSupport shows that any actual interior edge covered by the canonical rooted shared-edge image is covered by a peeled non-root face with a real shortest-path parent, and that the rooted shared edge on that face is exactly the original interior edge. Its nonempty-support version exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_canonicalParentCover_of_interiorEdgeSupport_nonempty makes the nondegenerate forest branch explicit. The closed-walk source specializations exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov of_mem_interiorEdgeSupport and exists_nonroot_peelFace_parent_rootedSharedInteriorEdge_eq_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalPa of_interiorEdgeSupport_nonempty instantiate the witness with the source boundary-face roots and source fallback edge. Therefore a nonempty canonical-parent cover cannot hide in the fallback case. The constructor layer now also has parent-side membership for canonical shared edges and canonicalParentChildMembership_of_canonicalParentCover, which turns the raw cover plus the two-incidence bound into the parent child-membership field. The remaining hard source task is the actual boundary-order proof of the pairwise-unique shared-edge selector input, root cover/separation, boundary-free peeled faces, and rooted shared-edge cover. The nondegenerate raw-cover constructor has now been tested directly on the larger chained-diamonds honest source. The regression chainedDiamondsWithTriangle_interiorEdgeSupport_nonempty records genuine interior support, and chainedDiamondsWithTriangleForcingInteriorEdgeWitness packages cdt12 as an interior edge all of whose incident faces already touch selected-boundary support. The theorem false_of_chainedDiamondsClosedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeCover then calls interiorDualBoundaryRootParentPeelDataOfClosedWalkAnnulusBoundarySourceBoundaryFac on the canonical chained-diamonds source and derives a contradiction from the forcing-edge obstruction. Thus this concrete nonempty-interior closed-walk benchmark cannot instantiate the source-fixed raw parent constructor: the missing datum is a genuine boundary-free incident peeled face, not another downstream selector or quotient wrapper. The obstruction is now source-level rather than merely benchmark-specific. PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalk.three_le_faceBoundary_card proves that every honest nonempty simple facial closed walk has at least three boundary edges. Combining that lower bound with the terminal-face theorem for PlanarBoundaryHeightOrderedFacePeelWitnessData, Lean proves false_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_interiorEdgeSupport_nonempty. Indeed, the raw source-fixed constructor lowers to a height-ordered witness cover; any nonempty interior support gives a terminal peeled face whose non-witness boundary edges already lie in selected-boundary support. Since the raw cover also requires peeled faces to be disjoint from selected-boundary support, the terminal face boundary is contained in the singleton witness edge, contradicting the closed-walk lower bound. Therefore the current honest closed-walk source plus boundary-free peeled-face interface has no positive nondegenerate raw-cover instantiation; the sufficiency route must change this source-side contract or replace the terminal parent condition before a genuine nonempty benchmark can exist. The degeneracy is also exposed as a usable route theorem: `interiorEdgeSupport_eq_empty_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover` concludes `interiorEdgeSupport=\emptyset` from the same raw cover hypotheses. This makes the existing no-interior constructor the only possible honest closed-walk source-fixed instance of that raw-cover contract, rather than merely a base case next to an open nonempty branch. The obstruction has also been lifted to the adjacent-distance/root-parent interface itself. The theorem `InteriorDualBoundaryRootAdjDistancePeelData.false_of_mem_interiorEdgeSupport_of_faceBoundaryClosedWalkGeometry` instantiates the existing terminal-card obstruction from `PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalkGeometry`: a covered interior edge would force a peeled face with at most one boundary edge, but the honest closed-walk lower bound gives at least three boundary edges on every face. Consequently `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_interiorEdgeSupport_nonempty_of_faceBoundaryClosedWalkGeometry` rules out any IDBRAD package on an honest closed-walk embedding with genuine interior support. Thus the defect is not merely a bad source-fixed raw-cover constructor; the current boundary-free terminal peeling interface must be changed before it can support a nondegenerate sufficiency route. This base case is now instantiated concretely in `Theorem49PlanarBoundaryAnnulusRootInteriorDualPositiveRegression.lean`. The finite witness `twoTriangleAnnulusGraph` has two disjoint triangular closed-walk boundary components, source boundary-face roots equal to all faces by `twoTriangleAnnulusBoundaryFaceRoots_eq_univ`, an empty interior dual, and empty `interiorEdgeSupport`. Consequently `twoTriangleClosedWalkSourceInteriorDualBoundaryRootParentPeelData` and `admitsPlanarBoundaryInteriorDualBoundaryRootParentPeelData_twoTriangleAnnulusGraph` are genuine positive source-fixed sufficiency checks, complementary to the two-band obstruction where separation and boundary-free cover fail. The same base case now reaches the selector surface used by the replacement-construction lane: `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData` extracts a `BoundaryFreeIncidentFaceSelector` from any canonical-parent payload while retaining the parent cover data: `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_faceOf_mem_peelFaces` keeps every selected face inside the parent package's `peelFaces`, and `rootedSharedInteriorEdge_boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_faceOf_eq` identifies the selected face's rooted shared edge with the original interior edge. The construction-layer compatibility theorem `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_peelFaces_subset` shows that the selector's finite image introduces no extra peeled faces. `boundaryFreeIncidentFaceSelectorOfClosedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteri` is the source-fixed empty-support specialization, and `twoTriangleClosedWalkSourceBoundaryFreeIncident` instantiates that selector on the concrete two-triangle source. The bridge is now strict-descent compatible as well: `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_injective` proves injectivity on interior edges, `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_selectedWitnessEdge_eq` identifies the selector's inverted witness with the canonical rooted shared edge, and `boundaryFreeIncidentFaceSelector_of_interiorDualBoundaryRootParentPeelData_strictDescent` turns parent child-membership into selector `hrest` using root distance. The route theorem `theorem49HeightOrderedPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `via_boundaryFreeSelector`, together with `exists_theorem49HeightOrderedPositiveProjectedGeometr` `of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMem` `and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector`, already exposes the packaged height-ordered positive geometry surface on that shell, and `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `via_boundaryFreeSelector` therefore composes the source-fixed parent hypotheses through the selector-to-construction lane directly. The selector path now also has the same-embedding and graph-level full closures `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `via_boundaryFreeSelector`, the route-facing raw quotient/error corollary `theorem49BoundaryRawQuotientErrorPackage_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `via_boundaryFreeSelector` and the graph-level endpoint and raw quotient/error wrappers `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector`, as well as `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector`, `exists_theorem49BoundaryRawQuotientErrorPackage_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector`, where the carrier side is the local `HasUnblockedInteriorEndpoint` witness. So graph-level source data plus the local endpoint witness can use the selector path all the way to full theorem 4.9 synthesis and the graph-level raw quotient/error package without first repackaging the carrier as raw selected-boundary nonemptiness. The live successor-cycle selected-arc shell now inherits the same stricter face-membership selector package too via `theorem49HeightOrderedPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_via_boundaryFreeSelector`, `theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_via_boundaryFreeSelector`, `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_via_boundaryFreeSelector`, `theorem49BoundaryRawQuotientErrorPackage_` of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_via_boundaryFreeSelector, exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, and exists_theorem49HeightOrderedPositiveProjectedGeometryOn_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, exists_theorem49BoundaryRawQuotientErrorPackage_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector. But under the current honest-source semantics that stronger endpoint-consuming shell is also vacuous: not_hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov not_hasUnblockedInteriorEndpoint_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover, and the matching graph-level no-coexistence theorems not_exists_embedding_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov and_hasUnblockedInteriorEndpoint and not_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_hasUnblockedInteriorEndpoint show that the stricter child-membership shell cannot actually coexist with the named local endpoint witness on the same honest source surfaces. So these selector/projected face-membership wrappers are presently route-diagnostic, not inhabited. More sharply, the generic converse theorems not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov of_exists_embedding_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint and not_forall_some_successorCycleBoundaryFaceRootsCanonicalPa of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_hasUnblockedInteriorEndpoint, together with the seed-free shared-interior-pair instances not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint_sharedInteriorPair and not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentShared of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_hasUnblockedInteriorEndpoint_sharedInteriorPair, show that no Tait coloring or exact v23 residual seed is needed to refute a universal derivation of that stronger shell. The weaker raw canonical-parent shared-edge-cover shell is no longer only a projected endpoint either: theorem49HeightOrderedPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_ via_boundaryFreeSelector, theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector, theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector, theorem49BoundaryRawQuotientErrorPackage_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector, exists_theorem49HeightOrderedPositiveProjectedGeometryOn_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, and exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, exists_theorem49BoundaryRawQuotientErrorPackage_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector. So once the honest closed-walk shell carries that raw cover together with HasUnblockedInteriorEndpoint, the selector lane already closes at the packaged height-ordered positive geometry surface, full Theorem49BoundaryRootSynthesis, and the graph-level raw quotient/error package on the same embedding; the remaining source-facing selector burden there is only to derive the raw cover and endpoint witness themselves. The same shared-interior-pair benchmark now also bundles the whole local selector/canonical-parent burden at once: not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair and not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair show that honest source data or the live successor-cycle selected-arc shell, even after adding a Tait coloring, still do not universally force any of the local BoundaryFreeIncidentFaceSelector, raw canonical-parent shared-edge cover, stricter canonical-parent face-membership cover, or the previously known same-embedding residual, height-ordered, collar-layer, attached-certificate, well-founded, or annulus-collar geometry surfaces from HasUnblockedInteriorEndpoint alone. So this endpoint-only obstruction now blocks the whole local selector/parent ladder, not just one stronger shell at a time. And this is no longer only a benchmark-facing statement: not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint and not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint lift that whole bundled failure pattern to reusable witness-based converses, so any same-embedding example with honest-source or live successor-cycle data, a Tait coloring, and HasUnblockedInteriorEndpoint but without the local selector/canonical-parent/known-geometry package already refutes a universal derivation of that whole burden. Theorem49ResidualBoundaryObstruction.lean now tightens the same diagnosis on the exact one-collar/v23 shared-interior-pair shell itself via exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_sharedInteriorPair, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_sharedInteriorPair, not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair, and not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair. So even after adding exact one-collar current-boundary data and the exact v23 residual seed, the live shared-interior-pair benchmark still fails the whole bundled local selector/canonical-parent/known-geometry workaround class in one theorem, not merely its individual branches. And this exact-shell bundled obstruction is now reusable too: not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState and not_forall_some_selectorOrCanonicalParentOrKnownSameEmbeddingGeometry_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState lift the same exact one-collar/v23 bundled failure pattern to witness-based converses, so any same-embedding exact-shell example with a Tait coloring, HasUnblockedInteriorEndpoint, and nonempty V23ResidualBoundaryInitialState but without the selector/canonical-parent/known-geometry package already refutes a universal derivation of that exact-shell burden. The selector lane now also closes directly at full theorem 4.9 on its own construction surface: theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint and BoundaryFreeIncidentFaceSelector.theorem49BoundaryRootSynthesis_of_strictDescent_and_hasUnblockedInteriorEndpoint show that once an injective strict-descent selector and a local endpoint witness exist on an embedding, no further projected packaging remains below the selector-built annulus construction. The actual successor-cycle boundary-order shell now has the matching raw-cover selector route too: theorem49HeightOrderedPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector, theorem49BoundaryRootNonemptyProjectedSynthesis_ of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector, theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector, theorem49BoundaryRawQuotientErrorPackage_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector, and exists_theorem49HeightOrderedPositiveProjectedGeometryOn_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector, exists_theorem49BoundaryRawQuotientErrorPackage_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_boundaryFreeSelector. So once the live successor-cycle shell carries the raw canonical-parent shared-edge cover together with HasUnblockedInteriorEndpoint, it already reaches the selector-built annulus-construction lane, the packaged height-ordered positive geometry surface, full Theorem49BoundaryRootSynthesis, and the graph-level raw quotient/error package directly, without first detouring through residual-boundary packaging; the remaining source-facing selector burden is only to derive that injective strict-descent selector package itself on the live shell. The same raw-cover selector lane now also exposes the route-facing terminal selected-face break itself: exists_terminal_face_boundary_remainders_and_disjoint_of_interiorDualBoundaryRootParentPeelData_via_boundaryFreeSelector lifts the generic selector terminal-face theorem to any canonical-parent package, and the source/raw/live-shell specializations exists_terminal_face_boundary_remainders_and_disjoint_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov via_boundaryFreeSelector, exists_terminal_face_boundary_remainders_and_disjoint_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover via_boundaryFreeSelector and exists_terminal_face_boundary_remainders_and_disjoint_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_via_boundaryFreeSelector, together with exists_terminal_face_boundary_remainders_and_disjoint_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_boundaryFreeSelector show that both the stricter canonical-parent face-membership shell and the raw canonical-parent cover shell, plus HasUnblockedInteriorEndpoint, already yield a peeled face whose non-witness remainders are ambient-boundary edges while its full face boundary remains disjoint from `selectedBoundarySupport`. Thus the live selector compatibility lane now reaches an explicit same-shell terminal boundary-break witness before any residual-boundary or collar-layer repackaging, even before discarding the child-membership refinement. The weaker boundary-root adjacency-distance shell now also factors explicitly through this same selector lane: `exists_terminal_face_boundary_remainders_and_disjoint_of_interiorDualBoundaryRootAdjDistancePeelData_via_boundaryFreeSelector`, `theorem49HeightOrderedPositiveProjectedGeometryOn_of_interiorDualBoundaryRootAdjDistancePeelData_via_boundaryFreeSelector`, `theorem49HeightOrderedPositiveProjectedGeometryOn_of_planarBoundaryAnnulusRootAdjDistancePeelData_via_boundaryFreeSelector`, `theorem49BoundaryRootNonemptyProjectedSynthesis_of_interiorDualBoundaryRootAdjDistancePeel` `via_boundaryFreeSelector`, and `theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusRootAdjDistancePeelData_via_boundaryFreeSelector` show that once generic boundary-root adjacency-distance data and `HasUnblockedInteriorEndpoint` are present on an embedding, the canonical upgrade to the parent shell already recovers the selector terminal boundary-break, height-ordered positive geometry, and projected Theorem 4.9 endpoint without leaving the selector-built annulus-construction architecture. The parent-oriented source packet now also reaches the route endpoint: `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `direct` packages the fixed source, source boundary-face roots, canonical parent child-membership, and a nonempty purified carrier into `Theorem49BoundaryRootNonemptyProjectedSynthesis`; the graph-level counterpart is `exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov` `with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct`. The wheel-with-inner-triangle regression now instantiates this source-fixed parent bridge at the actual endpoint: `wheelWithInnerTriangle_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover` specializes the direct theorem to the concrete wheel embedding and its explicit Tait coloring. The companion theorem `wheelWithInnerTriangle_source_to_theorem49_and_height_cycle_contradiction_of_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover` lowers the same parent hypotheses back to height-ordered witness-cover data and therefore contradicts the existing wheel height-cycle obstruction. Thus `not_exists_wheelWithInnerTriangle_closedWalkSourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_and_theorem49BoundaryRootNonemptyProjectedSynthesis` is the endpoint-level soundness check for this parent route: the wheel keeps its honest source, Tait coloring, and unblocked endpoint, but cannot instantiate the source-fixed canonical-parent hypotheses. The coverage sub-obligation is now separately accounted for: `PlanarBoundaryClosedWalkAnnulusBoundarySource.rootSetCovers_boundaryFaceRoots_of_interiorDualBoundaryRootAdjDistancePeelData` proves that any generic boundary-root IDBRAD datum already makes the source boundary-face roots cover all interior-dual components. The separation sub-obligation is not automatic: `PlanarBoundaryClosedWalkAnnulusBoundarySource.not_rootSetSeparated_` boundaryFaceRoots_of_reachable_outerBoundaryFace_innerBoundaryFace shows that an outer-to-inner interior-dual reachability path refutes separated source boundary-face roots. The stronger source-side obstruction PlanarBoundaryClosedWalkAnnulusBoundarySource.not_reachable_outerBoundaryFace_innerBoundaryFace_of_interiorDualBoundaryRootAdjDistancePeelData shows that any generic IDBRAD datum on the closed-walk source already rules out such a path, and PlanarBoundaryClosedWalkAnnulusBoundarySource.not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_reachable_outerBoundaryFace_innerBoundaryFace packages this as nonexistence of the generic datum under outer-to-inner interior-dual reachability. The same obstruction is now exposed in route-facing form by not_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_reachable_outerBoundaryFace_innerBoundaryFace and not_forall_interiorDualBoundaryRootAdjDistancePeelData_of_exists_closedWalkAnnulusBoundarySource_and_reachable_outerBoundaryFace_innerBoundaryFace. The live successor-cycle shell now inherits even that earlier source boundary-face separation failure directly: not_rootSetSeparated_boundaryFaceRoots_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_reachable_outerBoundaryFace_innerBoundaryFace and not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_rootSetSeparated_boundaryFaceRoots_and_reachable_outerBoundaryFace_innerBoundaryFace show that if actual boundary reachability plus successor-cycle data still extract an outer-to-inner interior-dual path, then the route already dies at the source hsepRoots premise itself, before any rooted shared-edge cover or generic IDBRAD payload is introduced. The live successor-cycle shell now inherits the same source-side obstruction directly: not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_reachable_outerBoundaryFace_innerBoundaryFace and not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_reachable_outerBoundaryFace_innerBoundaryFace show that if the source extracted from actual boundary reachability and successor-cycle data still has an interior-dual path from one outer boundary face to one inner boundary face, then the route already fails before the source-fixed rooted-cover IDBRAD bridge or any theorem-4.9 endpoint is used. The concrete two-band triangular annulus benchmark now realizes this live successor-cycle failure on an actual dart-successor shell: twoBandAnnulusDartSuccessorCycleGeometry and twoBandAnnulusDartSuccessorCycleGeometry_selectedBoundaryArcOnFace instantiate the live surface, twoBandDartSuccessorClosedWalkAnnulusBoundarySource_boundaryData_eq and twoBandDartSuccessorBoundaryFaceRoots_eq_univ show that this live shell extracts the same annulus boundary split and the same full six-face boundary-root set as the closed-walk source, rootSetCovers_twoBandDartSuccessorBoundaryFaceRoots makes the coverage half explicit there too, and not_rootSetSeparated_twoBandAnnulusBoundaryFaceRoots_via_dartSuccessor_source_reachability and not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_twoBandAnnulus_via_dartSuccessor_source_reachability show that the same outer-to-inner interior-dual adjacency already kills only the source hsepRoots premise, and hence generic IDBRAD data, before any rooted-cover or theorem-4.9 packaging is introduced. The obstruction is now formalized one premise deeper on the live source-fixed rooted-cover shell as well: false_of_boundaryReachabilityData_ and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_and_reachable_outerBoundaryFace_innerBoundaryFace and not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_and_reachable_outerBoundaryFace_innerBoundaryFace show that if actual successor-cycle data still extract an outer boundary face reaching an inner boundary face, then no rooted shared-edge face-membership package can exist there at all. The same two-band benchmark now instantiates that sharper failure directly: not_exists_twoBandAnnulus_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_via_dartSuccessor_source_reachability rules out every boundary-free peeled-face set, rooted shared-edge cover, and strict child-membership witness package on the actual successor-cycle shell. Theorem49PlanarBoundaryAnnulusRootInteriorDualObstructionRegression.lean defines the two-band triangular annulus twoBandAnnulusGraph, proves its closed-walk source twoBandClosedWalkAnnulusBoundarySource, and proves that the selected outer face 0 and selected inner face 3 are adjacent in the interior dual through the middle edge tbaM01. Consequently not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_twoBandAnnulus_via_source_reachability instantiates the source-side IDBRAD obstruction on a finite graph with an honest closed-walk annulus source. The same concrete source now proves the fixed-root calibration directly: twoBandAnnulusBoundaryFaceRoots_eq_univ identifies the source boundaryFaceRoots with all six faces, rootSetCovers_twoBandAnnulusBoundaryFaceRoots proves coverage, and not_rootSetSeparated_twoBandAnnulusBoundaryFaceRoots proves separation false using the shared middle edge tbaM01. Any future derivation from the source shell must therefore add or prove an actual outer/inner face-separation condition. The same regression now proves the lower-level peel obstruction twoBandAnnulus_peelFaces_eq_empty_of_boundaryFree: every face touches selected boundary support, so a boundary-free peel-face set is empty. Since tbaM01 is still an interior edge, not_interiorEdgeSupport_subset_boundaryFreePeelImage_twoBandAnnulus rules out an interior-edge-support cover by any image of those faces. The same obstruction is now registered in the common forcing-edge interface: twoBandAnnulusForcingInteriorEdgeWitness packages tbaM01, not_nonempty_boundaryFreeIncidentFaceSelector_twoBandAnnulus rules out the selector surface, and not_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness rules out the annulus-root adjacency-distance package on this embedding. The same forcing-edge guard now covers the parent-oriented sufficiency target: everyInteriorEdgeHasBoundaryFreeIncidentFace_of_interiorDualBoundaryRootParentPeelData extracts a boundary-free incident face from any canonical-parent boundary-root package, while not_nonempty_interiorDualBoundaryRootParentPeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness and not_nonempty_planarBoundaryAnnulusRootParentPeelData_twoBandAnnulus_via_forcingInteriorEdgeWitness instantiate the failure on the same two-band source. The same forcing witness now blocks the live theorem-4.9 raw canonical-parent cover shell one layer earlier: false_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_forcingInteriorEdgeWitness and not_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_ `and_forcingInteriorEdgeWitness` show that once a successor-cycle shell carries a forcing interior edge, no raw canonical-parent shared-edge cover package can exist there at all. The two-band benchmark now instantiates that sharper failure directly as `not_exists_twoBandAnnulus_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_forcingInteriorEdgeWitness`. The same benchmark also kills the older raw carrier repair on that exact shell: `not_exists_twoBandAnnulus_boundaryFaceRootsCanonicalParentSharedEdgeCover_and_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport` shows that even endpoint-support separation plus a nonempty raw interior-edge endpoint carrier cannot rescue the live parent-cover lane on the actual successor-cycle benchmark. The generic IDBRAD layer now has a source-side terminal obstruction independent of the concrete annuli. In `Theorem49PlanarBoundaryPeeling.lean`, `PlanarBoundaryHeightOrderedFacePeelWitnessData.exists_terminal_face_selectedBoundary_remainders` shows that any nonempty height-ordered witness cover has a terminal peeled face whose non-witness remainders are all selected-boundary edges. The IDBRAD boundary-free peeled-face invariant then forces those remainders to be empty: `InteriorDualBoundaryRootAdjDistancePeelData.exists_terminal_peelFace_witness_erase_eq_empty_of_mem_interiorEdgeSupport`. Consequently `InteriorDualBoundaryRootAdjDistancePeelData.exists_peelFace_card_le_one_of_mem_interiorEdgeSupport` says that any IDBRAD datum covering even one interior edge contains a peeled face of boundary cardinality at most one. Future source-side derivations must therefore produce this terminal singleton-edge face, or replace IDBRAD by a different witness-ownership surface. The same criterion is now instantiated on the wheel-with-inner-triangle source without passing through the forcing-edge API. The regression proves `wheelWithInnerTriangleFaceBoundary_card_eq_three` and `wheelWithInnerTriangle_two_le_faceBoundary_card`; together with `wit01_mem_interiorEdgeSupport`, these give `not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_wheelWithInnerTriangle_via_terminal_card_lower_bound`. The packaged source bridge `wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData_via_terminal_card` keeps the honest closed-walk source, Tait coloring, and local unblocked endpoint while deriving IDBRAD nonexistence from the terminal-card lower-bound obstruction. This is now lifted to reusable failed converses on both live source shells: `exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData_wheelWithInnerTriangleGraph` and `not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangleGraph` show that honest closed-walk source data, a real Tait coloring, and `HasUnblockedInteriorEndpoint` still do not generically force same-embedding IDBRAD on the wheel graph. Likewise `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData_wheelWithInnerTriangleGraph` and `not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangleGraph` collapse the same converse on the actual successor-cycle boundary-order shell. This is now tied to the real projected Theorem 4.9 endpoint on the same benchmark. The wheel file supplies `wheelWithInnerTriangleGraph_edgeSet_fintype`, and the regression names `hasUnblockedInteriorEndpoint_wheelWithInnerTriangle`. With those facts, `wheelWithInnerTriangle_theorem49BoundaryRootNonemptyProjectedSynthesis_of_interiorDualBour` instantiates the closed-walk source–IDBRAD bridge for the concrete wheel Tait coloring, and `wheelWithInnerTriangle_theorem49BoundaryRawQuotientErrorPackage_of_interiorDualBoundaryRoo` instantiates the raw Definition 4.8 quotient/error endpoint. The theorem `wheelWithInnerTriangle_source_to_theorem49_and_terminal_card_contradiction_of_interiorDualBoundaryRootAdjDistance` then records that the same hypothetical IDBRAD datum would also contradict the terminal-card obstruction. The endpoint-level form is now explicit: `not_exists_wheelWithInnerTriangle_closedWalkSource_and_interiorDualBoundaryRootAdjDistancePeelData_and_theorem49BoundaryRoot` rules out coexistence of the closed-walk source, same-embedding IDBRAD datum, and projected Theorem 4.9 endpoint on the wheel. The packaged positive-source statement `wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeel` `based_projectedSynthesis` retains the honest source, Tait coloring, and unblocked endpoint while ruling out an IDBRAD-based projected endpoint instantiation. The same regression now performs the analogous check for the source-fixed canonical-parent route: `wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_boundaryFaceRootsCanonicalParent_based_projectedSynthesis` keeps those positive source facts while ruling out a projected endpoint obtained from the source boundary-face roots and canonical parent child-membership. With the resulting acyclic datum in hand, `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint` reaches the projected endpoint once the local endpoint witness is supplied. The successor-cycle source version is `theorem49BoundaryRootNonemptyProjectedSynthesi` `of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint`. The route file now also exposes the raw quotient/error form of this same source-side bridge: `theorem49BoundaryRawQuotientErrorPackage_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint` and `theorem49BoundaryRawQuotientErrorPackage_of_boundaryReachabilityData_and_dartSuccessor` `and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedIr` return the raw Definition 4.8 representative plus selected-boundary-kernel decomposition for any Kirchhoff chain on the same embedding. Lean now also exposes the corresponding graph-level existential wrappers on this clean source-facing lane: `exists_theorem49BoundaryRawQuotientEr` `of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_` `and_hasUnblockedInteriorEndpoint_direct` and `exists_theorem49BoundaryRawQuotientErrorPackage` `of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundar` `and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint_direct`. The same root-distance source layer now also accepts the older carrier-side repairs directly: endpoint-support separation plus raw `interiorEdgeEndpointSupport` nonemptiness, or peeled-face endpoint no-touch on `planarBoundaryAnnulusConstruction_of_boundaryReachabilityDat` `and_interiorDualBoundaryRootAdjDistancePeelData_rootDistance` plus the same raw nonemptiness. Thus the source/interior-dual projected endpoint no longer has to be called only with a prepackaged `HasUnblockedInteriorEndpoint`; it can consume the more geometric separation or root-distance no-touch obligations itself. That raw carrier can now be supplied at the edge-support level: `interiorEdgeEndpointSupport_nonempty_of_interiorEdgeSupport_nonempty` turns nonempty `interiorEdgeSupport` into nonempty raw endpoint support, and the closed-walk root-distance bridge has direct projected-synthesis variants for both endpoint-support separation and root-distance peeled-face no-touch with this edge-level carrier hypothesis. The same bridge family now also accepts `HasUnblockedInteriorEndpoint`, with fixed-embedding, graph-level, nonempty-package, and projected-synthesis constructors from annulus collar geometry plus a local unblocked endpoint. Thus the remaining source-side burden is to derive those geometry-plus-carrier hypotheses, or directly the unblocked endpoint witness, not to repack the downstream endpoint;

- `Theorem49PositiveGeometricSourceReplacementRoute.lean` connects the replacement surface to both the honest closed-walk annulus source shell and the successor-cycle boundary-order source shell. It defines `Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometry`, which carries an honest source, same-embedding annulus collar geometry, and nonempty purified carrier, and it defines `Theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometry`, which carries boundary reachability, successor-cycle facial data, selected-boundary arcs, same-embedding annulus collar geometry, and nonempty purified carrier. The successor-cycle package lowers to the closed-walk route package for traceability; both packages lower to the replacement graph packages and now also expose graph-level full theorem-4.9 synthesis directly, so the projected synthesis endpoint is no longer the top of those source-facing surfaces. The constructive source-facing target is now explicit without canonical-choice or one-collar fields. The same route file now also gives direct closed-walk and successor-cycle full theorem-4.9 constructors from `HasUnblockedInteriorEndpoint`, together with the matching graph-level existential closures from either a surviving carrier or one local endpoint disjoint from selected ambient-boundary support. The older endpoint-separation surface now supplies this witness directly: `hasUnblockedInteriorEndpoint_of_endpointSupport_disjoint_and_nonempty` turns raw `interiorEdgeEndpointSupport` nonemptiness plus disjointness from selected-boundary endpoint support into `HasUnblockedInteriorEndpoint`, and the route file now has closed-walk and successor-cycle full theorem-4.9 wrappers for that raw endpoint-support form. The same is now true for the canonical peeled-face endpoint no-touch criterion on annulus-collar data. The closed-walk and successor-cycle annulus-collar package constructors still accept the same raw endpoint-support-separated input directly, so this is available before choosing a Tait coloring;

- the same endpoint-support bridge is now connected to the concrete collar geometry layer. The theorem `PlanarBoundaryAnnulusCollarGeometry.hasUnblockedInteriorEndpoint_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` turns peeled-face endpoint no-touch on `planarBoundaryAnnulusConstruction_of_annulusCollarGeometrydata`, plus raw `interiorEdgeEndpointSupport` nonemptiness, into the named `HasUnblockedInteriorEndpoint` witness. The replacement positive package therefore has direct finite-collar, height-ordered, and projected-synthesis constructors from annulus collar geometry plus this construction-level no-touch surface. The route-facing replacement file now exposes the same input at the honest source layer: closed-walk and successor-cycle source data plus same-embedding annulus collar geometry, canonical peeled-face endpoint no-touch, and raw endpoint-support nonemptiness directly package the source-facing annulus-collar positive predicates and reach the nonempty projected synthesis endpoint. The same source-layer wrappers now also produce the graph-level finite-collar and height-ordered replacement packages directly, so the peeled-face no-touch input can be consumed by the replacement API without an intermediate route-package lowering step;

- `Theorem49PositiveGeometricSourceReplacementRegression.lean` calibrates the new direct target against the shared-interior-pair annulus. The same model has boundary reachability, successor-cycle selected arcs, a Tait coloring, and nonempty purified carrier, but it has neither `Theorem49HeightOrderedPositiveProjectedGeometryOn` nor `Theorem49CollarLayerPositiveProjectedGeometryOn`. The same file now also proves `not_theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn_sharedInteriorPair` and `not_theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn_sharedInteriorPair`, so those facts do not universally construct either the direct replacement surface or the source-facing closed-walk/successor-cycle annulus-collar packages. The same regression now strengthens the carrier side to the local endpoint witness: `hasUnblockedInteriorEndpoint_sharedInteriorPair` and `not_forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that even explicit `HasUnblockedInteriorEndpoint` data, together with the same source and coloring fields, does not force any direct or route-facing replacement package. The closed-walk-source-level theorem `not_forall_any_replacementPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` strengthens this further: the honest closed-walk annulus source itself, a Tait coloring, and the endpoint witness still do not force any current direct or route-facing replacement package. The companion theorem `not_forall_some_witnessCoverData_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` rules out an even earlier shortcut: those same hypotheses also do not universally produce either raw height-ordered or collar-layer witness-cover datum. The bundled theorem `not_forall_some_currentSufficientSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` then rules out every current same-embedding synthesis endpoint from those hypotheses: height-ordered data, collar-layer data, the attached certificate, well-founded witness data, and annulus-collar geometry. A positive theorem must add real annulus-collar, height-ordered, or collar-layer witness-cover geometry. The wheel-with-inner-triangle regression now gives the complementary cycle obstruction at the honest-source level: `wheelWithInnerTriangleClosedWalkAnnulusBoundarySource` inhabits the closed-walk annulus source interface on the wheel embedding, while `not_forall_some_acyclicWitnessCover_or_annulusCollarGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` shows that this source, a Tait coloring, and nonempty purified carrier still do not force well-founded, height-ordered, collar-layer, or weak annulus-collar geometry. The same strict height-cycle now reaches the direct replacement endpoints themselves: `not_theorem49HeightOrderedPositiveProjectedGeometryOn_wheelWithInnerTriangle` and `not_theorem49CollarLayerPositiveProjectedGeometryOn_wheelWithInnerTriangle` refute the two fixed-embedding positive replacement predicates, while `not_forall_some_replacementPositiveProjectedGeometry_or_previousBoundaryWitness_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` also rules out deriving either direct replacement package or repaired previous-boundary witness geometry from just honest source data, Tait colorability, and the purified carrier. The generic cycle-obstruction lemmas in `Theorem49PlanarBoundaryPeeling.lean` explain the proof obligation exposed by this example: every proposed source-to-witness theorem must either put one of the remainder witnesses on selected boundary support, identify two witness choices so the apparent finite cycle collapses, or provide a genuine well-founded orientation;

- `Theorem49PositiveGeometricSourceConstruction.lean` gives the first positive construction of the direct replacement surface. On the two-collar annulus benchmark, Lean proves `counterEmbedding_selectedBoundaryInteriorEdgeEndpointVertices_nonempty` from the intermediate interior edge. `Theorem49PositiveGeometricSourceConstruction.lean` now also identifies the carrier exactly: `counterEmbedding_interiorEdgeSupport_eq` says the intermediate edge is the whole interior support, `counterEmbedding_interiorEdgeEndpointSupport_eq` and `counterEmbedding_selectedBoundaryEndpointSupport_eq` compute the raw interior endpoints and selected-boundary endpoints, and `counterEmbedding_endpointSupport_disjoint_selectedBoundarySupport` proves the disjointness that yields the named endpoint witness `counterEmbedding_hasUnblockedInteriorEndpoint`. This calibrates the benchmark against the first-order carrier obligation used by the replacement API. Lean then combines the carrier with annulus collar geometry to inhabit both `Theorem49CollarLayerPositiveProjectedGeometryOn` and `Theorem49HeightOrderedPositiveProjectedGeometryOn`, as well as the corresponding graph-level packages. The benchmark now also has an explicit constant nonzero Tait coloring, and Lean proves `counterEmbedding_boundaryRootNonemptyProjectedSynthesis` through the minimal annulus-collar/carrier bridge; the concrete two-collar geometry therefore reaches the nonempty projected theorem-4.9 endpoint itself. Lean now also exposes this nonvacuous construction at the raw quotient/error endpoint: `counterEmbedding_collarGeometry_explicitTait_nonemptyCarrier_to_theorem49RawQuotientErrorPackage` packages the collar geometry, explicit Tait coloring, nonempty purified carrier, nonempty projected synthesis, and the raw Definition 4.8 representative plus selected-boundary-kernel error decomposition for every Kirchhoff chain on the same embedding. The same file now also proves that this benchmark is not an honest closed-walk annulus source: `counterGraph_isAcyclic` identifies the graph as a three-edge matching, `not_nonempty_closedWalkAnnulusBoundarySource_counterEmbedding` blocks source data on the fixed embedding, and `counterEmbedding_rawQuotientErrorPackage_without_closedWalkAnnulusBoundarySource` bundles that source-side obstruction with the raw quotient/error package. It now also exposes that diagnosis at graph level: `counterGraph_explicitTait_positiveProjectedGeometry_and_boundaryRootNonemptyProjectedSynthesis_without_closedWalkAnnulusBoundarySource` packages the explicit Tait coloring, both graph-level direct replacement packages, a graph-level nonempty projected theorem-4.9 endpoint, and failure of honest closed-walk annulus-source admission for `counterGraph` itself. So the current positive benchmark is now explicitly classified as off the honest-source route already at graph level, not only on one chosen embedding. The stronger theorem `counterGraph_explicitTait_positiveProjectedGeometry_and_boundaryRootNonemptyProjectedSynthesis_with_forcingInteriorEdgeWitness_without_boundaryFreeSelector_or_planarBoundaryAnnulusConstructionFaceLayerData` now shows that on the same fixed embedding the direct positive predicates and the projected Theorem 4.9 endpoint already coexist with a forcing interior edge and simultaneous failure of `BoundaryFreeIncidentFaceSelector` and PlanarBoundaryAnnulusConstructionFaceLayerData. So this branch is not only off the honest-source route; even after the projected Theorem 4.9 endpoint it still does not supply the same-embedding selector/construction geometry needed by the present positive route. It now also carries the residual-boundary calibration counterEmbedding_residualBoundaryPositiveProjectedGeometryOn_without_boundaryFreeSelector_or_planarBoundaryAnnulusConstructionFaceLayerData: the same embedding already inhabits Theorem49ResidualBoundaryPositiveProjectedGeometryOn while still failing both BoundaryFreeIncidentFaceSelector and PlanarBoundaryAnnulusConstructionFaceLayerData. So the natural residual/current-boundary wrapper does not repair this selector/construction gap either; on the current positive benchmark it is still strictly below the same-embedding construction shells actually needed downstream. The stronger counterGraph_explicitTait_residualBoundaryPositiveProjectedGeometry_and_boundaryRootNonemptyProjectedSynthesis_with_forcingInteriorEdgeWitness_without_boundaryFreeSelector_or_planarBoundaryAnnulusConstructionFaceLayerData now shows that on the same fixed embedding the residual positive shell and the projected Theorem 4.9 endpoint already coexist with a forcing interior edge and simultaneous failure of both BoundaryFreeIncidentFaceSelector and PlanarBoundaryAnnulusConstructionFaceLayerData. So even after the residual wrapper and the projected endpoint are already present, this branch still does not supply the same-embedding selector/construction geometry needed by the present positive route. It now also carries the repaired-witness calibration counterEmbedding_previousBoundaryWitness_nonemptyProjectedSynthesis_without_parentWitnessOrientation: repaired previous-boundary witness geometry and the explicit endpoint witness coexist with the nonempty projected endpoint, while the stricter construction-side parent-witness orientation still fails on the same collar. The direct replacement endpoint is therefore non-vacuous, but it is weaker than the parent-orientation route; what remains is a general boundary-order construction of the same geometry-plus-carrier package. The same file now also calibrates the forcing-edge lane against this positive benchmark. counterEmbeddingForcingInteriorEdgeWitness packages the intermediate edge em as a concrete ForcingInteriorEdgeWitness, and not_nonempty_boundaryFreeIncidentFaceSelector_counterEmbedding shows that no boundary-free selector exists on the same embedding. Yet counterEmbedding_directPositiveProjectedGeometryOn_and_forcingInteriorEdgeWitness proves that this forcing edge coexists with both direct replacement-positive predicates. The failed universals not_forall_not_nonempty_forcingInteriorEdgeWitness_of_theorem49HeightOrderedPositiveProjectedGeometryOn_counterGraph, not_forall_not_nonempty_forcingInteriorEdgeWitness_of_theorem49CollarLayerPositiveProjectedGeometryOn_counterGraph, and not_forall_not_nonempty_forcingInteriorEdgeWitness_of_directPositiveProjectedGeometryOn_counterGraph therefore record a genuine ceiling on the forcing-edge obstruction: it is a real blocker for the selector and construction layers, but it cannot be promoted to a generic same-embedding obstruction for the direct replacement surface itself. The same benchmark now also states that ceiling directly at those lower route layers: counterEmbedding_directPositiveProjectedGeometryOn_without_boundaryFreeSelector_or_planarBoundaryAnnulusConstructionFaceLayerData shows that both direct replacement-positive predicates coexist on one embedding with failure of the boundary-free selector shell and the downstream construction face-layer package. not_forall_nonempty_boundaryFreeIncidentFaceSelector_ of_directPositiveProjectedGeometryOn_counterGraph and not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_directPositiveProjectedGeometryOn_counterGraph therefore show that direct replacement-positive geometry is not a generic same-embedding source for either selector data or construction-face-layer data. The carrier side has now been factored into the named endpoint predicate HasUnblockedInteriorEndpoint, equivalent to nonempty purified carrier by hasUnblockedInteriorEndpoint_iff_selectedBoundaryInteriorEdgeEndpointVertices_nonempty: nonemptiness is equivalent to an interior face-incidence edge with one endpoint untouched by the selected ambient boundary. Endpoint-support disjointness plus raw interiorEdgeEndpointSupport nonemptiness is now a proved sufficient criterion for this local witness;

- this has now been lifted from an algebraic bridge to the actual purified theorem-4.9 endpoint. Theorem49Synthesis.lean proves theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstruction_of_parentWitness, which packages the full synthesis surface directly from PlanarBoundaryAnnulusConstruction plus ParentWitnessOrientation: raw-source image identity, projected/raw finite generator representations, raw Kirchhoff representatives, and the boundary-kernel decomposition are now available together with the already proved span/separation/kernel/finrank consequences. The companion graph-level theorem exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation means the live boundary-order task is no longer "reach some theorem-4.9 algebraic bridge" but specifically "produce honest parent-witness orientation on a real annulus construction";

- there is now a checked source-level obstruction showing that honest closed-walk annulus-boundary data still does not force even the weakest peeled-interior construction shell. In Theorem49InteriorVerticesEndpointObstruction.lean, the explicit model diamondWithTriangleEmbedding already carries PlanarBoundaryClosedWalkAnnulusBoundarySource, but the theorem false_of_planarBoundaryAnnulusConstruction_and_hpeelInterior_diamondWithTriangle shows that any annulus construction with the usual peeled-interior disjointness field is impossible on that embedding: the unique interior edge must be covered by a peeled witness face, yet every face containing it also meets the selected boundary support. Consequently closedWalkAnnulusBoundarySource_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle, not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle, not_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle and closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle record that the honest boundary-order source does not even reach the peeled interior-dual shell, let alone the weakest peeled-interior construction shell; moreover the same embedding already carries the stronger cyclic ordered face-arc witness diamondWithTriangleCyclicOrderedFaceArcEmbeddingData, so the theorems annulusBoundaryReachability_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle and annulusBoundaryReachability_and_cyclicOrderedFaceArcEmbeddingData_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle show that upgrading the source to cyclic ordered face-arc data still does not start the peeled interior-dual route or even the weakest peeled construction shell on this model; the same diamond witness now survives after weakening that source shell to the bundled `PlanarBoundarySelectedBoundaryArcGeometry`. The theorems `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` and `annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData_diamondWithTriangle` show that even annulus roots plus one contiguous selected-boundary arc on each chosen face-boundary run still do not start the peeled interior-dual route or its weakest construction shell on this model; the explicit two-face model `doubleStarEmbedding` now blocks the source route strictly earlier. It carries both `PlanarBoundaryAnnulusBoundaryReachabilityData` and `PlanarBoundaryCyclicOrderedFaceArcEmbeddingData`, but `not_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_doubleStarEmbedding` shows that the same embedding still admits no honest facial closed-walk annulus source. The abstract summary theorem `not_forall_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_of_nonempty_planarBoundaryAnnulusBoundaryReachabilityData_and_nonempty_planarBoundaryCyclicOrderedFaceArcEmbeddingData` therefore blocks the obvious attempt to derive honest closed walks from annulus roots plus cyclic selected-boundary face arcs;

- the same double-star witness also carries the weaker bundled shell `PlanarBoundarySelectedBoundaryArcGeometry`, and the theorems `exists_embedding_withPlanarBoundaryAnnulusBoundaryReachabilityData_andPlanarBoundarySelectedBoundaryArcGeometry_withoutClosedWalkAnnulusBoundarySource` and `not_forall_nonempty_planarBoundaryClosedWalkAnnulusBoundarySource_of_nonempty_planarBoundaryAnnulusBoundaryReachabilityData_and_nonempty_planarBoundarySelectedBoundaryArcGeometry` show that even annulus roots plus one contiguous selected-boundary arc on each ambient face still do not recover the honest closed-walk annulus source;

- the positive lowering is now explicit in the source file too: `admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_cyclicOrderedFaceArcEmbeddingData_of_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource` and `admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry_of_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource` show that the honest closed-walk annulus source really does imply those weaker annulus-rooted source shells on the same embedding, so the double-star counterexamples are precise failed converses rather than shell mismatches;

- this converse failure is now graph-level too, not only same-embedding: `doubleStarGraph_isAcyclic` and `not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_doubleStarGraph` show that the underlying graph itself cannot support the honest closed-walk annulus source on any embedding, while still admitting the paired shells 'annulus reachability + cyclic ordered face arcs' and 'annulus reachability + selected-boundary arc geometry'; accordingly the new theorems `not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_cyclicOrderedFaceArcEmbeddingData` and `not_forall_` admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry record those failed converses directly at graph level;

- the same graph-level double-star witness now blocks the natural rotation-side repairs too. The file proves not_admitsPlanarBoundaryDartCycleEmbeddingData_doubleStarGraph, not_admitsPlanarBoundaryPureDartCycleEmbeddingData_doubleStarGraph, and not_admitsPlanarBoundaryDartSuccessorCycleEmbeddingData_doubleStarGraph, then packages failed converses such as not_forall_admitsPlanarBoundaryDartCycleEmbeddingData_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry, not_forall_admitsPlanarBoundaryPureDartCycleEmbeddingData_of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry, and not_forall_admitsPlanarBoundaryDartSuccessorCycleEmbedding of_admitsPlanarBoundaryAnnulusBoundaryReachabilityData_and_selectedBoundaryArcGeometry. So replacing the honest closed-walk annulus source by a dart-cycle, pure dart-cycle, or successor-cycle source does not repair the paired annulus-reachability plus selected-arc shell;

- this graph-level statement is now abstracted at the source interface itself: PlanarBoundaryClosedWalkSource.lean proves generic acyclic positive-edge separation theorems for the paired shells 'annulus reachability + cyclic ordered face arcs' and 'annulus reachability + selected-boundary arc geometry', plus existential failed-converse criteria; the double-star results are now concrete instances of that reusable source lemma rather than the only place where the comparison is formalized;

- the shared walk layer now abstracts the acyclic source obstruction itself. The theorem not_nonempty_faceBoundaryClosedWalkGeometry_of_isAcyclic_of_nonempty_edgeSet in WalkPlanarEmbeddingWithBoundaryAcyclic.lean shows that any acyclic graph with at least one edge cannot carry honest facial closed-walk geometry on any boundary-aware embedding, and the FourColor wrapper not_nonempty_planarBoundaryClosedWalkEmbeddingData_of_isAcyclic_of_nonempty_edgeSet exposes the same fact on the route-facing source interface. So the path, star, and peeled-endpoint-touch closed-walk impossibility lemmas are now instances of one reusable theorem rather than repeated local proofs; the same abstraction now reaches the stronger source shells too, via not_admitsPlanarBoundaryDartCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet, not_admitsPlanarBoundaryPureDartCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet, not_admitsPlanarBoundaryDartSuccessorCycleEmbeddingData_of_isAcyclic_of_nonempty_edgeSet, and not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_isAcyclic_of_nonempty_edgeSet, so acyclic positive-edge models now refute the rotation-side and explicit annulus-source interfaces without new witness-specific proofs; the same witness pattern is now packaged at the universal failed-converse level by not_forall_admitsPlanarBoundaryClosedWalkEmbeddingData_of_exists_isAcyclic_of_nonempty_edgeSet_and_property and not_forall_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_of_exists_isAcyclic_of_nonempty_edgeSet_and_property, so the star/path failures for cyclic ordered arcs, selected-boundary arc geometry, cyclic face-boundary vertex walks, and ordered face arcs are now instances of one generic acyclic positive-edge witness criterion rather than separate local contradiction proofs; the interior-dual forcing-edge mechanism is now abstracted as `false_of_interiorDualBoundaryRootAdjDistancePeelData_of_forcing_interiorEdge`, and the weaker construction-shell version is abstracted as `false_of_planarBoundaryAnnulusConstruction_and_hpeelInterior_of_forcing_interiorEdge`, while the positive construction invariant `PlanarBoundaryAnnulusConstruction.exists_peelFace_witnessEdge_eq_and_faceBoundary_disjoint_selectedBoundarySupport_of_mem_interiorEdgeSupport` now states exactly what the weak boundary-support construction shell supplies: each interior edge is witnessed on some peeled face whose whole boundary avoids selected-boundary support. The route-facing obstruction file now packages this as the positive predicate `EveryInteriorEdgeHasBoundaryFreeIncidentFace` and proves `not_nonempty_forcingInteriorEdgeWitness_iff_everyInteriorEdgeHasBoundaryFreeIncidentFace`, so “no forcing interior edge” is an explicit local face-incidence obligation rather than only a failed nonemptiness conclusion. The theorem `everyInteriorEdgeHasBoundaryFreeIncidentFace_of_interiorDualBoundaryRootAdjDistancePeelData` now shows that the interior-dual boundary-root adjacency-distance package already supplies this local obligation directly from its peeled-edge cover and peeled-interior disjointness fields. Consequently `not_nonempty_forcingInteriorEdgeWitness_of_interiorDualBoundaryRootAdjDistancePeelData` is the source-side test for any proposed derivation of that package: a forcing interior edge must be impossible before the route reaches the interior-dual data surface. The theorem `everyInteriorEdgeHasBoundaryFreeIncidentFace_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData` then records that the weak boundary-support construction shell gives exactly that obligation. Its selector form `BoundaryFreeIncidentFaceSelector` is now also formalized, with `nonempty_boundaryFreeIncidentFaceSelector_iff_everyInteriorEdgeHasBoundaryFreeIncidentFace` showing that selector nonemptiness is exactly the same local no-forcing condition. The construction shell yields the selector through `boundaryFreeIncidentFaceSelector_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData`. The same duality is now named without the intermediate predicate: `nonempty_boundaryFreeIncidentFaceSelector_iff_not_nonempty_forcingInteriorEdgeWitness` and `not_nonempty_boundaryFreeIncidentFaceSelector_iff_nonempty_forcingInteriorEdgeWitness` identify boundary-free selector existence exactly with absence of a forcing interior edge. This is useful downstream because annulus-construction code can consume a function returning a selected-boundary-free incident face for each interior edge; it still leaves the harder compatibility requirements between those face choices for later work. The first compatibility bridge is now in `Theorem49BoundaryFreeSelectorConstruction.lean`. It forms the selected peel-face set `BoundaryFreeIncidentFaceSelector.peelFaces`, inverts the selector on that finite image through `BoundaryFreeIncidentFaceSelector.selectedWitnessEdge`, and proves that an injective selector with a strict-distance remainder condition lowers to `PlanarBoundaryAnnulusConstruction` by `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction`. The lemma `BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_hpeelInterior` then recovers the selected-boundary-free peeled-face invariant from the selector itself. The strict-descent part of this bridge now has a finite-cycle diagnostic on the selector surface: `BoundaryFreeIncidentFaceSelector.remainderDependency` names the relation where the selected witness of one peeled face appears as a non-witness, non-boundary remainder edge on another, `BoundaryFreeIncidentFaceSelector.faceDistance_lt_` of_remainderDependency turns each dependency into a strict face-distance inequality, and BoundaryFreeIncidentFaceSelector.false_of_remainderDependency_cycle rules out nonempty finite closed chains of those dependencies. The comparison theorem BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_remainderDependency_iff now identifies this selector relation with the construction-level PlanarBoundaryAnnulusConstruction.remainderDependency after lowering a strict-descent selector to PlanarBoundaryAnnulusConstruction; the construction's strict face-distance field follows from the selector hrest hypothesis. Consequently, selector-side cycle diagnostics are not merely analogous to the construction diagnostics: they transfer across the lowering bridge exactly. The no-dependency terminal condition transfers as well by BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstruction_no_remainderDependency_iff. The selector-side terminal condition now has its own iff, BoundaryFreeIncidentFaceSelector.no_remainderDependency_iff_boundary_remainders: for a selected face, no outgoing selector dependency is exactly the statement that every non-witness remainder edge lies on the outer-or-inner ambient boundary. The maximum-distance strengthening BoundaryFreeIncidentFaceSelector.false_of_total_remainderDependency rules out a nonempty finite selected-face set where every selected face has an outgoing non-boundary remainder dependency, while BoundaryFreeIncidentFaceSelector.exists_face_without_remainderDependency extracts a terminal selected face with no such outgoing dependency. Thus a source-to-selector construction must exhibit an actual ambient-boundary break at that terminal face or add independent well-founded orientation data. The boundary-break theorem BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders makes this constructive: for one selected face, every non-witness remainder edge is already on the outer-or-inner ambient annulus boundary. The live-edge variants BoundaryFreeIncidentFaceSelector.peelFaces_nonempty_of_mem_interiorEdgeSupport, BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_mem_interiorEdgeSupport, and BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_interiorEdgeSupport_nonempty remove a separate selected-face nonemptiness assumption whenever an interior edge remains to be covered. The construction core now also records the non-selector version, PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_boundary_remainders: any nonempty BFS-stratified annulus construction has a maximum-distance peeled face whose non-witness remainders are all on the ambient outer or inner boundary. This now factors through the central strict dependency relation PlanarBoundaryAnnulusConstruction.remainderDependency: non-boundary remainders produce dependencies via PlanarBoundaryAnnulusConstruction.exists_remainderDependency_of_nonboundary_remainder, finite closed dependency chains are refuted by PlanarBoundaryAnnulusConstruction.false_of_remainderDependency_cycle, and the total-outgoing obstruction PlanarBoundaryAnnulusConstruction.false_of_total_remainderDependency exposes a peeled face with no outgoing dependency through PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_no_remainderDependency. The no-dependency condition has also been tied back to the local boundary break: PlanarBoundaryAnnulusConstruction.no_remainderDependency_iff_boundary_remainders says that a peeled face has no outgoing construction dependency exactly when all of its non-witness remainders lie on the outer-or-inner ambient annulus boundary, with the forward direction named as PlanarBoundaryAnnulusConstruction.boundary_remainders_of_no_remainderDependency. The core construction file now proves `PlanarBoundaryAnnulusConstruction.peelFaces_nonempty_of_mem_interiorEdgeSupport` and `PlanarBoundaryAnnulusConstruction.exists_terminal_peelFace_boundary_remainders_of_mem_interiorEdgeSupport`, so a live interior edge covered by the construction is enough to trigger the terminal boundary break. This turns the boundary break into a direct invariant of the construction's strict-descent field, without carrying peeled-face nonemptiness as a separate side condition. The orientation-obstruction file now proves the matching guardrail `not_forall_live_planarBoundaryAnnulusConstruction_terminalNoDependency_implies_parentWitnessOrientation`: even live interior-edge support plus a terminal peeled face with no outgoing construction-level `remainderDependency` does not force `PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation`. Thus the terminal dependency break diagnoses unbroken strict remainder cycles, while the well-founded parent-witness orientation remains an independent geometric burden. The same concrete counter-construction now lowers to `PlanarBoundaryAnnulusConstructionPositiveFrontierData` and `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData` while still refuting parent orientation, through `not_forall_planarBoundaryAnnulusConstructionPositiveFrontierData_implies_parentWitnessOrientation` and `not_forall_planarBoundaryAnnulusConstructionBoundarySupportFac` `implies_parentWitnessOrientation`. So selected-boundary contact and positive current-frontier bookkeeping are not enough to supply the missing well-founded parent witness. The raw construction obstruction is sharper: `not_forall_planarBoundaryAnnulusConstruction_positive_faceDistance_implies_parentWitnessOrientation` uses a variant where every peeled face has positive stored distance but the bad witness still has no strictly earlier incident face. Conversely, positive stored distance is not a possible repair to the positive-frontier shell itself: `PlanarBoundaryAnnulusConstructionPositiveFrontierData.exists_peelFace_faceDistance_zero` and `PlanarBoundaryAnnulusConstructionPositiveFrontierData.not_forall_peelFaces_positive` show that positive-frontier data necessarily includes a zero-distance peeled stratum. The construction core now factors this as a stratum-level clash: `PlanarBoundaryAnnulusConstruction.not_forall_stratumFaces_nonempty_of_parentWitness` shows that parent orientation makes the zero stratum empty, while the current positive-frontier axioms inhabit every stratum. The core construction file now records the stronger incompatibility as `PlanarBoundaryAnnulusConstructionPositiveFrontierData.not_parentWitnessOrientation`, with corresponding face-partition and boundary-support variants. The missing parent-orientation datum is therefore lower-distance incidence, not scalar positivity or the current zero-based positive-frontier shell. The same file now adds `not_exists_planarBoundaryAnnulusConstructionPositiveFrontierData_with_rootDistanceConstruction_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData`, which makes the source-side mismatch explicit: the parent-oriented root-distance construction produced from boundary reachability plus interior-dual boundary-root adjacency-distance data cannot be wrapped in the current positive-frontier shell. Thus the root-distance parent-orientation route and the zero-based frontier bookkeeping route are different construction surfaces. The same construction file now records the exact reindexing: `rootDistance_faceDistance_succ_filter_eq_shifted_faceDistance_filter_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData` says that root-distance layer $n+1$ on peeled faces is exactly shifted construction layer n, with `rootDistance_faceDistance_one_filter_eq_shifted_faceDistance_zero_filter_of_boundaryReachabilityData_` and_interiorDualBoundaryRootAdjDistancePeelData as the root-distance layer 1 / shifted layer 0 specialization. rootDistance_faceDistance_one_filter_nonempty_iff_not_shifted_parentWitnessOrientation_of_boundaryReachabilityData_and_interiorDualBoundaryRootAdjDistancePeelData then identifies nonempty root-distance layer 1 exactly with failure of parent-witness orientation for the shifted construction. Separately, the selector-construction file adds BoundaryFreeIncidentFaceSelector.toPlanarBoundaryAnnulusConstructionBoundarySupportFaceData, which reaches the stronger boundary-support face-data shell when the remaining annulus-side obligations are supplied explicitly: non-peeled faces meet selected-boundary support, outer and inner boundary faces are separated, and the positive current outer frontiers are nonempty and pairwise disjoint. The companion theorem BoundaryFreeIncidentFaceSelector.not_nonempty_forcingInteriorEdgeWitness_of_boundarySupportFaceData_conditions then combines that stronger bridge with the forcing-edge obstruction. Thus the unresolved burden is no longer an unspecified witness-choice problem; it is the explicit face/frontier geometry needed by the annulus construction shell. The new positive selector endpoint in Theorem49BoundaryFreeSelectorPositiveRoute.lean makes the useful half of this lane explicit: BoundaryFreeIncidentFaceSelector.theorem49HeightOrderedPositiveProjectedGeometryOn_of_strictDescent turns a boundary-free selector with annulus boundary data, injectivity over interior edges, strict-descent remainders, and a nonempty purified carrier into Theorem49HeightOrderedPositiveProjectedGeometryOn, while BoundaryFreeIncidentFaceSelector.theorem49BoundaryRootNonemptyProjectedSynthesis_of_strictDescent adds a Tait coloring and reaches the nonempty projected synthesis endpoint. It therefore specifies a concrete positive target for a boundary-order source argument without claiming that the source data already supplies the selector, strict descent, or frontier side conditions. The selector endpoint now factors through the reusable construction-level bridge theorem49HeightOrderedPositiveProjectedGeometryOn_of_planarBoundaryAnnulusConstruction in Theorem49PositiveGeometricSourceReplacement.lean: any PlanarBoundaryAnnulusConstruction, together with a live purified carrier on the same embedding, already supplies the height-ordered replacement package, and theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstruction adds a Tait coloring to reach the corrected nonempty projected synthesis endpoint. Thus the selector route is one way to inhabit the annulus construction endpoint, not a separate positive surface. The same construction and live carrier hypotheses also yield theorem49TerminalPeelFaceBoundaryRemainders_of_planarBoundaryAnnulusConstruction, which gives a terminal peeled face whose non-witness remainders are already ambient-boundary edges. The same construction bridge now has local HasUnblockedInteriorEndpoint entry points: theorem49HeightOrderedPositiveProjectedGeometryOn_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint, theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint, and theorem49TerminalPeelFaceBoundaryRemainders_of_planarBoundaryAnnulusConstruction_and_hasUnblockedInteriorEndpoint. These keep the construction target in the same local-endpoint language used by the source obstructions, instead of forcing a separate carrier-nonemptiness repackaging step. The selector positive route now has the matching local forms: BoundaryFreeIncidentFaceSelector.theorem49HeightOrderedPositiveProjectedGeometryOn_of_strictDescent_and_hasUnblockedInteriorEndpoint, BoundaryFreeIncidentFaceSelector.

theorem49BoundaryRootNonemptyProjectedSynthesis_of_strictDescent_and_hasUnblockedInteriorEndpoint, and BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_of_strictDescent_and_hasUnblockedInteriorEndpoint. The auxiliary interiorEdgeSupport_nonempty_of_hasUnblockedInteriorEndpoint is the only bridge needed to feed terminal selected-face diagnostics from a named local endpoint witness. The strengthened terminal diagnostic BoundaryFreeIncidentFaceSelector.exists_terminal_face_boundary_remainders_and_disjoint_of_strictDescent_and_hasUnblockedInteriorEndpoint also preserves the selector's full selected-boundary-support avoidance on that same terminal face. Thus the strict-descent endpoint now exposes a single selected face carrying both the ambient-boundary remainder break and the boundary-free incident-face condition. The wheel-with-inner-triangle benchmark now calibrates that selector burden at the source level too. The theorem not_nonempty_boundaryFreeIncidentFaceSelector_wheelWithInnerTriangle shows that this honest-source, Tait-colorable, nonempty-carrier model admits no boundary-free incident-face selector by the direct selector/forcing duality, while wheelWithInnerTriangle_closedWalkSource_tait_hasUnblockedEndpoint_without_interiorDualBoundaryRootAdjDistancePeelData states the same failure at the raw source-IDBRAD bridge boundary: honest source data, the explicit Tait coloring, and 'HasUnblockedInteriorEndpoint' coexist on the fixed embedding, but generic InteriorDualBoundaryRootAdjDistancePeelData is impossible. The stronger not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangleGraph and not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangleGraph show that this is not merely a single-embedding accident: the generic converse to IDBRAD is false already on both the honest-source and successor-cycle source shells for the wheel graph. The theorem not_forall_some_boundaryFreeSelector_or_acyclicEndpoint_of_closedWalkSource_tait_carrier_wheel refutes any universal construction from honest closed-walk source data, a Tait coloring, and a nonempty purified carrier to a boundary-free selector or to any current acyclic annulus endpoint. The same abstract witness now also reaches the weaker paired source shell 'annulus reachability + selected-boundary arc geometry' via annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_interiorDualBoundaryRootAdjDistancePeelData and annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionBoundarySupportFaceData, and also the reduced construction-side theorem annulusBoundaryReachability_and_selectedBoundaryArcGeometry_does_not_imply_planarBoundaryAnnulusConstructionPositiveFronti
so this obstruction applies to any embedding with an interior edge that is forced to meet selected-boundary support on every incident face, even after weakening from cyclic ordered face arcs to the bundled selected-boundary arc shell. The same forcing-edge witness now has a universal failed-converse interface: not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness and not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_forcingInteriorEdgeWitness show

that any source property coexisting with such a forcing edge cannot by itself derive the weakest peeled-interior construction shell; the regression file now also phrases this at graph level in same-embedding existential form on `diamondWithTriangleGraph`: there is an embedding with the weaker source shells plus a forcing witness, but no embedding of that same graph can combine the witness with the peeled interior-dual package, the reduced positive-frontier shell, or the concrete `PlanarBoundaryAnnulusConstructionFaceLayerData` shell that lowers into annulus decomposition; the same forcing witness now also blocks the current two-sided annulus-root lane itself, via `closedWalkAnnulusBoundarySource_does_not_imply_planarBoundaryAnnulusRootAdjDistancePeelData` and `not_exists_embedding_closedWalkAnnulusBoundarySource_and_forcingInteriorEdgeWitness_and_planarBoundaryAnnulusRootAdjDistancePeelData`, so boundary-split repackaging does not rescue a forcing-edge source embedding; this now reaches one step earlier too, since the same forcing witness blocks `PlanarBoundaryAnnulusRootAdjDistancePeelData` already from `PlanarBoundaryAnnulusBoundaryReachabilityData+PlanarBoundaryCyclicOrderedFaceArcEmbeddingD` and from the weaker bundled `PlanarBoundarySelectedBoundaryArcGeometry` shell; and this is now packaged as a reusable failed-converse surface by `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness` and its three route-facing specializations, so the forcing-edge blocker is not merely a family of diamond-with-triangle regressions but a same-embedding "no universal converse" theorem surface for the current annulus-root lane; more sharply, the same file now also proves `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_exists_embedding_with_property_and_forcingInteriorEdgeWitness` and its route-facing specializations, so the current source-side shells do not even universally exclude forcing interior edges on one embedding; and `PlanarBoundaryClosedWalkSource.lean` now makes explicit that an honest closed-walk source already lowers on the same embedding to the cyclic ordered face-arc and bundled selected-boundary-arc shells, while `Theorem49ForcingInteriorEdgeObstruction.lean` shows that those stronger honest-source shells still do not imply annulus-root data in the presence of a forcing interior edge; the same file now also records the graph-level same-embedding obstructions for those strengthened honest-source conjunctions, so "closed-walk source + cyclic face-arc shell" and "closed-walk source + bundled selected-boundary arc shell" are both formally blocked from coexisting with annulus-root data when a forcing witness is present; `Theorem49ForcingInteriorEdgeObstruction.lean` now also records that the natural face-local repair is insufficient. The proved failed-converse theorem `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_exists_embedding_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_and_forcingInteriorEdgeWitness` is instantiated in `Theorem49ForcingInteriorEdgeObstructionRegression.lean` by `exists_embedding_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_and_forcingInteriorEdgeWitness_diamondWithTriangleGraph` and `not_forall_not_nonempty_forcingInteriorEdgeWitness_of_reachability_and_closedWalkEmbedding_and_atMostOneInteriorEdge_diamondWithTriangle`: on `diamondWithTriangleEmbedding`, annulus boundary reachability, honest facial closed walks, and at most one interior boundary edge on each face coexist with the forcing interior edge `dt12`. So the forcing-edge exclusion burden cannot be discharged merely by adding an at-most-one interior-edge condition to the closed-walk source surface. The same regression file now also records the direct route implication of this model through `diamondWithTriangle_source_` previousBoundaryWitness_and_witnessCover_without_interiorDual_rootDistance: the embedding has honest closed-walk annulus source data, repaired previous-boundary witness geometry, and both well-founded and height-ordered witness-cover data, while the forcing edge blocks both InteriorDualBoundaryRootAdjDistancePeelData and the two-sided PlanarBoundaryAnnulusRootAdjDistancePeelData. The failed-converse theorems not_forall_interiorDualBoundaryRootAdjDistancePeelData_of_previousBoundaryWitnessGeometry_diamondWithTriangle and not_forall_planarBoundaryAnnulusRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_diamondWithTriangle make the route consequence explicit: root-distance interior-dual data cannot be derived from the repaired previous-boundary witness surface, even with the honest source retained on the same graph;

- the later peeled endpoint-touch model is now calibrated against the source side too. Although peeledEndpointTouchEmbedding carries the current outer-boundary-rooted peeled annulus package, the graph peeledEndpointTouchGraph is acyclic, so not_nonempty_planarBoundaryClosedWalkEmbeddingData_peeledEndpointTouchGraph and not_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph show that it cannot carry the honest facial closed-walk source at all. Consequently admitsPlanarBoundaryOuterBoundaryRootAdjDistancePeelData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph and admitsPlanarBoundaryHeightOrderedFacePeelWitnessData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph and admitsPlanarBoundaryWellFoundedFacePeelWitnessData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph and the new admitsPlanarBoundaryAnnulusRootAdjDistancePeelData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph and admitsPlanarBoundaryAttachedRootedFacePeelCertificate_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph should be read as calibration theorems about later route layers, not as source-level counterexamples;

- the endpoint-separation package has now reached a natural positive completion point. In Theorem49InteriorVertices.lean, the theorem PlanarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation.selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport shows that the packaged endpoint-disjointness field makes selected-boundary purification lossless on the raw interior-edge endpoint carrier. Then Theorem49Synthesis.lean proves theorem49OuterBoundaryRootNonemptySynthesis_of_planarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation: endpoint-separated outer-boundary-rooted annulus data plus nonempty raw interiorEdgeEndpointSupport already yields the non-vacuous Theorem 4.9 synthesis package, with graph-level wrapper exists_theorem49OuterBoundaryRootNonemptySynthesis_of_exists_planarBoundaryOuterBoundaryRootAdjDistancePeelDataWithEndpointSeparation_with_nonempty_interiorEdgeEndpointSupport. The route now also has the matching construction/source bridge: theorem49OuterBoundaryRootNonemptySynthesis_of_planarBoundaryOuterBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport lowers peeled-face endpoint no-touch directly to that non-vacuous endpoint, and `PlanarBoundaryClosedWalkSourceRoute.lean` packages the honest closed-walk source analogue under explicit outer-root localization and raw carrier nonemptiness. However, the same file now also proves `not_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_and_rootsOuterAmbientBoundary_and_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport` and the endpoint-level form `not_exists_closedWalkAnnulusBoundarySource_and_theorem49OuterBoundaryRootNonemptySynthesis_sameBoundaryData`. So this outer-root repair line is not a same-split rescue of the honest closed-walk route: once the annulus boundary split is fixed to the source's own extracted split, the needed outer-root localization witness is already inconsistent. The replacement positive line is now the two-sided annulus-root surface instead. `Theorem49Synthesis.lean` proves `theorem49AnnulusRootNonemptySynthesis_of_planarBoundaryAnnulusRootAdjDistancePeelData_of_endpointSupport_disjoint` and `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_planarBoundaryAnnulusRootAdjDistancePeelData_with_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport`, and the peeled-face repair has now been lifted to the same generic surface. `Theorem49InteriorVerticesAnnulusConstruction.lean` defines `planarBoundaryAnnulusConstruction_of_planarBoundaryAnnulusRootAdjDistancePeelData` and proves the annulus-root lossless-purification bridge `PlanarBoundaryAnnulusRootAdjDistancePeelData.selectedBoundaryInteriorEdgeEndpointVertices_eq_interiorEdgeEndpointSupport_of_peelFaces_endpoint_disjoint_selectedBoundarySupport`. `Theorem49Synthesis.lean` then proves `theorem49AnnulusRootNonemptySynthesis_of_planarBoundaryAnnulusRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` and the graph-level wrapper `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_planarBoundaryAnnulusRootAdjDistancePeelData_with_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport`. `PlanarBoundaryClosedWalkSourceRoute.lean` packages the honest closed-walk source specializations `theorem49AnnulusRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_endpointSupport_disjoint` and `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_with_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport`. It also keeps the source-facing peeled-face forms `theorem49AnnulusRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_of_peelFaces_endpoint_disjoint_selectedBoundarySupport` and `exists_theorem49AnnulusRootNonemptySynthesis_of_exists_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData_with_peelFaces_endpoint_disjoint_selectedBoundarySupport_and_nonempty_interiorEdgeEndpointSupport` as specializations of the generic bridge, deriving global raw-carrier endpoint disjointness from `PlanarBoundaryAnnulusConstruction.endpointSupport_disjoint_of_peelFaces_endpoint_disjoint_selectedBoundarySupport`. So the current non-vacuous repair no longer depends on forcing every root to the extracted outer boundary, and the peeled-face no-touch route is available for any explicit two-sided annulus-root datum, not only data produced from an honest closed-walk source. This extra hypothesis is not automatic from current annulus-root data alone: `Theorem49InteriorVerticesEndpointObstruction.lean` now proves `annulusRootAdjDistancePeelData_does_not_imply_endpointSupport_disjoint_peeledEndpointTouch` and the fixed-embedding universal failure `not_forall_annulusRootAdjDistancePeelData_implies_endpointSupport_disjoint_peeledEndpointTouch`. So the new annulus-root repair theorem is formally honest about its missing geometric burden instead of hiding it inside the present route data. The honest annulus-geometry lane now calibrates the same point from the source side: `Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean` proves `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_with_nonempty_interiorEdgeEndpointSupport_without_endpointSupport_disjoint_diamondWithTriangle`, so on `diamondWithTriangleEmbedding` an honest closed-walk annulus source, weak one-collar annulus geometry, the attached certificate endpoint, and nonempty raw `interiorEdgeEndpointSupport` already coexist while endpoint-support disjointness still fails. Thus the current endpoint- separation repair sits strictly later than the honest collar/certificate lane too. The stronger repaired geometry surface also fails to force it: `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_wellFoundedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_without_endpointSupport_disjoint_diamondWithTriangle` shows that even previous-boundary witness geometry and the derived boundary-aware well-founded peel data can coexist with a nonempty raw carrier while endpoint-support disjointness still fails on the same honest source model. And the stronger theorem `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_wellFoundedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_and_empty_selectedBoundaryInteriorEdgeEndpointVertices_diamondWithTriangle` now shows that this is not just a failure of equality between the raw and purified carriers: on the same honest source model the raw `interiorEdgeEndpointSupport` is nonempty while the purified `selectedBoundaryInteriorEdgeEndpointVertices` is exactly empty. The same file now pushes this one layer further: the explicit collar-layer data `diamondWithTriangleOneCollarLayerFacePeelWitnessData` and its induced height-ordered peel witness `diamondWithTriangleOneCollarHeightOrderedFacePeelWitnessData` already exist on the same honest source model, yet `closedWalkAnnulusBoundarySource_and_previousBoundaryWitnessGeometry_and_heightOrderedFacePeelWitnessData_with_nonempty_interiorEdgeEndpointSupport_and_empty_selectedBoundaryInteriorEdgeEndpointVertices_diamondWithTriangle` shows that the purified carrier is still empty there. So the endpoint-touch obstruction and the endpoint-separation repair line now meet cleanly on the same theorem surface;

- the theorem-4.9 synthesis layer is now generic above the boundary-aware certificate surface. `Theorem49Synthesis.lean` proves `theorem49BoundaryRootSynthesis_of_boundaryZeroAnnihilatorTrivial`, then packages the full purified endpoint directly from `PlanarBoundaryAttachedRootedFacePeelCertificate` via `theorem49BoundaryRootSynthesis_of_planarBoundaryAttachedRootedFacePeelCertificate` and from `PlanarBoundaryWellFoundedFacePeelWitnessData` via `theorem49BoundaryRootSynthesis_of_planarBoundaryWellFoundedFacePeelWitnessData`, with matching graph-level wrappers. The older interior-dual and annulus-construction synthesis theorems now reduce through the same generic bridge, so once an honest route reaches the attached certificate or well-founded peel witness surface, no further theorem-4.9 algebra remains. More sharply, `Theorem49Synthesis.lean` now proves `theorem49BoundaryRootSynthesis_of_planarBoundaryHeightOrderedFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryCollarLayerFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryLocalRemainderBoundaryCollarLayerFacePeelWitnessData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusWitnessAssignment`, and the graph-level wrappers `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryHeightOrderedFacePeelWitnessData`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryCollarLayerFacePeelWitnessData`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryLocalRemainderBoundaryCollarL` and `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusWitnessAssignment`, so theorem-4.9 synthesis already starts at the boundary-aware height-ordered witness-cover surface and even at the weaker finite collar-layer witness-cover surface, hence also at the weakest explicit local remainder annulus surface and at the split "annulus decomposition + witness assignment" surface. `Theorem49PositiveGeometricSourceReplacement.lean` now also proves `theorem49BoundaryRootSynthesis_of_collarLayerPositiveProjectedGeometryOn`, `Theorem49HeightOrderedPositiveProjectedGeometry.boundaryRootSynthesis`, `Theorem49HeightOrderedPositiveProjectedGeometry.exists_boundaryRootSynthesis`, and the matching collar-layer graph-package pair. So once the route already reaches either of those named positive witness-cover packages, full `Theorem49BoundaryRootSynthesis` is immediate; the projected endpoint and raw quotient/error package are now explicit projections of that stronger endpoint rather than the top of those surfaces. `admitsPlanarBoundaryAttachedRootedFacePeelCertificate_of_admitsPlanarBoundaryAnnulusConstructionParentWitnessOrientation`, so the repaired parent-orientation route already lands on the certificate surface explicitly, and the direct honest-source synthesis corollaries in `PlanarBoundaryClosedWalkSourceRoute.lean` now factor through that certificate surface rather than bypassing it. The same source file now also proves `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_heightOrderedFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_collarLayerFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusWitnessAssignment_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_heightOrderedFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_collarLayerFacePeelWitnessData_direct`, `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusWitnessAssignment_direct`, so the honest source shell already reaches theorem-4.9 synthesis as soon as the same embedding carries either boundary-aware height-sorted witness-cover data, the weaker finite collar-layer witness-cover package, or the later pure annulus decomposition together with witness-edge ownership. The same route file now also proves the matching direct Step 2 vanishing corollaries `zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_heightOrderedFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily`, `zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_collarLayerFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily`, `zero_on_allEdges_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_heightOrderedFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily`, and `zero_on_allEdges_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_collarLayerFacePeelWitnessData_direct_of_annihilatesKempeClosureGeneratorFamily`. So the positive algebraic boundary now matches the positive synthesis boundary: once an honest source shell reaches boundary-aware height-sorted witness-cover data, or even just the finite collar-layer witness-cover package, no later annulus-geometry packaging is needed either for Step 2 vanishing or for the purified theorem-4.9 endpoint. `Theorem49Synthesis.lean` also proves `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusCollarGeometry`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry`, and the graph-level wrappers `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusCollarGeometry`, `exists_theorem49BoundaryRootSynthesis_of_admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeomet` and `exists_theorem49BoundaryRootSynthesis_of_exists_planarBoundaryAnnulusCollarGeometry_and_witnessOnCurrentBoundary`. So the synthesis burden itself no longer waits even for bundled collar geometry: once the route produces the boundary-aware height-ordered witness-cover surface, or even the weaker finite collar-layer witness-cover package, the certificate layer and the full current theorem-4.9 endpoint are automatic; the live positive burden is now upstream of explicit remainder bookkeeping and witness ownership themselves; the geometry file `Theorem49PlanarBoundaryAnnulusGeometry.lean` now makes that burden explicit by constructing the witness-cover surfaces directly from annulus geometry itself: `PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryCollarLayerFacePeelWitnessData`, `PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryHeightOrderedFacePeelWitnessData`, and the graph-level wrappers from admitted collar geometry, plus the repaired lane `admitsPlanarBoundaryWellFoundedFacePeelWitnessData_of_exists_planarBoundaryAnnulusCollarGeometry_and_witnessOnCurrentBoundary`. Then `PlanarBoundaryClosedWalkSourceRoute.lean` adds the matching honest closed-walk and successor-cycle wrappers from same-embedding annulus collar geometry to collar-layer witness-cover, height-ordered witness-cover, and, with current-boundary witness placement, the well-founded peel witness surface. So the remaining live route burden is cleanly geometric: derive same-embedding annulus collar geometry from honest source hypotheses, and add the repaired witness placement only if one wants the well-founded lane;

- Lean now isolates one explicit sufficient criterion for that first geometric step on the canonical one-collar route. `Theorem49PlanarBoundaryAnnulusWitness.lean` defines `PlanarBoundaryCanonicalWitnessChoiceboundaryData`, a facewise choice of one boundary edge on each ambient face of a fixed annulus boundary split such that all interior edges are chosen by some face and every non-witness boundary edge already lies on the outer or inner ambient annulus boundary. `PlanarBoundaryCanonicalWitnessChoice.`

`toPlanarBoundaryAnnulusWitnessAssignment` then upgrades the canonical decomposition `planarBoundaryAnnulusDecomposition_of_boundaryDataboundaryData` to a full annulus witness assignment, and `PlanarBoundaryClosedWalkSourceRoute.lean` packages the honest-source corollary as `admitsPlanarBoundaryAnnulusCollarGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct`. So the live one-collar construction burden is now stated on a concrete facewise object rather than an opaque bundled collar-geometry shell;

- the stronger cyclic ordered face-arc source object is now equivalent, on one embedding and at graph level, to that split cyclic walk-plus-contiguity shell, so future honest embedding-side work may target either theorem surface without changing the downstream annulus route;

- on the theorem-4.9 annulus side, the bundled collar geometry object is now also exposed as an exact packaging of the split surface "pure annulus decomposition + witness assignment", both on one embedding and at graph level, so downstream work may use whichever form is more convenient without changing the annihilator route;

- the newer residual/current-boundary wrapper is now calibrated just as sharply on the positive side. The file `Theorem49ResidualBoundaryPeeling.lean` proves `nonempty_planarBoundaryResidualBoundaryLayerFacePeelWitnessData_iff_nonempty_planarBoundaryCollarLayerFacePeelWitnessData`, `theorem49ResidualBoundaryPositiveProjectedGeometryOn_iff_collarLayerPositiveProjectedGeometryOn`, and `theorem49ResidualBoundaryPositiveProjectedGeome` `iff_heightOrderedPositiveProjectedGeometryOn`. So at the present theorem-4.9 endpoint the residual/current-boundary surface is formally conservative: it is exactly the old collar-layer / height-ordered witness-cover strength on a fixed embedding. What is new is that `Theorem49ResidualBoundaryPeeling.lean` now also lowers the source-fixed canonical-parent raw-cover package, and its successor-cycle boundary-order shell, directly to this residual endpoint via `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_via_residualBoundary`, `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFaceRootsCanonicalParentSharedEdgeCover_via_residualBoundary`, `theorem49ResidualBoundaryPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint`, `theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_and_hasUnblockedInteriorEndpoint_via_residualBoundary`, and their successor-cycle analogues. The generic parent and boundary-free-selector route files expose the same raw-cover lowering to their own theorem-4.9 endpoints, so the downstream side of this stronger source-fixed package is now completely explicit: this shell already reaches the full Theorem 4.9 synthesis endpoint on the same embedding, and `HasUnblockedInteriorEndpoint` is only needed for the weaker projected / raw quotient corollaries. The same benchmark-side ceiling is now explicit on this residual surface too: `counterEmbedding_residualBoundaryPositiveProjectedGeometryOn_and_boundaryRootNonemptyProjectedSynthesis_and_forcingInteriorEdgeWitness_without_boundaryFreeSelector_or_planarBoundaryAnnulusConstructionFaceLayerData`, `counterGraph_explicitTait_residualBoundaryPositiveProjectedGeometry_and_boundaryRootNonemptyProjectedSynthesis_with_forcingInteriorEdgeWitness_` without_boundaryFreeSelector_or_planarBoundaryAnnulusConstructionFaceLayerData, not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_ theorem49ResidualBoundaryPositiveProjectedGeometryOn_counterGraph and not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_ theorem49ResidualBoundaryPositiveProjectedGeometryOn_counterGraph show that even after the residual/current-boundary shell already reaches the projected Theorem 4.9 endpoint on the same embedding, it still does not generically force either the selector shell or the downstream construction face-layer shell there. However, Theorem49ForcingInteriorEdgeObstruction.lean now proves exists_ closedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges_of_exists_ closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover, interiorEdgeSupport_eq_empty_of_boundaryReachabilityData\ _and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc\ _and_boundaryFaceRootsCanonicalParentSharedEdgeCover, not_ hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalP not_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_of_ closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover, not_endpointSupport_disjoint_and_nonempty_interiorEdgeEndpointSupport_of_ closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover, and the matching successor-cycle / graph-level boundary-order variants. The same raw-cover hypotheses force interiorEdgeSupport=\emptyset, so they already collapse to the previously isolated NoInteriorEdges branch and cannot coexist with either HasUnblockedInteriorEndpoint, the older raw endpoint-support-disjoint carrier shell, or a nonempty purified selected-boundary carrier on the same embedding. Therefore the raw canonical-parent cover lowerings to the parent, selector, and residual theorem-4.9 endpoints are presently vacuous on the live endpoint semantics. The open route question is no longer how to derive that package together with HasUnblockedInteriorEndpoint from honest source data or from the exact v23 seed: even before any live carrier is added, the shell has already degenerated to empty interior support. So the route must change the source interface / endpoint notion or prove impossibility directly on that live shell;

- the same negative calibration now reaches the exact v23 Step 2 seed itself. In Theorem49ResidualBoundaryObstruction.lean, nonempty_ sharedInteriorPair_v23ResidualBoundaryInitialState_sipFace0Boundary builds V23ResidualBoundaryInitialState on a genuine face boundary of the shared-interior-pair source, while sharedInteriorPair_closedWalkSource_tait_ hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_ residualBoundaryGeometry shows that this exact algebraic seed, together with honest source data, a Tait coloring, and the purified endpoint witness, still does not force residual positive geometry on the same embedding. So the live missing theorem is a real source-to-residual geometric bridge, not better packaging of the existing single-component purification algebra. The sharper theorems exists_embedding_closedWalkAnnulusBoundarySource_ and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_ hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_ without_residualBoundaryPositiveProjectedGeometryOn_sharedInteriorPair and not_forall_residualBoundaryPositiveProjectedGeometryOn_of_ closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_ and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_ hasUnblockedInteriorEndpoint_sharedInteriorPair show that even exact one-collar current-boundary data preserving the same source boundary split still adds no forcing here;

- the same exact route already fails one step earlier at the still-stronger source-fixed canonical-parent shared-edge face-membership shell. In Theorem49ResidualBoundaryObstruction.lean, exists_embedding_
closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint_
without_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_
sharedInteriorPair and exists_embedding_boundaryReachabilityData_
and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_
hasUnblockedInteriorEndpoint_without_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembers
sharedInteriorPair, together with the matching closed-walk / successor-cycle failed-universal theorems, show that the honest source shell and the actual successor-cycle shell plus HasUnblockedInteriorEndpoint already do not force this stronger face-membership package, with no Tait coloring and no exact V23ResidualBoundaryInitialState needed. The theorem sharedInteriorPair_closedWalkSource_tait_
hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_
boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover packages the honest closed-walk source, concrete Tait coloring, HasUnblockedInteriorEndpoint, exact v23 seed, and failure of that child-membership refinement on the same embedding. The failed universal not_forall_some_
closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov
of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_
taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that the exact Step 2 algebraic seed does not even recover this stronger source-fixed canonical-parent face-membership package on the honest closed-walk surface. The stronger exact-shell refinement pair exists_embedding_closedWalkAnnulusBoundarySource_
and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_
hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_
without_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_
sharedInteriorPair and not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanc
of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_
and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_
hasUnblockedInteriorEndpoint_sharedInteriorPair shows that this remains false even after adding exact one-collar current-boundary data preserving that same honest source boundary split. Theorem49ResidualBoundaryObstruction.lean now upgrades both exact-shell failures to generic converse theorems: not_forall_some_
closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov
of_exists_embedding_closedWalkAnnulusBoundarySource_and_
taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_
v23ResidualBoundaryInitialState and not_forall_some_
closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCov
of_exists_embedding_closedWalkAnnulusBoundarySource_and_
oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_
hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState. So any same-embedding honest-source exact-shell witness with HasUnblockedInteriorEndpoint already refutes a universal derivation of that stricter face-membership shell; the shared-interior-pair benchmark is just one instantiation;

- the same exact v23 seed also fails to force even the stronger upstream source-fixed raw canonical-parent shared-edge-cover shell. In `Theorem49ResidualBoundaryObstruction.lean`, the new witness theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair`, together with the matching closed-walk / successor-cycle failed-universal theorems, show that the honest source and actual successor-cycle shells plus `HasUnblockedInteriorEndpoint` already do not force this raw canonical-parent cover package, with no Tait coloring and no exact `V23ResidualBoundaryInitialState` needed. The theorem `sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover` packages the honest closed-walk source, concrete Tait coloring, `HasUnblockedInteriorEndpoint`, exact v23 seed, and failure of the closed-walk raw-cover shell on the same embedding. The failed universal `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that the exact Step 2 algebraic seed does not even recover the present source-fixed canonical-parent raw-cover package on the honest closed-walk surface. The stronger exact-shell refinement pair `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair` and `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover`
`of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that this remains false even after adding exact one-collar current-boundary data preserving that same honest source boundary split. `Theorem49ResidualBoundaryObstruction.lean` now also upgrades both exact-shell failures to generic converse theorems: `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`. So any same-embedding honest-source exact-shell witness with `HasUnblockedInteriorEndpoint` already refutes a universal derivation of that raw canonical-parent shell; the shared-interior-pair benchmark is just one instantiation;

- the same exact v23 seed also fails to force the weaker degenerate `NoInteriorEdges` shell to which that raw-cover package collapses. The new witness theorems `exists_` embedding_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsNoInteriorEdges_sharedInteriorPair and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsNoInteriorEdges_sharedInteriorPair, together with the matching closed-walk / successor-cycle failed-universal theorems, show that the honest source and actual successor-cycle shells plus HasUnblockedInteriorEndpoint already do not force even this degenerate branch, with no Tait coloring and no exact V23ResidualBoundaryInitialState needed. sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsNoInteriorEdges packages the exact seed together with failure of the closed-walk boundary-root plus interiorEdgeSupport=\emptyset shell on the same embedding, while the failed universal not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that the exact Step 2 seed does not even recover this already-degenerate source branch on the honest closed-walk surface. The matching exact-shell refinement pair exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsNoInteriorEdges_sharedInteriorPair and not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsNoInteriorEdges_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that exact one-collar current-boundary data still does not recover even this degenerate closed-walk source branch;

- the same exact v23 seed also fails already at the strictly weaker local boundary-free-selector surface. The new witness theorems exists_embedding_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint_without_boundaryFreeIncidentFaceSelector_sharedInteriorPair and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_hasUnblockedInteriorEndpoint_without_boundaryFreeIncidentFaceSelector_sharedInteriorPair, together with the matching closed-walk / successor-cycle failed-universal theorems, show that the honest source and actual successor-cycle shells plus HasUnblockedInteriorEndpoint already do not force even a boundary-free selector, with no Tait coloring and no exact V23ResidualBoundaryInitialState needed. The theorem sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector packages the exact seed together with failure of BoundaryFreeIncidentFaceSelector on the same honest embedding, while the failed universal not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that the exact Step 2 seed does not even recover the local no-forcing selector surface that any selector/construction descent route would need before source packaging questions arise;

- the same sharp failure now holds already on the route-facing successor-cycle boundary-order shell. The theorem `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_residualBoundaryPositiveProjectedGeometryOn_sharedInteriorPair` packages explicit annulus boundary reachability, selected-boundary arcs from successor cycles, the concrete Tait coloring, 'HasUnblockedInteriorEndpoint', and the exact v23 residual seed together with failure of residual positive geometry on the same embedding. Its universal converse `not_forall_residualBoundaryPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that the missing bridge is already absent at the actual boundary-order route surface, before any lowering or repackaging to closed-walk source form. The stronger exact-shell pair `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_residualBoundaryPositiveProjectedGeometryOn_sharedInteriorPair` and `not_forall_residualBoundaryPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that even exact one-collar current-boundary data preserving the extracted boundary split still adds no forcing on this live route-facing shell;

- the same route-facing shell now already fails at the sharper source-fixed rooted shared-edge face-membership package itself. `Theorem49ResidualBoundaryObstruction.lean` proves `not_exists_sharedInteriorPairClosedWalkSourceBoundaryFaceRootsCanonicalRootedSharedEdgeFaceMembership` so on the shared-interior-pair honest source this concrete rooted-cover package is impossible outright: it would immediately construct generic `InteriorDualBoundaryRootAdjDistancePeelData`, contradicting the already formalized shared-interior-pair IDBRAD obstruction. This exact-seed obstruction is now known to be strictly upstream: the new witness theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_sharedInteriorPair` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembers` `sharedInteriorPair`, together with the matching closed-walk / successor-cycle failed-universal theorems, show that the same benchmark already refutes any universal derivation of this rooted-cover package from the honest source or actual successor-cycle shell plus `HasUnblockedInteriorEndpoint`, with no Tait coloring and no exact `V23ResidualBoundaryInitialState` needed. The exact-shell witness theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_` `hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_sharedInteriorPair` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover_sharedInteriorPair`, together with the matching closed-walk / successor-cycle failed-universal theorems, show that even exact one-collar current-boundary data preserving the honest boundary split still does not force this sharper route-facing rooted cover package. So the repaired exact shell already fails before the older canonical-parent cover refinements are even reached;

- the same route-facing shell also fails already at the stricter canonical-parent shared-edge face-membership refinement. The theorem `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeFaceMemb` `sharedInteriorPair` packages successor-cycle source data, the exact v23 seed, the concrete Tait coloring, and `HasUnblockedInteriorEndpoint` together with failure of that stronger face-membership shell, and the failed universal `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that the same exact-seed gap already appears before any attempt to lower to the weaker route-facing raw-cover shell. The parallel exact-shell pair `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_sharedInteriorPair` and `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentShared` `of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` shows that exact one-collar current-boundary data preserving the extracted boundary split still adds no forcing even at that stronger route-facing surface. The same file now lifts these to generic exact-shell converse failures: `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeFaceMembershipCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_` v23ResidualBoundaryInitialState. So any same-embedding successor-cycle exact-shell witness with HasUnblockedInteriorEndpoint already refutes a universal derivation of that stronger route-facing face-membership shell;

- the same route-facing shell also fails already at the level of the raw canonical-parent cover package itself. The theorem exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair packages successor-cycle source data, the exact v23 seed, the concrete Tait coloring, and HasUnblockedInteriorEndpoint together with failure of the boundary-order raw-cover shell, and the failed universal not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that the same exact-seed gap already appears before any attempt to lower from the actual boundary-order shell to a closed-walk source. The parallel exact-shell pair exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_sharedInteriorPair and not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that exact one-collar current-boundary data preserving the extracted boundary split still adds no forcing at that stronger route-facing raw-cover surface either. The same file now lifts these to generic exact-shell converse failures: not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState and not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState. So any same-embedding successor-cycle exact-shell witness with HasUnblockedInteriorEndpoint already refutes a universal derivation of that route-facing raw-cover shell;

- the same route-facing shell also fails already at the weaker degenerate boundary-root plus interiorEdgeSupport=\emptyset surface. The theorem exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_ v23ResidualBoundaryInitialState_without_boundaryFaceRootsNoInteriorEdges_sharedInteriorPair packages successor-cycle source data, the exact v23 seed, the concrete Tait coloring, and HasUnblockedInteriorEndpoint together with failure of the boundary-order NoInteriorEdges shell, and the failed universal not_forall_some_successorCycleBoundaryFaceRootsNoInteriorEdges_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that the exact seed does not even force the degenerate branch to which the raw-cover shell already collapses. The matching one-collar refinement pair exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsNoInteriorEdges_sharedInteriorPair and not_forall_some_successorCycleBoundaryFaceRootsNoInteriorEdges_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that even exact one-collar current-boundary data still does not force that weaker route-facing degenerate branch;

- the same route-facing shell also fails already at the weaker boundary-free-selector layer. The theorem exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector_sharedInteriorPair packages successor-cycle source data, the exact v23 seed, the concrete Tait coloring, and HasUnblockedInteriorEndpoint together with failure of the boundary-order selector shell, and the failed universal not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair shows that the exact seed does not even force a route-facing boundary-free selector on the same embedding;
- the exact v23 residual seed also fails to recover any previously accepted same-embedding geometry surface. The theorem sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_knownSameEmbeddingGeometry packages the honest source, concrete Tait coloring, 'HasUnblockedInteriorEndpoint', and exact seed together with failure of residual witness data, residual selector data, height-ordered witness data, collar-layer witness data, the attached boundary-rooted certificate, well-founded witness data, and weak annulus-collar geometry on the same embedding. The stronger theorem exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_

`without_knownSameEmbeddingGeometry_sharedInteriorPair` shows that even exact one-collar current-boundary data preserving the same source boundary split does not repair any of those known same-embedding surfaces. More sharply, the theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair` already isolate the weak annulus-collar geometry shell itself: even that single route-facing same-embedding burden is not forced by the exact one-collar plus exact-v23 shell. The same exact shell now also fails at the central acyclic witness burden itself: the theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryWellFoundedFacePeelWitnessData_sharedInteriorPair` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryWellFoundedFacePeelWitnessData_sharedInteriorPair` show that even same-embedding well-founded parent-peeling is not recovered on either exact-shell source surface. The failed universals `not_forall_some_knownSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` and `not_forall_some_knownSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, `not_forall_some_knownSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair`, and `not_forall_some_knownSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair` show that this exact-seed gap already persists on both the honest closed-walk and the actual successor-cycle boundary-order shells, even after adding exact one-collar current-boundary data preserving the extracted annulus boundary split. The same exact-shell diagnosis now also lifts to generic converse theorems: `not_forall_some_knownSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_some_knownSameEmbeddingGeometry_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_` and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_ v23ResidualBoundaryInitialState. So any explicit same-embedding exact-shell witness carrying simultaneous failure of the currently known geometry surfaces already refutes a universal derivation of that whole disjunctive burden; the shared-interior-pair shell is just one concrete instantiation. So the live v23.5 obligation is a genuinely new source-to-residual bridge, not a return to any existing same-embedding geometry ladder, to the upstream raw-cover shell, or even to its boundary-free-selector / degenerate NoInteriorEdges shadows;

- the same exact one-collar plus exact-v23 shell still does not even recover the current direct or route-facing replacement-positive packages. The theorems exists_embedding_closedWalkAnnulusBoundarySource_ and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_ hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_ any_replacementPositiveProjectedGeometryOn_sharedInteriorPair and exists_ embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_ and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_ and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_ v23ResidualBoundaryInitialState_without_any_replacementPositiveProjectedGeometryOn_ sharedInteriorPair, together with the failed universals not_forall_any_ replacementPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_ and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_ and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair and not_forall_any_replacementPositiveProjectedGeometryOn_of_ boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_ and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_ and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_ hasUnblockedInteriorEndpoint_sharedInteriorPair, show that even exact one-collar current-boundary data preserving the same extracted annulus boundary split still does not recover the direct or route-facing replacement-positive endpoints from the exact Step 2 seed. The same exact-shell obstruction now also lifts to generic converse theorems: not_forall_any_replacementPositiveProjectedGeometryOn_ of_exists_embedding_closedWalkAnnulusBoundarySource_and_ oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_ hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState and not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_ embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_ and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_ and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_ v23ResidualBoundaryInitialState. So any explicit same-embedding exact-shell witness carrying simultaneous failure of the current direct and route-facing replacement-positive endpoints already refutes a universal derivation of that whole positive disjunction; the shared-interior-pair shell is again just one concrete instantiation. The same shared-interior-pair successor-cycle shell is now also packaged directly with the local two-interior-edge burden at that replacement-positive boundary: exists_ embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_ and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_ and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_ v23ResidualBoundaryInitialState_and_twoDistinctInteriorEdgesOnFaceBoundary_ without_any_replacementPositiveProjectedGeometryOn_sharedInteriorPair shows that the exact one-collar/v23 benchmark already carries a concrete face boundary with two distinct interior edges while still failing every current direct and route-facing replacement-positive package on that same embedding;

- the same exact one-collar plus exact-v23 shell also still does not recover even the downstream positive construction face-layer package. The theorems exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusConstructionFaceLayerData_sharedInteriorPair and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusConstructionFaceLayerData_sharedInteriorPair, together with the failed universals not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair and not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair, show that even exact one-collar current-boundary data preserving the same extracted annulus boundary split still does not force PlanarBoundaryAnnulusConstructionFaceLayerData from the exact Step 2 seed. Since the selected-boundary, face-partition, and positive-frontier construction shells all lower to this face-layer package, the exact-shell gap already blocks the whole positive construction stack. The same exact-shell obstruction now also lifts to generic forcing-edge converse theorems: not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness and not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness. So any explicit same-embedding exact-shell witness carrying a forcing interior edge already refutes a universal derivation of PlanarBoundaryAnnulusConstructionFaceLayerData from that shell; the shared-interior-pair benchmark is only one concrete instantiation. The same shared-interior-pair successor-cycle shell is now also packaged directly with the local two-interior-edge burden at that face-layer boundary: exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_ v23ResidualBoundaryInitialState_and_twoDistinctInteriorEdgesOnFaceBoundary_without_planarBoundaryAnnulusConstructionFaceLayerData_sharedInteriorPair shows that the exact one-collar/v23 benchmark already carries a concrete face boundary with two distinct interior edges while still failing PlanarBoundaryAnnulusConstructionFaceLayerData on that same embedding. The same obstruction file now also states the repaired local burden already on that exact one-collar shell: not_forall_not_nonempty_forcingInteriorEdgeWitness_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair and not_forall_not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair show that even this exact one-collar v23/v23.5 source package still does not universally exclude forcing interior edges. So any sound repair on that shell must add or derive a precise local no-forcing hypothesis before it can reach the downstream face-layer package. The same obstruction file now also makes the repaired local burden explicit on that shell itself: everyInteriorEdgeHasBoundaryFreeIncidentFace_of_planarBoundaryAnnulusConstructionFaceLayerData, boundaryFreeIncidentFaceSelector_of_planarBoundaryAnnulusConstructionFaceLayerData, and not_nonempty_forcingInteriorEdgeWitness_of_planarBoundaryAnnulusConstructionFaceLayerData show that any sound v23.5 theorem reaching the face-layer package must already build the boundary-free selector, equivalently exclude forcing interior edges, before any later annulus-decomposition lowering; Theorem49ResidualBoundaryRoute.lean now states the repaired positive direction on that exact shell too. theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint and its successor-cycle counterpart theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint say exactly what the current strongest sound v23/v23.5 repair is: if the exact one-collar shell already carries annulus-root parent peel data on the same boundary split, then the local unblocked-endpoint witness closes the route all the way to Theorem49BoundaryRootNonemptyProjectedSynthesis;

- the constructive selector/construction lane now advances one real shell further in Theorem49BoundaryFreeSelectorConstruction.lean. planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary and the successor-cycle version planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_planarBoundaryAnnulusRootParentPeelData_and_ `selectorDeficitSelectedBoundary` show that same-split annulus-root parent peel data already lowers to the real `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData` shell once the exact local gap is discharged: peeled faces missed by the selector image must still meet selected-boundary support, and the resulting positive `currentOuterBoundary` layers must be nonempty and pairwise disjoint. Thus the parent-peel-to-construction burden is now stated honestly and sharply. It is not a generic requirement on all non-peeled faces, but the strictly smaller selector-deficit peeled-face condition. Lean now also proves `selectorDeficitSelectedBoundary_of_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary`, so whenever an older route already establishes the stronger hypothesis that every non-peeled face of the same parent-peel construction meets selected-boundary support, this repaired selector-deficit premise is obtained for free. The same exact one-collar/v23 shell now also has generic forcing-edge converse failures already at that stronger construction surface and the next two intermediate construction shells: `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`. `not_forall_nonempty_planarBoundaryAnnulusConstructionFacePartitionData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`, `not_forall_nonempty_planarBoundaryAnnulusConstructionFacePartitionData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`, `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFrontierData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`, and `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFrontierData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` now show the same obstruction one layer later inside the construction lane. So any explicit same-embedding exact-shell witness carrying a forcing interior edge already refutes universal derivations of `PlanarBoundaryAnnulusConstructionBoundarySupportFaceData`, `PlanarBoundaryAnnulusConstructionFacePartitionData`, and `PlanarBoundaryAnnulusConstructionPositiveFrontierData`; the live gap therefore persists before the final lowering to face-layer data;

- the same shared-interior-pair successor-cycle shell is now also packaged directly at that positive-frontier boundary: `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_twoDistinctInteriorEdgesOnFaceBoundary_without_planarBoundaryAnnulusConstructionPositiveFrontierData_sharedInteriorPair` shows that the exact one-collar/v23 benchmark already carries a concrete face boundary with two distinct interior edges while still failing `PlanarBoundaryAnnulusConstructionPositiveFrontierData` on that same embedding;

- that same repaired exact-shell package now reaches the downstream construction face-layer shell used by annulus decomposition. `Theorem49ResidualBoundaryRoute.lean` proves `nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary` and `nonempty_planarBoundaryAnnulusConstructionFaceLayer` `of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary`. So once the exact one-collar shell carries same-split parent peel data together with the selector-deficit selected-boundary condition and the positive `currentOuterBoundary` side conditions, the route no longer stops at selected-boundary support data: it already produces real `PlanarBoundaryAnnulusConstructionFaceLayerData` on that same embedding. So the remaining issue is not any further lowering below parent-peel itself, but whether the current exact shell forces that package at all;

- that same repaired exact-shell package now also reaches a real annulus decomposition on the same embedding. `Theorem49ResidualBoundaryRoute.lean` proves `nonempty_planarBoundaryAnnulusDecomposition_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary` and `nonempty_planarBoundaryAnnulusDecomposition_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary`. So once the exact shell carries the same-split parent-peel and selector-deficit package, the route already passes through annulus decomposition itself. The remaining burden is therefore not another decomposition-side shell, but whatever stronger geometry or witness data is needed downstream of that repaired local package. Lean now also proves `nonempty_planarBoundaryAnnulusDecomposition_of_planarBoundaryAnnulusConstructionFaceLayerData`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionFaceLayerData_and_resultingDecompositionWitnessAssignment`, `theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionFaceLayerData_and_inducedDecompositionWitnessAssignment`, `nonempty_planarBoundaryAnnulusDecomposition_` of_planarBoundaryAnnulusConstructionBoundarySupportFaceData, and theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData and_inducedDecompositionWitnessAssignment, theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData_and_resultingDecompositionWitnessAssignment, so once the route reaches PlanarBoundaryAnnulusConstructionFaceLayerData, this decomposition step is automatic and the full Theorem 4.9 endpoint reduces exactly to witness ownership on the exact annulus decomposition canonically induced by that shell. The stronger selected-boundary PlanarBoundaryAnnulusConstructionBoundarySupportFaceData shell is now only an upstream route into that face-layer package, not part of the downstream closure theorem itself. The same file now also proves the graph-level route wrappers exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_planarBoundaryAnnulusConstructionFaceLayerData_and_resultingDecompositionWitnessAssignment_direct, the stronger selected-boundary-support analogue, and the matching successor-cycle pair. So once the honest-source or successor-cycle route reaches either same-embedding construction shell and the resulting decomposition carries witness ownership, full Theorem49BoundaryRootSynthesis already follows directly at graph level; no further repackaging step remains below that shell. Theorem49PositiveGeometricSourceReplacement.lean now also proves theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionFaceLayerData, theorem49BoundaryRootNonemptyProjectedSynth of_planarBoundaryAnnulusConstructionFaceLayerData, theorem49BoundaryRawQuotientErrorPackage of_planarBoundaryAnnulusConstructionFaceLayerData, theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData, theorem49BoundaryRootNonemptyProjectedSynthesis_of_planarBoundaryAnnulusConstructionBounda and theorem49BoundaryRawQuotientErrorPackage_of_planarBoundaryAnnulusConstructionBoundaryS so once either construction shell exists on the same embedding, full Theorem49BoundaryRootSynthesis is already automatic directly in that positive-replacement file itself, while a surviving purified carrier is only needed to reach the newer projected endpoint and its raw quotient/error package. The remaining burden above those shells is therefore to preserve the purified carrier, or else derive stronger earlier geometry, not to find another replacement-side lowering. Theorem49PositiveGeometricSourceReplacementRoute.lean now also proves the graph-package methods Theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometry.boundaryRootSynthesis, Theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeom exists_boundaryRootSynthesis, and the matching successor-cycle pair. So once those named construction-face-layer route packages themselves carry witness ownership on the same derived annulus decomposition, full Theorem49BoundaryRootSynthesis follows immediately; there is no remaining repackaging gap between those graph-level route objects and the Theorem 4.9 endpoint. Theorem49PositiveGeometricSourceReplacementRoute.lean now also packages the stronger selected-boundary-contact shell itself as the fixed-embedding predicates Theorem49ClosedWalkAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn and Theorem49SuccessorCycleAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn, together with the graph-level structures Theorem49ClosedWalkAnnulusConstructionBoundarySupportPosi and Theorem49SuccessorCycleAnnulusConstructionBoundarySupportPositiveProjectedGeometry. Those graph packages already expose boundaryRootSynthesis and exists_boundaryRootSynthesis on the induced annulus decomposition, so callers no longer need to first repackage this shell as construction-face-layer route data just to state the direct Theorem 4.9 closure. The same file still packages the derived face-layer closure as the fixed-embedding predicates Theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometryOn and Theorem49SuccessorCycleAnnulusConstructionFaceLayerPositiveProjectedGeometryOn, together with the graph-level structures Theorem49ClosedWalkAnnulusConstructionFaceLayerPositivePr and Theorem49SuccessorCycleAnnulusConstructionFaceLayerPositiveProjectedGeometry. The stronger selected-boundary-contact shell still lowers directly into those derived face-layer route surfaces via PlanarBoundaryAnnulusConstructionBoundarySupportFaceData.toPlanarBoundaryAnnulusConstructionFaceLayerData. The same file now also proves not_theorem49ClosedWalkAnnulusConstructionFaceLayerPositiveProjectedGeometryOn_of_facewiseAtMostOneInteriorEdge, not_theorem49SuccessorCycleAnnulusConstructionFaceLayerPo of_facewiseAtMostOneInteriorEdge, and exists_two_distinct_interior_edges_on_faceBoundary_of_successorCycleAnnulusConstructionFaceLayerPositiveProjectedGeometryOn. So the stronger source-facing construction face-layer shell itself already sits outside the canonical at-most-one branch, and the successor-cycle shell now directly exposes the same local two-interior-edge obstruction there too. So once the honest-source or successor-cycle route reaches same-embedding PlanarBoundaryAnnulusConstructionFaceLayerData and still carries HasUnblockedInteriorEndpoint, the current projected Theorem 4.9 endpoint, the raw quotient/error package, and the replacement-positive height/collar packages already follow directly, without any further collar-geometry or well-founded-witness detour. That diagnosis is now repaired again. Theorem49BoundaryFreeSelectorConstruction.lean proves parentWitnessOrientation_of_boundaryData_and_interiorDualBoundaryRootParentPeelData, parentWitnessOrientation_of_planarBoundaryAnnulusRootParentPeelData, and not_exists_planarBoundaryAnnulusConstructionBoundarySupportFaceData_with_planarBoundaryAnnulusRootParentPeelConstruction. So the exact selector-driven parent-peel construction is already PlanarBoundaryAnnulusConstruction.ParentWitnessOrientation on the same embedding, while any same-construction selected-boundary-support shell still forbids parent orientation. The same construction file now isolates the resulting contradiction at the actual shell that creates it: false_of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary and false_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary. The same construction file now also packages this as raw boundary-order graph-level nonexistence theorems: not_exists_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary and not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary. It also now gives the fixed-embedding local nonexistence forms not_exists_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace and not_exists_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_ and_currentOuterBoundary_of_boundaryReachabilityData_and_ dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc, which are the direct reusable statements downstream route arguments should consume. So the repaired selector-deficit plus positive-current-boundary package is already globally impossible before any exact one-collar or exact-v23 residual hypotheses are added. Theorem49ResidualBoundaryRoute.lean then re-exports this at the repaired exact one-collar shell itself with false_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary and false_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary. So the old positive currentOuterBoundary side conditions are not merely another local burden to derive on the same parent-peel construction: they are incompatible with that construction. The earlier exact-shell face-layer, decomposition, and synthesis theorems with those hypotheses remain valid but vacuous. Theorem49ResidualBoundaryRoute.lean now also packages this as direct graph-level nonexistence theorems: not_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary and not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary. So this repaired selector-deficit plus positive-current-boundary shell is not merely unforced by the live route: even the stricter exact-shell witness form is globally impossible, as an immediate corollary of the boundary-order contradiction above. Theorem49ResidualBoundaryObstruction.lean now upgrades this to direct exact-shell failed universals: not_forall_exists_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair and not_forall_exists_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair, with the matching forcing-edge abstract converse failures. It now also exposes the exact-shell counterexamples themselves as exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_sharedInteriorPair and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_ v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_currentOuterBoundary_sharedInteriorPair. So even the full repaired selector-deficit plus positive-current-boundary package is not forced by the live exact shell on the shared-interior-pair benchmark. The live route must therefore avoid that positive-frontier shell entirely and continue through the already parent-oriented construction / well-founded-parent lane. Theorem49ResidualBoundaryRoute.lean now also proves the direct exact-shell endpoint theorems theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData and theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData. So once the exact one-collar shell carries same-split annulus-root parent peel data at all, the full Theorem 4.9 synthesis endpoint is already immediate on that embedding; the selector-deficit and positive-current-boundary detour is not just unnecessary but inconsistent on that same construction. The live burden is therefore sharper once again: derive the actual PlanarBoundaryAnnulusRootParentPeelData package from the residual shell, rather than any construction-shell refinement beyond it. The same file now also proves the even earlier exact-shell theorems theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalParentSharedEdgeCover and theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalParentSharedEdgeCover, together with the matching graph-level direct existence forms. So once the exact one-collar shell already carries the raw source-fixed canonical-parent shared-edge cover on the extracted boundary-face roots, full Theorem49BoundaryRootSynthesis is already immediate on that embedding, with no HasUnblockedInteriorEndpoint and no separate parent-peel repackaging step. The live exact-shell burden is therefore sharper again: force that concrete source-fixed shared-edge cover itself, or prove that the repaired shell cannot support it. The same file now also proves the sharper exact-shell rooted shared-edge face-membership theorems theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover and theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_boundaryFaceRootsCanonicalRootedSharedEdgeFaceMembershipCover, together with the matching graph-level direct existence forms. So once the repaired exact shell already carries the concrete source-fixed rooted shared-edge face-membership cover package on the extracted boundary-face roots, full Theorem49BoundaryRootSynthesis is immediate on that same embedding, with no further descent either through canonical-parent cover packaging or through the generic IDBRAD shell. The live exact-shell burden can therefore be stated at that sharper source-facing rooted-cover layer itself. The same file now also proves the exact-shell theorems theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_ oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_interiorDualBoundaryRootAdjDistancePeelData and theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_interiorDualBoundaryRootAdjDistancePeelData. So on the exact one-collar v23/v23.5 shell, even the annulus-root parent-peel package is no longer the minimal positive burden: it already suffices to derive generic InteriorDualBoundaryRootAdjDistancePeelData on the same embedding, or any equivalent cycle-breaking parent witness strong enough to construct it. The same file now also proves the exact-shell route-surface lowerings theorem49ClosedWalkAnnulusRootParentPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint, theorem49SuccessorCycleAnnulusRootParentPositiveProjected of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint, theorem49ClosedWalkAnnulusRootAdjDistancePositiveProjecte of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint, and theorem49SuccessorCycleAnnulusRootAdjDistancePositive of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint. So the live exact shell no longer merely has endpoint corollaries: once it carries the parent-peel or generic adj-distance package together with HasUnblockedInteriorEndpoint, it already lands on the named root-parent or root-distance positive route surfaces used elsewhere in the replacement pipeline. The same file now also packages the corresponding graph-level existential forms through nonempty_theorem49ClosedWalkAnnulusRootParentPositiveProjectedGeometry_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint, its successor-cycle analogue, and the matching root-distance pair. So downstream graph-level route consumers can now use the live exact shell directly, without rebuilding those positive packages by hand. Theorem49PositiveGeometricSourceReplacementRoute.lean now also gives those four graph-level positive route packages their own boundaryRootSynthesis and exists_boundaryRootSynthesis methods, so once the exact shell has lowered into the named root-parent or root-distance surfaces, no further manual descent through collar-layer packages is needed to recover full Theorem 4.9 synthesis. Theorem49PositiveGeometricSourceReplacementJointObstruction.lean now adds a more durable diagnosis on exactly these four named surfaces: each of the closed-walk/successor-cycle root-parent and root-distance positive route packages simultaneously forces a face boundary with two distinct interior edges and excludes forcing interior-edge witnesses on the same embedding. So this parent-oriented positive lane is no longer only an endpoint wrapper; it now carries an explicit combined local burden that any constructive derivation into the lane must satisfy. More sharply, these same four packages now also prove EveryInteriorEdgeHasBoundaryFreeIncidentFace and furnish concrete BoundaryFreeIncidentFaceSelector data on the same embedding, so the local boundary-free selector shell is no longer a downstream burden below those positive route packages. The same file now closes one step lower still: it proves the direct exact-shell graph-level endpoint and raw quotient/error wrappers exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_hasUnblockedInteriorEndpoint_direct, its successor-cycle analogue, the matching generic adj-distance pair, and the four corresponding exists_theorem49BoundaryRawQuotientErrorPackage... theorems. So once the exact shell already carries parent-peel or generic adj-distance data together with HasUnblockedInteriorEndpoint, the route no longer needs even an explicit graph-level positive-surface unpacking step before reaching the current projected endpoint or raw quotient/error package. The same file now also proves the direct graph-level full-synthesis closures exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_direct, its successor-cycle analogue, and the matching generic adj-distance pair. So once the live exact shell already carries same-embedding parent-peel data or generic adj-distance data at all, the route now reaches full Theorem49BoundaryRootSynthesis directly; even the projected-endpoint or positive-surface packaging steps are no longer downstream burdens below that shell. The residual interface now closes one step lower too. Theorem49ResidualBoundaryPeeling.lean proves theorem49BoundaryRootSynthesis_of_residualBoundarySelectorData, theorem49BoundaryRootSynthesis_of_residualBoundaryLayerFacePeelWitnessData, and theorem49BoundaryRootSynthesis_of_residualBoundaryPositiveProjectedGeometryOn, while Theorem49ResidualBoundaryRoute.lean proves theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_residualBoundarySelectorData and theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_residualBoundaryLayerFacePeelWitnessData, the successor-cycle analogue theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_residualBoundaryLayerFacePeelWitnessData, and the graph-level direct closures exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_residualBoundaryLayerFacePeelWitnessData_direct and exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_residualBoundaryLayerFacePeelWitnessData_direct. So once the route reaches residual/current-boundary selector data itself, the full Theorem 4.9 synthesis endpoint is already immediate on that embedding; the lower residual witness package and HasUnblockedInteriorEndpoint are no longer downstream burdens beyond that surface, and even the route-facing successor-cycle shell now exports that same residual-layer synthesis package directly at graph level. Theorem49ResidualBoundaryRoute.lean also proves the route-facing projected-endpoint lift theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_residualBoundaryLayerFacePeelWitnessData_and_hasUnblockedInteriorEndpoint and its graph-level direct closure exists_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_residualBoundaryLayerFacePeelWitnessData_and_hasUnblockedInteriorEndpoint_direct. Theorem49ResidualBoundaryPeeling.lean now also proves nonempty_residualBoundarySelectorData_of_closedWalkAnnulusBoundarySource_and_interiorDualBoundaryRootAdjDistancePeelData and nonempty_residualBoundarySelectorData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_interiorDualBoundaryRootAdjDistancePeelData. So the residual selector surface is no longer the first unresolved layer above the live parent-oriented shell: generic InteriorDualBoundaryRootAdjDistancePeelData already lowers to that surface on the same embedding. The live constructive burden therefore sharpens again to deriving that adjacency-distance package, or any equivalent cycle-breaking parent witness strong enough to build residual/current-boundary selector data. Theorem49ResidualBoundaryPeeling.lean now also proves theorem49ResidualBoundaryPositiveProjectedGeometryOn_of_boundaryData_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint and theorem49CurrentBoundaryRemainders_of_boundaryData_and_interiorDualBoundaryRootAdjDistancePeelData_and_hasUnblockedInteriorEndpoint, while Theorem49ResidualBoundaryRoute.lean adds the exact one-collar closed-walk and successor-cycle wrappers. So once the same embedding carries generic InteriorDualBoundaryRootAdjDistancePeelData together with HasUnblockedInteriorEndpoint, the route already recovers both the residual/current-boundary positive projected geometry surface and a concrete layer-face current-boundary remainder witness there. These lower residual surfaces therefore are not any additional burden below the adjacency-distance package; they already follow from it. More sharply, Theorem49ForcingInteriorEdgeObstruction.lean now also exposes explicit BoundaryFreeIncidentFaceSelector constructors from both generic InteriorDualBoundaryRootAdjDistancePeelData and PlanarBoundaryAnnulusRootAdjDistancePeelData, while Theorem49ResidualBoundaryRoute.lean adds the exact one-collar closed-walk and successor-cycle wrappers for both same-split annulus-root parent-peel data and generic adjacency-distance data. So once the live exact shell reaches either of those parent-oriented packages at all, the local boundary-free selector shell is already present on the same embedding; that selector burden is no longer downstream of the parent/adjacency-distance stage;

- Theorem49ResidualBoundaryObstruction.lean now calibrates that same live burden directly on the exact one-collar/v23 shared-interior-pair shell. It proves sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_without_interiorDualBoundaryRootAdjDistancePeelData and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_interiorDualBoundaryRootAdjDistancePeelData_sharedInteriorPair, together with the closed-walk and successor-cycle failed-universal forms. So the current live adjacency-distance burden is not merely unproved on that benchmark: even exact one-collar current-boundary data, a real Tait coloring, `HasUnblockedInteriorEndpoint`, and the exact v23 residual initial state still do not force any same-embedding generic `InteriorDualBoundaryRootAdjDistancePeelData` there. The same file now also proves `not_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_sharedInteriorPair`, `sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootAdjDistancePeelData`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootAdjDistancePeelData_sharedInteriorPair`, and the matching closed-walk / successor-cycle failed-universal theorems. The same obstruction is now lifted to reusable forcing-edge converse failures too: `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness`. So the same repaired exact one-collar/v23 shell does not merely fail to force generic interior-dual peel data: it already fails to force the source-preserving two-sided annulus-root adjacency-distance route package on the same embedding. The same file now also proves `not_nonempty_planarBoundaryAnnulusRootParentPeelData_sharedInteriorPair`, `sharedInteriorPair_closedWalkSource_tait_hasUnblockedInteriorEndpoint_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_sharedInteriorPair`, and the matching closed-walk / successor-cycle failed-universal theorems. So the same repaired exact one-collar/v23 shell does not merely fail at the stronger generic adjacency-distance package: it already fails to force the actual same-embedding `PlanarBoundaryAnnulusRootParentPeelData` route target itself. The live diagnosis is therefore sharper than “derive parent-peel from the residual shell”: that implication is false on the current shell, so the route must either strengthen the upstream same-embedding hypotheses or turn the residual shell into an impossibility theorem instead of continuing to lower downstream. The same obstruction is now lifted from the benchmark instance to reusable route-level failed-universal theorems too: `not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness` and `not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_` embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_forcingInteriorEdgeWitness. So any same-embedding exact-shell witness carrying a forcing interior edge now formally refutes the parent-peel converse; the obstruction is no longer stated only as a single shared-interior-pair counterexample;

- Theorem49InteriorVerticesEndpointObstruction.lean now exposes the opposite concrete calibration: nonempty_interiorDualBoundaryRootAdjDistancePeelData_peeledEndpointTouch and admitsPlanarBoundaryInteriorDualBoundaryRootAdjDistancePeelData_withoutClosedWalkAnnulusBoundarySource_peeledEndpointTouchGraph show that generic boundary-root adjacency-distance data is itself realizable on a concrete benchmark, but only off the live source-preserving route. The endpoint-touch graph is acyclic, so it cannot satisfy the honest closed-walk annulus source at all. Thus the shared-interior-pair exact-shell failure is a source-compatible obstruction on the live route, not an absolute impossibility of InteriorDualBoundaryRootAdjDistancePeelData itself;

- Theorem49ForcingInteriorEdgeObstructionRegression.lean now strengthens that diagnosis on an honest source-compatible benchmark too. The chained-diamonds source already inhabits the stronger local package exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph, which is enough elsewhere in the development to derive canonical witness choice and weak collar geometry. But the new theorems not_nonempty_interiorDualBoundaryRootAdjDistancePeelData_chainedDiamondsWithTriangle_via_forcingInteriorEdgeWitness and not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph show that even this stronger honest closed-walk source package still does not force generic boundary-root adjacency-distance data on the same embedding. So the adj-distance obstruction is not peculiar to the repaired exact one-collar/v23 shell; it already appears higher up on a source-compatible local source surface. The same file now also proves not_nonempty_planarBoundaryAnnulusRootParentPeelData_chainedDiamondsWithTriangle_via_forcingInteriorEdgeWitness and not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph, so even that stronger honest-source package still does not force the current parent-oriented annulus-root target on the same embedding. This sharpens the live burden again: the route now needs genuinely stronger source-side structure than the chained-diamonds at-most-one package, not just more downstream lowering from it. Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean now also proves exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_annulusWitnessAssignment_chainedDiamondsWithTriangleGraph_with_empty_carrier_and_without_taitEdgeColoring, so this same honest source-compatible benchmark already reaches exact one-collar current-boundary data and split-annulus witness assignment on the same embedding. Even there the purified carrier stays empty and no Tait coloring exists. The stronger theorem exists_closedWalkAnnulusBoundarySource_ and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_annulusWitnessAssignment_and_theorem49BoundaryRootSynthesis_for_any_taitEdgeColoring_chainedDiamondsWithTriangleGraph_without_any_taitEdgeColoring now shows that any hypothetical Tait coloring would already close to full Theorem49BoundaryRootSynthesis on that same embedding. Thus on the chained-diamonds at-most-one branch the live blocker for the Theorem 4.9 endpoint is Tait colorability itself, not more exact-shell or witness-assignment packaging;

- that same repaired exact-shell package now also has an honest full Theorem 4.9 synthesis endpoint once the exact construction induced by that package carries witness ownership. Theorem49ResidualBoundaryRoute.lean proves theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_inducedDecompositionWitnessAssignment, theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_inducedDecompositionWitnessAssignment, and also the older chosen-decomposition forms theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_resultingDecompositionWitnessAssignment and theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary_and_resultingDecompositionWitnessAssignment. So once the exact shell carries the same-split parent-peel and selector-deficit package, no further Theorem 4.9 algebra remains after the derived decomposition acquires witness ownership. More sharply, the live downstream burden is now to build PlanarBoundaryAnnulusWitnessAssignment on the exact same-embedding annulus decomposition canonically induced by that repaired parent-peel construction, rather than on an arbitrary chosen decomposition or through another selector wrapper. theorem49BoundaryRootSynthesis_of_planarBoundaryAnnulusConstructionBoundarySupportFaceData_and_resultingDecompositionWitnessAssignment, so once the concrete selected-boundary boundary-support shell is present, the full Theorem 4.9 endpoint reduces exactly to witness ownership on the derived decomposition. The new theorem theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_nonPeelSelectedBoundary_and_resultingDecompositionWitnessAssignment records this downward derivation on the live successor-cycle shell: under the older stronger non-peeled-face contact hypothesis, the selector-deficit burden is discharged internally and the remaining exact-shell work is to reach that boundary-support shell and then supply witness ownership;

- that same repaired local package now already discharges the forcing-edge obstruction on the exact one-collar v23/v23.5 shell itself. `Theorem49ResidualBoundaryRoute.lean` proves `not_nonempty_forcingInteriorEdgeWitness_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary` and `not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_planarBoundaryAnnulusRootParentPeelData_and_selectorDeficitSelectedBoundary`. So once the exact shell carries same-split parent peel data together with the selector-deficit selected-boundary condition and the positive `currentOuterBoundary` side conditions, forcing interior edges are excluded on that same embedding. The remaining exact-shell burden is therefore no longer an unspecified no-forcing theorem, but precisely to derive this explicit selector-deficit construction package from the actual residual shell. Lean now also records that this burden can be stated one layer more directly on the actual boundary-order selector package itself: `not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_closedWalkEmbeddingData_and_selectedBoundaryArcOnFace_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions_and_inducedDecompositionWitnessAssignment`. The same route is now also named directly on the actual successor-cycle shell by `not_nonempty_forcingInteriorEdgeWitness_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions`, `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions_and_inducedDecompositionWitnessAssignment`, and the graph-level closure `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryFreeIncidentFaceSelector_and_boundarySupportFaceData_conditions_and_inducedDecompositionWitnessAssignment_direct`. So once boundary reachability together with facewise selected-boundary arcs already supports a boundary-free selector whose nonselected faces meet selected-boundary support and whose induced positive `currentOuterBoundary` layers are nonempty and pairwise disjoint, both the forcing-edge obstruction and the remaining Theorem 4.9 algebra disappear on that same embedding. More sharply, the remaining burden is witness ownership on the exact selector-induced annulus decomposition, not merely on an unspecified chosen decomposition. Thus the live exact-shell burden is better stated directly at this selector-side local package, and then at that induced decomposition, rather than at another intermediate construction shell; and once that exact successor-cycle selector package is available at graph level, no further route-level repackaging step remains before full `Theorem49BoundaryRootSynthesis`;

- this does not mean the exact Step 2 seed is intrinsically incompatible with the degenerate no-interior source-fixed branch. The new module `Theorem49ResidualBoundaryPositiveRegression.lean` equips the existing two-triangle no-interior source with a proper nonzero Tait edge coloring `twoTriangleTaitEdgeColoring`, proves nonempty_twoTriangleOuterFace_v23ResidualBoundaryInitialState, and then packages that exact seed together with the honest closed-walk source, the root-cover and root-separation facts, empty interiorEdgeSupport, the resulting parent package InteriorDualBoundaryRootParentPeelData, and even a BoundaryFreeIncidentFaceSelector on the same embedding. The new coexistence theorems twoTriangle_closedWalkSource_tait_and_v23ResidualBoundaryInitialState_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis and twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis now make that selector compatibility explicit on both the honest closed-walk shell and the live successor-cycle shell. The stronger coexistence theorems twoTriangle_closedWalkSource_tait_and_v23ResidualBoundaryInitialState_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells and twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells now show that even this selector compatibility still coexists with failure of every currently formalized positive-stage construction shell on both source interfaces. In fact that same concrete benchmark already reaches full theorem-4.9 synthesis on the no-interior parent branch: theorem49BoundaryRootSynthesis_twoTriangleAnnulusEmbedding_for_explicitTaitColoring_via_noInteriorParentRoute proves the endpoint on the actual embedding, and twoTriangle_closedWalkSource_tait_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis plus twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis package the same coexistence on the honest closed-walk and live successor-cycle shells. So the shared-interior-pair obstruction family is not exposing an outright inconsistency between v23 Step 2 algebra and the degenerate no-interior surfaces; it isolates a genuine non-forcing gap on the live endpoint benchmark;

- the same positive exact-seed benchmark now also shows where that compatibility stops on the construction side. The file still proves nonempty_twoTriangleClosedWalkSourcePlanarBoundaryAnnulusConstruction and the actual successor-cycle package twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_planarBoundaryAnnulusConstruction, but it also proves not_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_twoTriangleAnnulusEmbedding, not_nonempty_planarBoundaryAnnulusConstructionFacePartitionDat twoTriangleAnnulusEmbedding, not_nonempty_planarBoundaryAnnulusConstructionPositiveFrontier twoTriangleAnnulusEmbedding, and not_nonempty_planarBoundaryAnnulusConstructionFaceLayerDat twoTriangleAnnulusEmbedding. So the exact Step 2 seed is compatible with the degenerate no-interior branch up through the base annulus construction and residual witness surfaces, and indeed even through full Theorem49BoundaryRootSynthesis on the same embedding via the no-interior parent route, yet it still cannot supply any genuinely positive-stage frontier package there: every positive `currentOuterBoundary` is forced inside `interiorEdgeSupport=\emptyset`. More sharply, this is now stated directly as a synthesis-side failed converse on that benchmark: `twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_not_nonempty_planarBoundaryAnnulusConstructionPositiveF` packages the live successor-cycle shell together with the exact Step 2 seed, full `Theorem49BoundaryRootSynthesis`, and failure of positive-frontier data on one embedding; then `SomePositiveStageConstructionShell`, `NoPositiveStageConstructionShells`, `not_somePositiveStageConstructionShell_of_noPositiveStageConstructionShells`, and `noPositiveStageConstructionShells_of_interiorEdgeSupport_eq_empty` in `PlanarBoundaryAnnulusConstructionCore.lean`, as well as `noPositiveStageConstructionShells_twoTriangleAnnulusEmbedding`, `twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells`, and `exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells_twoTriangleAnnulusGraph` show that this is not merely a last-shell failure: on the actual successor-cycle benchmark, full theorem-4.9 synthesis already coexists with simultaneous failure of all four currently formalized positive-stage construction shells at once, namely boundary-support face data, face-partition data, face-layer data, and positive-frontier data. The separate witness theorems `exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionPositiveFrontierData_twoTriangleAnnulusGraph` and `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFr of_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph` show that even the final theorem-4.9 endpoint under a real Tait coloring does not universally force the positive-frontier construction shell. This is now also lifted to a reusable witness-based converse in `Theorem49Synthesis.lean`: `not_forall_nonempty_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without` abstracts the witness-to-failed-universal pattern itself, while `not_forall_somePositiveStageConstructionShell_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells` packages the whole disjunctive positive-stage surface at once, and `not_forall_nonempty_planarBoundaryAnnulusConstructionBoundarySupportFaceData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionBoundarySupportFaceData`, `not_forall_nonempty_planarBoundaryAnnulusConstructionFacePartitionData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionFacePartitionData`, `not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusConstructionFaceLayerData`, and `not_forall_nonempty_planarBoundaryAnnulusConstructionPositiveFrontierData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_` `planarBoundaryAnnulusConstructionPositiveFrontierData` says any same-embedding example carrying a Tait coloring, full theorem-4.9 synthesis, and failure of one of those positive-stage construction packages already refutes the corresponding universal implication. The exact two-triangle benchmark now instantiates that same reusable converse pattern not only for positive-frontier data but also for boundary-support face data, face-partition data, the stronger face-layer shell, and the single aggregated `SomePositiveStageConstructionShell` surface. It also now states that failed converse directly on the exact-v23 source surfaces themselves: `twoTriangle_closedWalkSource_tait_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells`, `exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_without_somePositiveStageConstructionShell_twoTriangleAnnulusGraph`, and `not_forall_somePositiveStageConstructionShell_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph` do this on the honest closed-walk shell, while `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_without_somePositiveStageConstructionShell_twoTriangleAnnulusGraph` and `not_forall_somePositiveStageConstructionShell_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph` do the same on the live successor-cycle selected-arc shell. These are now reusable route-side patterns rather than only benchmark-local proofs: `Theorem49ResidualBoundaryRoute.lean` also provides `not_forall_somePositiveStageConstructionShell_of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells` and `not_forall_somePositiveStageConstructionShell_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells`, and the two-triangle failed universals now instantiate those shell-specific exact-v23 source-shell converses directly from the stronger witness package `NoPositiveStageConstructionShells`. The same route file now also pushes that converse architecture one step closer to the repaired live shell through the exact one-collar current-boundary versions `not_forall_target_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_without_target`, `not_forall_target_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_without_target`, `not_forall_somePositiveStageConstructionShell_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells`, and not_forall_somePositiveStageConstructionShell_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells. So once a future benchmark carries Theorem 4.9 synthesis together with a real exact one-collar residual shell on the same embedding, the failed-converse layer no longer needs benchmark-local proof scripts. The positive two-triangle benchmark now also supplies one concrete inhabitant of that exact-shell witness side below Theorem 4.9 itself: twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData shows that the live successor-cycle exact one-collar/v23 shell already reaches a real PlanarBoundaryResidualBoundaryLayerFacePeelWitnessData witness on the same embedding. The strengthening twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData adds the full Theorem 4.9 synthesis endpoint to that same exact shell. The new blocker theorem twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_positiveProjectedGeometryOn then makes the next obstruction explicit on that same shell: because the interior-edge support is empty, the benchmark still has not_hasUnblockedInteriorEndpoint_twoTriangleAnnulusEmbedding, not_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_twoTriangleAnnulusEmbedding, not_theorem49ResidualBoundaryPositiveProjectedGeometryOn_twoTriangle, not_theorem49CollarLayerPositiveProjectedGeometryOn_twoTriangle, and not_theorem49HeightOrderedPositiveProjectedGeometryOn_twoTriangle. So the two-triangle exact shell is now formalized as a concrete positive benchmark together with a Lean-visible reason it cannot yet promote to the current positive projected-geometry lane. The further package twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_positiveProjectedGeometryOn shows that the same blocker survives even after adjoining the full Theorem 4.9 endpoint to that exact one-collar residual witness shell. Theorem49ResidualBoundaryPositiveRegression.lean now also lifts both benchmark-local blockers to reusable failed-converse theorems: not_forall_any_positiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_twoTriangle and not_ forall_any_positiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_twoTriangle. So the two-triangle exact-v23 no-interior shell is no longer only a positive benchmark plus local blocker; it now formally refutes any universal derivation from that exact successor-cycle current-boundary shell, with PlanarBoundaryResidualBoundaryLayerFacePeelWitnessData and even after full Theorem49BoundaryRootSynthesis, to any of the current residual / collar-layer / height-ordered positive projected-geometry targets. The same Theorem49ResidualBoundaryPositiveRegression.lean now also lifts that diagnosis to reusable witness-based converses: not_forall_any_positiveProjectedGeometryOn_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData and not_forall_any_positiveProjectedGeometryOn_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData. So once any graph-level exact one-collar/v23 successor-cycle shell carries the same residual-boundary layer witness plus same-embedding failure of those direct positive endpoints, the universal converse is already refuted without replaying the concrete two-triangle proof. The same Theorem49ResidualBoundaryPositiveRegression.lean now also lifts the route-facing replacement-positive obstruction on that shell: not_theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn_twoTriangle, not_theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn_twoTriangle, the bundled blockers twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_replacementPositiveProjectedGeometryOn and twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_replacementPositiveProjectedGeometryOn, and the failed universals not_forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_twoTriangle and not_forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_ nonempty_residualBoundaryLayerFacePeelWitnessData_twoTriangle. So even after adjoining exact one-collar current-boundary data, the exact v23 seed, same-embedding residual-boundary layer witness data, and full Theorem49BoundaryRootSynthesis, the two-triangle successor-cycle shell still does not force any of the current replacement-positive targets either, including the route-facing honest closed-walk and successor-cycle annulus-collar packages. The same Theorem49ResidualBoundaryPositiveRegression.lean now also packages that replacement-positive diagnosis in reusable witness-based form: not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData and not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData. So the exact-v23 residual-layer obstruction is no longer tied to the concrete two-triangle graph: any graph-level witness of that same shell, together with same-embedding failure of the current direct and route-facing replacement-positive endpoints, already collapses the universal converse. The same live-shell diagnosis is now also exhibited by concrete graph-level successor-cycle witness packages themselves: exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_positiveProjectedGeometryOn_twoTriangleAnnulusGraph, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_positiveProjectedGeometryOn_twoTriangleAnnulusGraph, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_replacementPositiveProjectedGeometryOn_twoTriangleAnnulusGraph, and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_replacementPositiveProjectedGeometryOn_twoTriangleAnnulusGraph. So the same two-triangle graph now witnesses that exact one-collar/v23 residual-layer obstruction not only on the fixed twoTriangleAnnulusEmbedding, but as a graph-level existential package directly on the live successor-cycle selected-arc shell too. The same successor-cycle exact-shell branch is now also blocked at the projected nonempty theorem-4.9 endpoint itself, not only at the positive geometry packages above. not_theorem49BoundaryRootNonemptyProjectedSynthesis_ twoTriangle records the fixed-embedding carrier obstruction; the bundled benchmark theorem twoTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_theorem49BoundaryRootNonemptyProjectedSynthesis shows that exact one-collar current-boundary data, the exact v23 seed, a same-embedding residual-boundary layer witness, and full Theorem49BoundaryRootSynthesis already coexist on the live successor-cycle shell while Theorem49BoundaryRootNonemptyProjectedSynthesis still fails. This now comes both as a benchmark-local failed universal not_forall_theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_twoTriangle and as graph-level witness/converse packaging: exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_theorem49BoundaryRootNonemptyProjectedSynthesis_twoTriangleAnnulusGraph and not_forall_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData. So even after theorem-4.9 synthesis itself already closes on that exact residual-layer shell, the live successor-cycle source does not yet force the nonvacuous projected endpoint. The same exact-v23 residual-layer diagnosis now also exists on the honest closed-walk source presentation. The witness packages exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_positiveProjectedGeometryOn_twoTriangleAnnulusGraph, exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_positiveProjectedGeometryOn_twoTriangleAnnulusGraph, exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_replacementPositiveProjectedGeometryOn_twoTriangleAnnulusGraph, and exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_any_ replacementPositiveProjectedGeometryOn_twoTriangleAnnulusGraph show that the same two-triangle exact one-collar/v23 residual witness shell already fails the direct and route-facing positive endpoints on the honest source too. The corresponding reusable converses not_forall_any_positiveProjectedGeometryOn_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData, not_forall_any_positiveProjectedGeometryOn_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData, not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_nonempty_residualBoundaryLayerFacePeelWitnessData, and not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData put the same failed-converse diagnosis on both exact-v23 one-collar source presentations. The honest closed-walk exact-v23 residual-layer shell is now also blocked at the projected nonempty theorem-4.9 endpoint itself. The fixed-embedding benchmark theorem twoTriangle_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_theorem49BoundaryRootNonemptyProjectedSynthesis shows that exact one-collar current-boundary data, the exact v23 seed, a same-embedding residual-boundary layer witness, and full Theorem49BoundaryRootSynthesis already coexist on the honest closed-walk shell while Theorem49BoundaryRootNonemptyProjectedSynthesis still fails. This now comes both as a benchmark-local failed universal not_forall_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_twoTriangle and as graph-level witness/converse packaging: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData_without_theorem49BoundaryRootNonemptyProjectedSynthesis_twoTriangleAnnulusGraph and not_forall_theorem49BoundaryRootNonemptyProjectedSynthesis_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_theorem49BoundaryRootSynthesis_and_nonempty_residualBoundaryLayerFacePeelWitnessData. So even after theorem-4.9 synthesis itself already closes on that exact residual-layer shell, the honest closed-walk source does not yet force the nonvacuous projected endpoint either. The same exact-v23 source-shell diagnosis is now also stated after adding an explicit local selector. The two-triangle benchmark theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_without_somePositiveStageConstructionShell_twoTriangleAnnulusGraph` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_without_somePositiveStageConstructionShell_twoTriangleAnnulusGraph`, together with the failed universals `not_forall_somePositiveStageConstructionShell_of_closedWalkAnnulusBoundarySource_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph` and `not_forall_somePositiveStageConstructionShell_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_twoTriangleAnnulusGraph`, show that even exact-v23 source shells carrying a real `BoundaryFreeIncidentFaceSelector` and the full theorem-4.9 endpoint still do not force any currently formalized positive-stage construction shell. `Theorem49ResidualBoundaryRoute.lean` now packages the corresponding reusable selector-strengthened exact-v23 route converses `not_forall_somePositiveStageConstructionShell_of_exists_embedding_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells` and `not_forall_somePositiveStageConstructionShell_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_and_boundaryFreeIncidentFaceSelector_and_theorem49BoundaryRootSynthesis_and_noPositiveStageConstructionShells`. So on this degenerate branch the live gap is now calibrated even more sharply: it begins strictly after the selector layer and remains open even after full theorem-4.9 synthesis is already in hand on the same exact-v23 source shells;

- the same exact Step 2 seed is now also calibrated on a live nondegenerate carrier benchmark. In `Theorem49ForcingInteriorEdgeObstructionRegression.lean`, `nonempty_wheelWithInnerTriangle_v23ResidualBoundaryInitialState_wheelFace0Boundary` instantiates the exact seed on one of the wheel faces, and `wheelWithInnerTriangle_closedWalkSource_tait_nonempty_carrier_and_v23ResidualBoundaryInitialState_without_naturalResidualSameEmbeddingGeometry` packages that seed together with the honest closed-walk source, the concrete wheel Tait coloring, the nonempty purified selected-boundary interior carrier, and simultaneous failure of residual witness data, residual selector data, height-ordered witness data, and collar-layer witness data on the same embedding. The failed universal `not_forall_some_naturalResidualSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_wheelWithInnerTriangle` shows that honest source data, Tait colorability, a live purified carrier, and the exact v23 seed still do not force any of the current residual same-embedding witness packages on the nondegenerate wheel benchmark. So the exact seed is now calibrated on both sides: it is compatible with the degenerate no-interior branch, yet it remains non-forcing on a live nondegenerate carrier source. The same file now also defines the actual successor-cycle boundary-order geometry `wheelWithInnerTriangleDartSuccessorCycleGeometry` with `wheelWithInnerTriangleDartSuccessorCycleGeometry_selectedBoundaryArcOnFace` and packages the exact seed on that route-facing shell in `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_naturalResidualSameEmbeddingGeometry_wheelWithInnerTriangle`. The failed universal `not_forall_some_naturalResidualSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_wheelWithInnerTriangle` shows that even actual successor-cycle boundary-order data, together with the same Tait coloring, live purified carrier, and exact v23 seed, still does not force any of the current residual same-embedding witness packages on the wheel benchmark. The stronger exact-shell theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_naturalResidualSameEmbeddingGeometry_wheelWithInnerTriangle` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_naturalResidualSameEmbeddingGeometry_wheelWithInnerTriangle`, together with the failed universals `not_forall_some_naturalResidualSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_wheelWithInnerTriangle` and `not_forall_some_naturalResidualSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_wheelWithInnerTriangle`, show that even exact one-collar current-boundary data preserving the extracted annulus boundary split still adds no forcing on this live nondegenerate carrier benchmark. So the exact-seed gap now persists all the way up to the real boundary-order shell, even after the exact one-collar refinement. The same wheel natural residual obstruction is now lifted to the reusable converses `not_forall_some_naturalResidualSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState` and `not_forall_some_naturalResidualSameEmbeddingGeometry_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_` taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState, so both the honest closed-walk and successor-cycle exact-shell natural residual obstructions are now available in reusable witness-based form. Any same-embedding exact one-collar/v23 example with a real Tait coloring, nonempty purified carrier, and nonempty V23ResidualBoundaryInitialState, but without any of the current natural residual same-embedding endpoints, already refutes any universal derivation of that natural residual burden. The same file now also shows that adjoining a real local endpoint witness does not rescue that exact-shell natural residual lane either. It proves the endpoint-bearing witnesses exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_naturalResidualSameEmbeddingGeometry_wheelWithInnerTriangle and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_naturalResidualSameEmbeddingGeometry_wheelWithInnerTriangle, their failed universals not_forall_some_naturalResidualSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle and not_forall_some_naturalResidualSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle, and the reusable converses not_forall_some_naturalResidualSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState and not_forall_some_naturalResidualSameEmbeddingGeometry_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState. So even on the endpoint-bearing exact-v23 one-collar honest closed-walk and successor-cycle shells, a real Tait coloring plus HasUnblockedInteriorEndpoint still does not force any current natural residual same-embedding witness package. The same file now also packages that exact-shell residual obstruction against the graph-level Theorem 4.9 endpoint itself: wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_naturalResidualSameEmbeddingGeometry shows that the explicit Tait coloring still has a graph-level synthesis witness on some embedding even while the honest exact-v23 successor-cycle one-collar wheel shell with the same coloring and live purified carrier still lacks residual boundary witness data, residual selector data, height-ordered witness data, and collar-layer witness data on the fixed wheel embedding. The same benchmark now also carries its local forcing burden on that exact Theorem 4.9-adjoined wheel shell: wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_twoDistinctInteriorEdgesOnFaceBoundary_without_naturalResidualSameEmbeddingGeometry adds a concrete face boundary with two distinct interior edges to the same successor-cycle package, and the new honest-source graph-level companion wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_naturalResidualSameEmbeddingGeometry does the same on the closed-walk shell. So this nondegenerate wheel obstruction is now visible not only as a witness-package failure on the fixed exact-v23 shell, but already as a witness-package failure coexisting with the local two-interior-edge face burden on both source presentations of that same shell. The same file now pushes that exact-v23 one-collar wheel benchmark one layer higher, to the direct and route-facing replacement-positive surface. The theorems exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_any_replacementPositiveProjectedGeometryOn_wheelWithInnerTriangle and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_any_replacementPositiveProjectedGeometryOn_wheelWithInnerTriangle, together with the failed universals not_forall_any_replacementPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle and not_forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle, show that even exact one-collar current-boundary data, together with a real Tait coloring, a live purified carrier, and the exact v23 residual seed, still does not force either direct replacement-positive package or the route-facing annulus-collar replacement packages on the nondegenerate wheel benchmark. The same file now also records the local obstruction directly on that successor-cycle exact-v23 shell: exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_replacementPositiveProjectedGeometryOn_wheelWithInnerTriangle shows that the same fixed wheel shell already carries a concrete face boundary with two distinct interior edges while still failing every current direct and route-facing replacement-positive package. So this upstream positive obstruction is now visible directly at the benchmark geometry itself, not only through the downstream failed universal. The same exact-shell replacement-positive wheel obstruction is now also lifted to the reusable converses not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState and not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_ and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState, so both the honest closed-walk and successor-cycle exact-shell replacement-positive obstructions are now available in reusable witness-based form. Any same-embedding exact one-collar/v23 witness with a real Tait coloring, nonempty purified carrier, and nonempty V23ResidualBoundaryInitialState, but without one of the current direct or route-facing replacement-positive endpoints, already refutes a universal derivation of that whole positive burden. The same file now also pushes that exact-v23 one-collar wheel shell down to the positive construction face-layer package. The theorems exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusConstructionFaceLayerData_wheelWithInnerTriangle and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusConstructionFaceLayerData_wheelWithInnerTriangle, together with the failed universals not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle and not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle, show that even this stronger live benchmark still does not force the downstream positive construction face-layer package on either the honest closed-walk or successor-cycle shell. The same wheel face-layer obstruction is now lifted to the reusable converses not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState and not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState, so both the honest closed-walk and successor-cycle exact-shell face-layer obstructions are now available in reusable witness-based form. Any same-embedding exact one-collar/v23 example with a real Tait coloring, nonempty purified carrier, and nonempty V23ResidualBoundaryInitialState, but without construction face-layer data, already refutes a universal derivation of that downstream positive face-layer burden. The same file also now records the local obstruction directly on that successor-cycle exact-v23 shell: exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_

taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_
v23ResidualBoundaryInitialState_and_twoDistinctInteriorEdgesOnFaceBoundary_
without_planarBoundaryAnnulusConstructionFaceLayerData_
wheelWithInnerTriangle shows that the same fixed wheel shell already carries a concrete face boundary with two distinct interior edges while still failing the upstream face-layer package. So the wheel obstruction is now visible directly at the benchmark geometry itself, not only through the downstream failed universal. The same file now also packages that face-layer obstruction against the graph-level Theorem 4.9 endpoint itself: wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_
fixedEmbedding_without_planarBoundaryAnnulusConstructionFaceLayerData shows that the explicit Tait coloring still has a graph-level synthesis witness on some embedding even while the honest exact-v23 successor-cycle one-collar wheel shell fails the same-embedding positive construction face-layer package;

- the same file now also shows that this exact-v23 one-collar wheel shell still does not force any same-embedding annulus decomposition with witness ownership. The fixed-embedding theorem not_exists_planarBoundaryAnnulusDecomposition_
and_witnessAssignment_wheelWithInnerTriangle, the honest closed-walk exact-shell package exists_embedding_closedWalkAnnulusBoundarySource_and_
oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_
selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_
without_any_planarBoundaryAnnulusWitnessAssignment_wheelWithInnerTriangle, its failed universal not_forall_exists_planarBoundaryAnnulusDecomposition_
and_witnessAssignment_of_closedWalkAnnulusBoundarySource_and_
oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_
and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_
wheelWithInnerTriangle, its reusable converse not_forall_exists_
planarBoundaryAnnulusDecomposition_and_witnessAssignment_
of_exists_embedding_closedWalkAnnulusBoundarySource_and_
oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_
selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState, the route-facing successor-cycle exact-shell package exists_embedding_boundaryReachabilityData_
and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_
oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_
selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_
any_planarBoundaryAnnulusWitnessAssignment_wheelWithInnerTriangle, and its failed universal not_forall_exists_planarBoundaryAnnulusDecomposition_
and_witnessAssignment_of_boundaryReachabilityData_and_
dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_
oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_
and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_
wheelWithInnerTriangle show that even this stronger live benchmark still does not force any annulus decomposition carrying PlanarBoundaryAnnulusWitnessAssignment on the same embedding. The same successor-cycle exact-shell witness obstruction is now also lifted to the reusable converse not_forall_exists_
planarBoundaryAnnulusDecomposition_and_witnessAssignment_of_exists_
embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_
and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_

taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState, so both the honest closed-walk and successor-cycle exact-shell obstructions are now available in reusable witness-based form. Any same-embedding exact one-collar/v23 example with a real Tait coloring, nonempty purified carrier, and nonempty V23ResidualBoundaryInitialState, but without annulus witness ownership on any same-embedding decomposition, already refutes a universal derivation of that downstream decomposition-and-witness burden. The same file now also shows that adjoining a real local endpoint witness does not rescue that exact-shell annulus witness-assignment lane either. It proves the endpoint-bearing witnesses exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_any_planarBoundaryAnnulusWitnessAssignment_wheelWithInnerTriangle and exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_any_planarBoundaryAnnulusWitnessAssignment_wheelWithInnerTriangle, their failed universals not_forall_exists_planarBoundaryAnnulusDecomposition_and_witnessAssignment_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle and not_forall_exists_planarBoundaryAnnulusDecomposition_and_witnessAssignment_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle, and the reusable converses not_forall_exists_planarBoundaryAnnulusDecomposition_and_witnessAssignment_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState and not_forall_exists_planarBoundaryAnnulusDecomposition_and_witnessAssignment_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState. So even on the endpoint-bearing exact-v23 one-collar honest closed-walk and successor-cycle shells, a real Tait coloring plus HasUnblockedInteriorEndpoint still does not force any same-embedding annulus decomposition carrying witness ownership. The same file now also packages that endpoint-bearing witness-assignment failure directly at graph-vs-fixed-embedding level by wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_without_any_planarBoundaryAnnulusWitnessAssignment and wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_without_any_planarBoundaryAnnulusWitnessAssignment, so both the successor-cycle and honest-source endpoint-bearing shells now have explicit graph-level separation theorems against same-embedding witness ownership. The same file had already packaged the non-endpoint version of that obstruction against the graph-level Theorem 4.9 endpoint itself: `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_any_planarBoundaryAnnulusWitnessAssignment` shows that the explicit Tait coloring still has a graph-level synthesis witness on some embedding even while both the honest exact-v23 closed-walk shell and the successor-cycle one-collar wheel shell fail every same-embedding witness assignment. So the remaining repaired-shell burden is sharper again: before the new exact-shell synthesis theorem can apply, the route must derive witness ownership itself, not just decomposition or face-layer data. The same witness-assignment lane now also retains that explicit local two-interior-edge burden at graph-vs-fixed-embedding level through `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_planarBoundaryAnnulusWitnessAssignment`, `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_planarBoundaryAnnulusWitnessAssignment`, `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_planarBoundaryAnnulusWitnessAssignment`, and `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_planarBoundaryAnnulusWitnessAssignment`. So even the annulus witness-assignment obstruction itself is now packaged with the concrete local two-interior-edge face witness on both the non-endpoint and endpoint-bearing successor-cycle and honest-source shells. The same file now also proves that this same exact-v23 one-collar wheel shell already fails to force the earliest same-embedding boundary-free selector package even without any additional endpoint hypothesis: `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector_wheelWithInnerTriangle`, `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle`, `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector_wheelWithInnerTriangle`, and `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle`. So even before adjoining `HasUnblockedInteriorEndpoint`, the live exact one-collar/v23 shell with real Tait coloring and surviving purified carrier still does not generically force even the boundary-free selector that every later canonical-parent repair would need on the same embedding. The same wheel selector obstruction is now lifted to the reusable converse `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState`, so both the honest closed-walk and successor-cycle exact-shell selector obstructions are now available in reusable witness-based form. Any same-embedding exact one-collar/v23 example with a real Tait coloring, nonempty purified carrier, and nonempty `V23ResidualBoundaryInitialState`, but without a boundary-free selector, already refutes a universal derivation of that earliest same-embedding selector burden. The same file now also shows that adjoining a real local endpoint witness does not rescue that exact-shell selector lane either. It proves the endpoint-bearing witnesses `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector_wheelWithInnerTriangle` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFreeIncidentFaceSelector_wheelWithInnerTriangle`, their failed universals `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle` and `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle`, and the reusable converses `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_nonempty_boundaryFreeIncidentFaceSelector_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`. So even on the endpoint-bearing exact-v23 one-collar honest closed-walk and successor-cycle shells, a real Tait coloring plus `HasUnblockedInteriorEndpoint` still does not force even the earliest same-embedding boundary-free selector burden. The same file now also packages this selector failure directly at graph-vs-fixed-embedding level by `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_boundaryFreeIncidentFaceSelector`, `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_without_boundaryFreeIncidentFaceSelector`, `wheelWithInnerTriangleGraph_explicitTait_` graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_without_boundaryFreeIncidentFaceSelector, and wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_without_boundaryFreeIncidentFaceSelector. So the earliest exact-v23 selector obstruction is now graph-level source-symmetric in both its non-endpoint and endpoint-bearing forms, instead of only appearing shell-wise before the later bundled no-endpoint separation. The same file then proves that this exact-v23 one-collar wheel shell already fails to force the still-later same-embedding raw canonical-parent shared-edge-cover package under the same no-endpoint shell: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_wheelWithInnerTriangle, not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_wheelWithInnerTriangle, and not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentS of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle. So the raw canonical-parent cover is not the earliest obstruction on this branch either. The same wheel canonical-parent obstruction is now lifted to the reusable converses not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState and not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState, so both the honest closed-walk and successor-cycle exact-shell raw canonical-parent obstructions are now available in reusable witness-based form. Any same-embedding exact one-collar/v23 example with a real Tait coloring, nonempty purified carrier, and nonempty V23ResidualBoundaryInitialState, but without that raw canonical-parent cover, already refutes a universal derivation of the whole source-fixed cover burden. The same file now also lifts this raw canonical-parent obstruction to the endpoint-bearing exact-v23 shells: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_wheelWithInnerTriangle, not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_

and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_boundaryFaceRootsCanonicalParentSharedEdgeCover_wheelWithInnerTriangle, and not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentS
of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle.
The endpoint-bearing reusable converses not_forall_some_closedWalkAnnulusBoundarySourceBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState and not_forall_some_successorCycleBoundaryFaceRootsCanonicalParentSharedEdgeCover_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState therefore show that even on the honest closed-walk and successor-cycle endpoint-bearing exact one-collar/v23 shells, a real Tait coloring plus HasUnblockedInteriorEndpoint still does not force the same source-fixed raw canonical-parent shared-edge-cover package. So adding an unblocked endpoint alone does not rescue this canonical-parent lane either. The same file now also packages this raw canonical-parent failure directly at graph-vs-fixed-embedding level by wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_boundaryFaceRootsCanonicalParentSharedEdgeCover, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_without_boundaryFaceRootsCanonicalParentSharedEdgeCover, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsCanonicalParentSharedEdgeCover, and wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_without_boundaryFaceRootsCanonicalParentSharedEdgeCover. So the raw canonical-parent shared-edge-cover obstruction is now graph-level source-symmetric in both its non-endpoint and endpoint-bearing forms, instead of only appearing shell-wise before the later parent-peel and bundled no-endpoint separations. The same file then also proves that this exact-v23 one-collar wheel shell already fails to force the later same-embedding annulus-root parent-peel package under the same no-endpoint shell: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_wheelWithInnerTriangle, not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_

selectedBoundaryInteriorCarrier_wheelWithInnerTriangle, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_wheelWithInnerTriangle, and not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle. So the no-endpoint parent-peel failure is not the earliest obstruction on this branch either. The same wheel parent-peel obstruction is now lifted to the reusable converses not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState and not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState, so both the honest closed-walk and successor-cycle exact-shell parent-peel obstructions are now available in reusable witness-based form. Any same-embedding exact one-collar/v23 example with a real Tait coloring, nonempty purified carrier, and nonempty V23ResidualBoundaryInitialState, but without annulus-root parent-peel data, already refutes a universal derivation of that whole source-fixed parent-peel burden. The same file now also lifts this parent-peel obstruction to the endpoint-bearing exact-v23 shells: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_wheelWithInnerTriangle, not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusRootParentPeelData_wheelWithInnerTriangle, and not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle. The endpoint-bearing reusable converses not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState and not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_

embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_
and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_
and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_
v23ResidualBoundaryInitialState therefore show that even on the honest closed-walk and successor-cycle endpoint-bearing exact one-collar/v23 shells, a real Tait coloring plus HasUnblockedInteriorEndpoint still does not force the same source-fixed annulus-root parent-peel package. So adding an unblocked endpoint alone does not rescue this parent-peel lane either. The same file now also packages this parent-peel failure directly at graph-vs-fixed-embedding level by wheelWithInnerTriangleGraph_
explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_
planarBoundaryAnnulusRootParentPeelData, wheelWithInnerTriangleGraph_
explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_
without_planarBoundaryAnnulusRootParentPeelData, wheelWithInnerTriangleGraph_
explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_
hasUnblockedInteriorEndpoint_without_planarBoundaryAnnulusRootParentPeelData,
and wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_
fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_without_
planarBoundaryAnnulusRootParentPeelData. So the annulus-root parent-peel obstruction is now graph-level source-symmetric in both its non-endpoint and endpoint-bearing forms, instead of only appearing shell-wise before the later construction-face-layer and bundled no-endpoint separations. The same file now also lifts the construction face-layer obstruction to the endpoint-bearing exact-v23 shells: exists_embedding_closedWalkAnnulusBoundarySource_
and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_
hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_
planarBoundaryAnnulusConstructionFaceLayerData_wheelWithInnerTriangle,
not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_of_
closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_
and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_
hasUnblockedInteriorEndpoint_wheelWithInnerTriangle, exists_embedding_
boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_
and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_
and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_
v23ResidualBoundaryInitialState_without_planarBoundaryAnnulusConstructionFaceLayerData_
wheelWithInnerTriangle, and not_forall_nonempty_planarBoundaryAnnulusConstructionFaceLayerD
of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_
and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_
and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_
hasUnblockedInteriorEndpoint_wheelWithInnerTriangle. The endpoint-bearing reusable converses not_forall_nonempty_
planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_
closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_
and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_
v23ResidualBoundaryInitialState and not_forall_nonempty_
planarBoundaryAnnulusConstructionFaceLayerData_of_exists_embedding_
boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_
and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_
and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_

`v23ResidualBoundaryInitialState` therefore show that even on the honest closed-walk and successor-cycle endpoint-bearing exact one-collar/v23 shells, a real Tait coloring plus `HasUnblockedInteriorEndpoint` still does not force the downstream positive construction face-layer package. The graph-level package `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_without_planarBoundaryAnnulusConstructionFaceLayerData` now records the same diagnosis alongside explicit Theorem 4.9 synthesis on some embedding. So adding an unblocked endpoint alone does not rescue this face-layer lane either. The same file now also proves that this burden can be reduced to a smaller honest local lemma on the stricter facewise at-most-one branch: `theorem49BoundaryRootSynthesis_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge` and `theorem49BoundaryRootSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge` show that once the exact-v23 one-collar shell already satisfies facewise interior-edge cardinality at most one on the same embedding, full `Theorem49BoundaryRootSynthesis` follows directly. The route file now also exports the corresponding graph-level closures `exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge_direct` and `exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge_direct`, so any exact-v23 one-collar witness already carrying that honest local bound now closes directly to full Theorem 4.9 synthesis at graph level for the supplied Tait coloring. But the same wheel route now also records that this branch is vacuous on the present live endpoint semantics: `not_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge`, `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge`, `not_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_theorem49BoundaryRootNonemptyProjectedSynthesis_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge`, and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_theorem49BoundaryRootNonemptyProjectedSynthesis_and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge` show that once facewise at-most-one holds, neither `HasUnblockedInteriorEndpoint` nor the projected nonempty theorem-4.9 endpoint can coexist with the repaired exact-v23 one-collar shell, on either the honest closed-walk or successor-cycle presentation. So this branch now has a fully formalized split diagnosis: exact-v23 plus facewise at-most-one is sufficient for same-embedding full synthesis, but

incompatible with the current endpoint-bearing semantics. This is now known to be
a genuine inhabited branch rather than a merely conditional one: the new benchmark
theorems `nonempty_diamondWithTriangleFace0_v23ResidualBoundaryInitialState`,
`diamondWithTriangle_closedWalkSource_tait_and_oneCollarAnnulusCurrentBoundaryData_
and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge_and_theorem49Boundar
without_hasUnblockedInteriorEndpoint_or_projectedEndpoint`, and `diamondWithTriangle_
boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_
and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_v23ResidualBoundaryInitia
and_facewiseAtMostOneInteriorEdge_and_theorem49BoundaryRootSynthesis_without_
hasUnblockedInteriorEndpoint_or_projectedEndpoint` show that the explicit-Tait
diamond benchmark already carries the repaired exact-v23 seed on the same fixed
one-collar embedding where facewise at-most- one closes to full synthesis, while both
the local endpoint witness and the projected nonempty theorem-4.9 endpoint still
fail there. The same wheel The same exact-v23 repaired branch is now also shown
not to force the heavier peel-data routes: the consolidated benchmark theorems
`diamondWithTriangle_closedWalkSource_exactV23_facewiseAtMostOne_consolidatedRouteDiagnosis`
and `diamondWithTriangle_successorCycle_exactV23_facewiseAtMostOne_consolidatedRouteDiagnos`
show that on those same honest closed-walk and successor-cycle shells, the explicit-
Tait diamond still fails generic `InteriorDualBoundaryRootAdjDistancePeelData`,
source-fixed `PlanarBoundaryAnnulusRootAdjDistancePeelData`, and source-fixed
`PlanarBoundaryAnnulusRootParentPeelData`. So even after adjoining the repaired
exact-v23 seed and honest facewise-at-most-one cardinality on the same embed-
ding, these annulus-root peel lanes remain sufficient routes only, not necessary
consequences. The same wheel The same repaired exact-v23 at-most-one branch
is not blocked at witness ownership either. The new fixed-embedding theorems
`diamondWithTriangle_closedWalkSource_tait_and_oneCollarAnnulusCurrentBoundaryData_
and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge_and_annulusWitnessAs
and_theorem49BoundaryRootSynthesis_without_hasUnblockedInteriorEndpoint_or_projectedEndpoi`
and `diamondWithTriangle_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_
and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_
and_v23ResidualBoundaryInitialState_and_facewiseAtMostOneInteriorEdge_and_annulusWitnessAs
and_theorem49BoundaryRootSynthesis_without_hasUnblockedInteriorEndpoint_or_projectedEndpoi`
show that the exact-v23 seed, facewise at-most-one, same-embedding annulus witness as-
signment, and full Theorem 4.9 synthesis already coexist on the explicit-Tait diamond
benchmark, on both the honest closed-walk and successor-cycle one-collar shells, before the
branch runs into the live endpoint impossibility. So on this repaired local branch the surviving
blockers are not raw annulus witness ownership either, but the endpoint semantics and the
heavier peel-data repairs killed just above. The same wheel regression file now also proves
`exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_
and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_without_atMostOneInteriorEdgePerF
wheelWithInnerTriangle`, `not_forall_atMostOneInteriorEdgePerFace_of_closedWalkAnnulusBoundar
and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_
taitEdgeColoring_wheelWithInnerTriangle`, `not_forall_atMostOneInteriorEdgePerFace_
of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryDat
and_taitEdgeColoring_and_v23ResidualBoundaryInitialState`, `exists_embedding_boundaryReachabi
and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBou
and_taitEdgeColoring_and_v23ResidualBoundaryInitialState_without_atMostOneInteriorEdgePerF
wheelWithInnerTriangle`, `not_forall_atMostOneInteriorEdgePerFace_of_boundaryReachabilityData`

`and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBou`
`and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_wheelWithInnerTriangle`,
`not_forall_atMostOneInteriorEdgePerFace_of_exists_embedding_boundaryReachabilityData_`
`and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBou`
`and_taitEdgeColoring_and_v23ResidualBoundaryInitialState`, and `wheelWithInnerTriangleGraph_`
`explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_atMostOneInteriorEdgePerFace`.
So this smaller honest local burden is not derivable from the live exact-v23 one-collar shell either: the graph still has a Theorem 4.9 synthesis witness on some embedding, while both the honest closed-walk shell and the successor-cycle exact shell fail face-wise at-most-one on the fixed wheel embedding. The endpoint-bearing graph-level package
`wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_`
`and_hasUnblockedInteriorEndpoint_without_atMostOneInteriorEdgePerFace` now records the same local-cardinality failure alongside explicit Theorem 4.9 synthesis on some embedding. Both the honest closed-walk and successor-cycle exact-shell obstructions are now lifted to reusable converses, so any same-embedding exact one-collar/v23 successor-cycle example with a real Tait coloring and nonempty `V23ResidualBoundaryInitialState`, but without facewise at-most-one interior-edge cardinality, already refutes a universal derivation of that local-cardinality burden, embedding. The same local-cardinality lane now also retains that explicit local two-interior-edge burden at graph-vs-fixed-embedding level through
`wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_`
`and_twoDistinctInteriorEdgesOnFaceBoundary_without_atMostOneInteriorEdgePerFace`,
`wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_`
`and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_atMostOneInteriorE`
`wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_`
`and_hasUnblockedInteriorEndpoint_and_twoDistinctInteriorEdgesOnFaceBoundary_without_`
`atMostOneInteriorEdgePerFace`, and `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthe`
`with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_and_twoDistinctInter`
`without_atMostOneInteriorEdgePerFace`. So even the local-cardinality obstruction itself is now packaged with the concrete local two-interior-edge face witness on both the non-endpoint and endpoint-bearing successor-cycle and honest-source shells. Any surviving repair on this branch therefore has to add new same-embedding local geometry strictly stronger than the present exact-v23 one-collar package, and stronger than the no-endpoint parent-peel lane already killed on the same wheel benchmark. The same regression file now also packages that diagnosis as a single bundled no-endpoint collapse on both exact-v23 one-collar shells:
`exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_`
`and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryI`
`without_currentNoEndpointRepairGeometry_wheelWithInnerTriangle` and `exists_embedding_`
`boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_`
`and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundary`
`and_v23ResidualBoundaryInitialState_without_currentNoEndpointRepairGeometry_wheelWithInner`
their failed universals `not_forall_some_currentNoEndpointRepairGeometry_of_closedWalkAnnulusBoun`
`and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_`
`taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle`
and `not_forall_some_currentNoEndpointRepairGeometry_of_boundaryReachabilityData_`
`and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBou`
`and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_nonempty_selectedBoundaryInte`
`wheelWithInnerTriangle`, and the route-facing graph-level package `wheelWithInnerTriangleGraph_`
`explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_currentNoEndpointRepairGeomet`

So the honest closed-walk and route-facing exact-v23 one-collar wheel shells now simultaneously fail selector, raw canonical-parent cover, parent-peel, construction face-layer, residual selector packages, acyclic/annulus-collar endpoints, and facewise at-most-one, while the explicit Tait coloring still reaches Theorem 4.9 synthesis on some embedding. Any surviving repair on this branch must therefore introduce genuinely new same-embedding geometry outside the current no-endpoint family. The same bundled wheel obstruction is now lifted to the reusable converses `not_forall_some_currentNoEndpointRepairGeometry_of_exists_embedding_closedWalkAnnu` `and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundary` `and_v23ResidualBoundaryInitialState` and `not_forall_some_currentNoEndpointRepairGeometry_` `of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_` `and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_` `and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState`, so any same-embedding exact one-collar/v23 honest closed-walk or successor-cycle example with a real Tait coloring, nonempty purified carrier, and nonempty `V23ResidualBoundaryInitialState`, but without one of the currently known no-endpoint repairs, already refutes a universal derivation of that whole bundled burden. The same regression file now also shows that adjoining a real local endpoint witness does not rescue that bundled no-endpoint lane either. It proves the endpoint-bearing witnesses `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnul` `and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_` `without_currentNoEndpointRepairGeometry_wheelWithInnerTriangle` and `exists_embedding_` `boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_` `and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpo` `and_v23ResidualBoundaryInitialState_without_currentNoEndpointRepairGeometry_wheelWithInner` their failed universals `not_forall_some_currentNoEndpointRepairGeometry_of_closedWalkAnnulusBoun` `and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_` `taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle` and `not_forall_some_currentNoEndpointRepairGeometry_of_boundaryReachabilityData_and_` `dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundar` `and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_` `wheelWithInnerTriangle`, and the reusable converses `not_forall_some_currentNoEndpointRepairGeomet` `of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryDat` `and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_some_currentNoEndpointRepairGeometry_of_exists_embedding_boundaryReachabili` `and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBou` `and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState`. So even on the endpoint-bearing exact-v23 one-collar honest closed-walk and successor-cycle shells, a real Tait coloring plus `HasUnblockedInteriorEndpoint` still fails every currently bundled no-endpoint repair. Adding an unblocked endpoint alone therefore does not escape the wheel obstruction on this lane either. The new graph-level package `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_` `and_hasUnblockedInteriorEndpoint_without_currentNoEndpointRepairGeometry` now records that same endpoint-bearing bundled failure alongside explicit Theorem 4.9 synthesis on some embedding. So this branch is now calibrated at graph-vs-fixed-embedding level too: even after adjoining a real unblocked endpoint on the live exact-v23 one-collar wheel shell, the graph-level positive witness still sits strictly upstream of every currently bundled no-endpoint same-embedding repair. The honest-source side is now packaged at the same graph-vs-fixed-embedding level too by `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_` `with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_without_` currentNoEndpointRepairGeometry. So both endpoint-bearing exact-v23 one-collar source presentations now have explicit graph-level separation theorems against the full bundled no-endpoint repair family. The same graph-vs-fixed-embedding honest-source lift is now also available for the earlier endpoint-bearing wheel obstruction layers wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInte without_any_replacementPositiveProjectedGeometryOn, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInte without_currentSufficientSameEmbeddingGeometry, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_ without_planarBoundaryAnnulusConstructionFaceLayerData, and wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInte without_atMostOneInteriorEdgePerFace. So the full endpoint-bearing wheel diagnostic stack is now source-symmetric at graph level from replacement-positive, through current-sufficient, construction-face-layer, and local-cardinality failure, all the way up to the bundled no-endpoint repair family. The same bundled no-endpoint wheel obstruction now also retains the explicit local two-interior-edge face witness at graph-vs-fixed-embedding level through wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_twoDistinctInteriorEdgesOnFaceBoundary_without_currentNoEndpointRepairGeometry, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_currentNoEndpointR wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_and_twoDistinctInteriorEdgesOnFaceBoundary_without_currentNoEndpointRepairGeometry, and wheelWithInnerTriangleGraph_explicitTait_graphLevelSyn with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_and_twoDistinctInter without_currentNoEndpointRepairGeometry. So even the full bundled no-endpoint collapse is now packaged together with the concrete local two-edge reason that blocks canonical witness choice and facewise-at-most-one collapse, not only as a coarse disjunction of missing repair layers. The same regression file now also carries the analogous graph-vs-fixed-embedding honest-source lift one layer earlier on the non-endpoint wheel stack: wheelWithInnerTriangleGra explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_without_naturalResidualSameEmbeddingGeometry, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_without_any_replacementPositi wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_replacementPos wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_without_planarBoundaryAnnulusConstructionFaceLayerData, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_planarBoundaryAnnu wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_without_any_planarBoundaryAnnulusWitnessAssignment, wheelWithInnerTria explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_planarBoundaryAnnulusWitnessAssignment, wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_and_twoDistinctInteriorEdgesOnFaceBoundary_without_atMostOneInteriorE and wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_without_atMostOneInteriorEdgePerFace. So the non-endpoint exact-v23 wheel diagnostic stack is now source-symmetric at graph-vs-fixed-embedding level from natural-residual failure, through the earlier replacement-positive obstruction together

with its explicit two-interior-edge face witness, construction-face-layer together with its explicit
two-interior-edge face witness, and witness-assignment failure, up to the local-cardinality
obstruction immediately below the bundled no-endpoint collapse. The same endpoint-bearing construction-
face-layer lane now also retains that explicit two-interior-edge face burden at graph level through
`wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_`
`and_hasUnblockedInteriorEndpoint_and_twoDistinctInteriorEdgesOnFaceBoundary_without_`
`planarBoundaryAnnulusConstructionFaceLayerData` and `wheelWithInnerTriangleGraph_`
`explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInte`
`and_twoDistinctInteriorEdgesOnFaceBoundary_without_planarBoundaryAnnulusConstructionFaceLa`
So the endpoint-bearing nondegenerate face-layer wheel obstruction is now source-symmetric at
graph-vs-fixed-embedding level too, not only through the coarser endpoint-bearing face-layer sep-
arations without the explicit local two-edge witness. The same endpoint-bearing witness-assignment
lane now also retains that explicit local two-interior-edge burden at graph-vs-fixed- embedding level
through `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_`
`and_hasUnblockedInteriorEndpoint_and_twoDistinctInteriorEdgesOnFaceBoundary_without_`
`any_planarBoundaryAnnulusWitnessAssignment` and `wheelWithInnerTriangleGraph_explicitTait_`
`graphLevelSynthesis_with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_`
`and_twoDistinctInteriorEdgesOnFaceBoundary_without_any_planarBoundaryAnnulusWitnessAssignm`
So the endpoint-bearing nondegenerate witness-assignment obstruction is now source-symmetric
at graph-vs-fixed-embedding level too, not only through the coarser endpoint-bearing witness-assignment
separations without the explicit local two-edge burden. The same endpoint-bearing local-cardinality
lane now also retains that explicit local two-interior-edge burden at graph-vs-fixed- embedding level
through `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_`
`and_hasUnblockedInteriorEndpoint_and_twoDistinctInteriorEdgesOnFaceBoundary_without_`
`atMostOneInteriorEdgePerFace` and `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthes`
`with_fixedEmbedding_and_closedWalkSource_hasUnblockedInteriorEndpoint_and_twoDistinctInter`
`without_atMostOneInteriorEdgePerFace`. So the endpoint-bearing nondegenerate local-cardinality
obstruction is now source-symmetric at graph-vs-fixed-embedding level too, not only through the
coarser endpoint-bearing local-cardinality separations without the explicit local two-edge burden.
The same regression file now also shows that simply adjoining the honest local endpoint witness
does not rescue the replacement-positive lane. It proves `exists_embedding_closedWalkAnnulusBoundarySo`
`and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpo`
`and_v23ResidualBoundaryInitialState_without_any_replacementPositiveProjectedGeometryOn_`
`wheelWithInnerTriangle`, `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmb`
`and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_`
`and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_and_twoDistinctInteri`
`without_any_replacementPositiveProjectedGeometryOn_wheelWithInnerTriangle`, its failed
universals `not_forall_any_replacementPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundar`
`and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_`
`taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle` and `not_`
`forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_`
`and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBou`
`and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_`
`wheelWithInnerTriangle`, and the graph-level package `wheelWithInnerTriangleGraph_explicitTait_`
`graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_without_`
`any_replacementPositiveProjectedGeometryOn`. So even the endpoint-bearing exact-v23 one-
collar wheel shell still fails every current direct or route-facing replacement-positive package on both
the honest closed-walk and successor-cycle boundary-order shells, and on the successor-cycle shell

this coexists with a concrete two-interior-edge face boundary, although the explicit Tait coloring reaches Theorem 4.9 synthesis on some embedding. Adding `HasUnblockedInteriorEndpoint` alone therefore does not escape the wheel obstruction either; any surviving endpoint-bearing positive repair must add genuinely new same-embedding geometry beyond the present replacement-positive family. The same wheel obstruction is now lifted to the reusable converses `not_forall_any_replacementPosi` `of_exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryDat` `and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState` and `not_forall_any_replacementPositiveProjectedGeometryOn_of_exists_embedding_` `boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_` `and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpo` `and_v23ResidualBoundaryInitialState`, so any same-embedding exact one-collar/v23 example with a real Tait coloring, `HasUnblockedInteriorEndpoint`, and nonempty `V23ResidualBoundaryInitia` but without one of the current direct or route-facing replacement-positive endpoints, already refutes a universal derivation of that whole burden. The larger endpoint-bearing `currentSufficientSameEmbed` target on this wheel shell now collapses on both the honest closed-walk and successor-cycle exact-v23 shells. The theorems `currentSufficientSameEmbeddingGeometry_wheelWithInnerTriangle_` `only_if_attachedBoundaryRootedFacePeelCertificate`, `exists_embedding_closedWalkAnnulusBounda` `and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpo` `and_v23ResidualBoundaryInitialState_reducing_currentSufficientSameEmbeddingGeometry_` `to_attachedBoundaryRootedFacePeelCertificate_wheelWithInnerTriangle`, and `exists_` `embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundar` `and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpo` `and_v23ResidualBoundaryInitialState_reducing_currentSufficientSameEmbeddingGeometry_` `to_attachedBoundaryRootedFacePeelCertificate_wheelWithInnerTriangle` show that on the live endpoint-bearing exact-v23 wheel shells, the height-ordered, collar-layer, well-founded, and annulus-collar branches were already ruled out, so any future proof on either shell reduces to the raw attached certificate on the same embedding. The wheel certificate obstruction `not_nonempty_` `attachedBoundaryRootedFacePeelCertificate_wheelWithInnerTriangle` then kills that last branch as well, and the bundled witnesses `exists_embedding_closedWalkAnnulusBoundarySource_` `and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpo` `and_v23ResidualBoundaryInitialState_without_currentSufficientSameEmbeddingGeometry_` `wheelWithInnerTriangle` and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycle` `and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_` `and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState_without_` `currentSufficientSameEmbeddingGeometry_wheelWithInnerTriangle`, together with the failed universals `not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_closedWalkAnnulu` `and_oneCollarAnnulusCurrentBoundaryData_and_v23ResidualBoundaryInitialState_and_` `taitEdgeColoring_and_hasUnblockedInteriorEndpoint_wheelWithInnerTriangle` and `not_` `forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_` `and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBou` `and_v23ResidualBoundaryInitialState_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_` `wheelWithInnerTriangle` show that even the endpoint-bearing exact-v23 honest closed-walk and successor-cycle shells still do not force any of the currently sufficient same-embedding geometry endpoints on the fixed wheel embedding. The new generic converses `not_forall_exists_some_` `currentSufficientSameEmbeddingGeometry_of_exists_embedding_closedWalkAnnulusBoundarySource` `and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpo` `and_v23ResidualBoundaryInitialState` and `not_forall_exists_some_currentSufficientSameEmbeddi` `of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_` and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_ and_hasUnblockedInteriorEndpoint_and_v23ResidualBoundaryInitialState now lift that same exact-shell current-sufficient failure to reusable witness-based refutations: any same-embedding exact one-collar/v23 honest closed-walk or successor-cycle example with a Tait coloring, HasUnblockedInteri and nonempty V23ResidualBoundaryInitialState, but without one of the current sufficient geometry endpoints, already kills any universal derivation of that whole burden. The live-carrier predecessor shell now has the same reusable converse: not_forall_some_naturalResidualSameEmbeddingGeom of_exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_ and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_ and_nonempty_selectedBoundaryInteriorCarrier_and_v23ResidualBoundaryInitialState. So any same-embedding exact one-collar/v23 successor-cycle example with a Tait coloring, nonempty purified carrier, and nonempty V23ResidualBoundaryInitialState, but without one of the current natural residual same-embedding endpoints, already refutes any universal derivation of that whole live-carrier burden as well. The graph-level package wheelWithInnerTriangleGraph_explicitTait_ graphLevelSynthesis_with_fixedEmbedding_and_hasUnblockedInteriorEndpoint_without_ currentSufficientSameEmbeddingGeometry now records the same diagnosis alongside explicit Theorem 4.9 synthesis on some embedding for the same Tait coloring. So any surviving endpoint-bearing Theorem 4.9 repair must add genuinely new same-embedding geometry outside the present current-sufficient family, not merely revive one of its existing branches;

- the annulus witness file now exposes the actual parent-carrying geometry that had previously been hidden inside the previous-boundary witness endpoint. PlanarBoundaryAnnulusDecomposition.layerOf gives the canonical collar index of an ambient face, and PlanarBoundaryAnnulusWitnessAssignment.exists_parentFace_of_ previousBoundaryWitness proves that every positive-collar witness edge placed on the previous annulus boundary has an adjacent face in a strictly earlier collar layer. The chosen map PlanarBoundaryAnnulusWitnessAssignment.parentFaceOfPreviousBoundaryWitness is then used directly in the well-founded witness construction. So the live geometric burden is now explicit: prove that previous-boundary witness placement, or an honest stronger substitute such as annular no-touch, from real boundary-order data;

- even this repaired witness-placement surface should not be sent back through the canonical BFS construction as if it implied the construction-side ParentWitnessOrientation shell. The new regression file Theorem49PlanarBoundaryAnnulusGeometryRegression.lean gives a two-face model carrying PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry, equivalently WitnessOnCurrentBoundary, whose canonical lowering planarBoundaryAnnulusConstruction_of_annulusCollarGeometry still fails ParentWitnessOrientation because the outer collar face itself has construction distance 0. The same repaired model already carries the attached boundary-rooted face-peel certificate reached by the direct annulus route, so the canonical construction shell is now explicitly shown to be strictly unnecessary on that witness. More sharply, the live Lean route now shows that weak collar geometry already reaches the attached certificate surface, while the repaired previous-boundary witness condition is only needed for the well-founded parent-peeling branch. So the honest ground-up annulus route is the direct geometry $\rightarrow$ certificate bridge together with the repaired-geometry $\rightarrow$ well-founded-witness bridge, not "repaired geometry $\rightarrow$ canonical construction $\rightarrow$ parent orientation";

- that route picture is now witnessed on an honest source model too. The new file Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean equips the genuine closed-walk source model `diamondWithTriangleEmbedding` with a weak one-collar annulus geometry `diamondWithTriangleOneCollarGeometry`. On that same embedding the weak geometry already yields the attached certificate `diamondWithTriangleOneCollarGeometry_hasAttachedRootedFacePeelCertificate`, while the canonical lowering `diamondWithTriangleOneCollarConstruction` still fails `ParentWitnessOrientation` by `not_diamondWithTriangleOneCollarConstructionParentWitnessOrientation`. Even more sharply, the theorem `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_without_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangle` shows that the honest facial closed-walk source, weak annulus geometry, and attached certificate coexist on one embedding even though the current interior-dual boundary-root adjacency-distance package still fails there. So the direct certificate lane is not merely earlier than canonical parent orientation; it is already earlier than the present interior-dual route on a genuine cyclic source model. The same file now also proves `closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_without_planarBoundaryAnnulusRootParentPeelData_diamondWithTriangle`, so even that honest source plus weak one-collar annulus geometry and attached certificate still does not force the current parent-oriented annulus-root target on the same cyclic source model. Thus the direct certificate lane is already strictly earlier than the live parent-peel burden too. The same file now also proves `exists_embedding_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangleGraph`, `exists_embedding_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusRootAdjDistancePeelData_diamondWithTriangleGraph`, `exists_embedding_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusRootParentPeelData_diamondWithTriangleGraph`, and the converse failures `not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph`, `not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph`, and `not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph`. So this is no longer merely a benchmark-local fact about the weak certificate shell in isolation: even after that same honest collar/certificate shell already carries an explicit Tait coloring and the full Theorem 4.9 synthesis endpoint on one embedding, none of the current generic, source-preserving, or live parent-peel adj-distance burdens are necessary consequences there. The direct certificate lane therefore remains strictly earlier

than all three of those parent-oriented routes even post-synthesis. The same file now also proves diamondWithTriangle_explicitTait_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_theorem49BoundaryRootSynthesis_blocks_nonemptyProjectedSynthesis, exists_embedding_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangleGraph, and not_forall_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_and_attachedRootedFacePeelCertificate_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph. So the same weak collar/certificate shell is already too weak not only for the parent-oriented adj-distance targets, but even for the projected nonempty Theorem 4.9 endpoint itself: after full synthesis closes on that same embedding, the selected-boundary-purified carrier is still empty. Thus the direct certificate lane is strictly earlier than the projected endpoint route too, not merely than the parent-peel derivations lying above it. The same file now also proves diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_consolidatedRouteDiagnosis, which packages the whole same-embedding calibration on that genuine cyclic source model: the full current positive geometry stack and explicit-Tait Theorem 4.9 synthesis coexist with a nonempty raw interior-edge endpoint carrier, an empty purified carrier, failure of the projected nonempty endpoint, failure of generic InteriorDualBoundaryRootAdjDistancePeelData, and failure of the live parent-peel package. The more specialized theorems diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_without_interiorDualBoundaryRootAdjDistancePeelData and diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_without_planarBoundaryAnnulusRootAdjDistancePeelData, and diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_without_planarBoundaryAnnulusRootParentPeelData remain available as focused projections of that same diagnosis. The same file now also proves exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_without_interiorDualBoundaryRootAdjDistancePeelData_diamondWithTriangleGraph, exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry
and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusRootAdjDistancePeelData_diamondWithTriangleGraph, exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry
and_theorem49BoundaryRootSynthesis_without_planarBoundaryAnnulusRootParentPeelData_diamondWithTriangleGraph, and the converse failures not_forall_nonempty_interiorDualBoundaryRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph and not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph and

not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph. So even the full current same-embedding positive geometry stack plus full Theorem 4.9 synthesis is still not enough to force generic interior-dual adj-distance data, the source-preserving annulus-root adj-distance package, or the live parent-peel package on that genuine cyclic source benchmark. The same file now also proves diamondWithTriangle_explicitTait_synthesis_without_interiorDualBoundaryRootAdjDistancePeelData, diamondWithTriangle_explicitTait_synthesis_without_planarBoundaryAnnulusRootAdjDistancePeelData, and diamondWithTriangle_explicitTait_synthesis_without_planarBoundaryAnnulusRootParentPeelData, so even the bare Theorem 4.9 synthesis endpoint under the explicit Tait coloring coexists with failure of the generic interior-dual adj-distance package, the source-preserving annulus-root adj-distance package, and the current parent-peel package on that same embedding. Thus none of those parent-oriented burdens is necessary even for a genuine cyclic positive synthesis model. This is now lifted to a reusable converse-failure theorem too: Theorem49ForcingInteriorEdgeObstruction.lean proves not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_forcingInteriorEdgeWitness, not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_forcingInteriorEdgeWitness, and Theorem49ForcingInteriorEdgeObstructionRegression.lean instantiates them on the diamond model via exists_embedding_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_and_forcingInteriorEdgeWitness_diamondWithTriangleGraph and not_forall_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_of_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangle_via_forcingInteriorEdgeWitness and not_forall_nonempty_planarBoundaryAnnulusRootParentPeelData_of_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangle_via_forcingInteriorEdgeWitness. So any synthesis model carrying a forcing interior edge now formally refutes the universal converse from full Theorem 4.9 synthesis back to either same-embedding annulus-root adjacency-distance data or same-embedding parent-peel data. The same file now also proves diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry_without_nonemptyProjectedSynthesis, which seals that comparison on the same genuine cyclic source model: the full current same-embedding positive geometry stack, together with explicit-Tait Theorem 4.9 synthesis, already exists on diamondWithTriangleEmbedding, yet the newer projected nonempty endpoint still fails there because the selected-boundary-purified carrier is empty. This is now lifted to the honest-source graph level by exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_without_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangleGraph and not_forall_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph, which show that even honest closed-walk source data together with canonical witness choice, explicit Tait coloring, annulus collar geometry, repaired previous-boundary witness geometry, all currently sufficient peel witnesses, and full Theorem 4.9 synthesis still do not generically force the projected nonempty endpoint on that graph. `Theorem49ForcingInteriorEdgeObstructionRegression.lean` now also records the complementary graph-vs-embedding separation on the wheel benchmark. `exists_theorem49BoundaryRootSynthesis_wheelWithInnerTriangleGraph_via_degenerateAttachedCertificate` shows that the wheel graph still admits the full Theorem 4.9 synthesis endpoint for the explicit Tait coloring on some embedding via the degenerate all-boundary certificate route, while `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_IDBRAD_obstruction` packages that graph-level positive fact together with the honest source-preserving wheel embedding's Tait coloring, explicit unblocked interior endpoint, and failure of generic same-embedding `InteriorDualBoundaryRootAdjDistancePeelData`. So the current wheel obstructions are genuinely source-preserving same-embedding route failures, not graph-level impossibility of Theorem 4.9 synthesis for that colored graph. The new theorem `wheelWithInnerTriangleGraph_explicitTait_degenerateSynthesis_blocks_nonemptyProjectedSynthesis` sharpens the positive side of that comparison: the current graph-level witness is exactly the degenerate all-boundary embedding, so it reaches full synthesis only on an embedding whose selected-boundary-purified interior-edge endpoint carrier is empty and therefore cannot witness `Theorem49BoundaryRootNonemptyProjectedSynthesis`. Thus the currently known wheel graph-level positive theorem still sits entirely off the projected nonempty lane. The new theorem `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_requires_embedding_change` sharpens that diagnosis one step further: the currently known graph-level witness is provably not the honest source-preserving wheel embedding at all, because the degenerate witness has empty selected-boundary-purified carrier while the honest wheel embedding has a surviving purified carrier. So the present wheel positivity result necessarily changes embeddings before it can fire. The stronger theorem `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_selector_or_acyclicEndpoint_or_IDBRAD` shows that this same separation already covers every currently sufficient same-embedding selector/acyclic endpoint on the honest wheel embedding too: with the same explicit Tait coloring, honest source, nonempty purified carrier, and explicit unblocked interior endpoint, the fixed wheel embedding still has no boundary-free selector, no well-founded/height-ordered/collar-layer witness surface, no weak annulus- collar geometry, and no generic same-embedding `InteriorDualBoundaryRootAdjDistancePeelData`. The new theorem `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_sameEmbedding_projectedSourceRoutes` pushes that same wheel diagnosis all the way to the currently formalized projected endpoint routes: the graph still admits full Theorem 4.9 synthesis on some embedding, but on the honest source-preserving wheel embedding the same explicit Tait coloring still cannot instantiate either the IDBRAD-based projected endpoint route or the source-fixed canonical-parent projected endpoint route. So the benchmark now separates graph-level Theorem 4.9 positivity not only from the live same-embedding sufficient shells, but from both concrete same-embedding projected source routes already formalized in Lean. The same projected-route wheel obstruction now also retains the explicit local two-interior-edge burden through `wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_and_twoDistinctInteriorEdgesOnFaceBoundary_without_` sameEmbedding_projectedSourceRoutes. So even this route-level projected-endpoint failure is now packaged together with the concrete local reason that blocks the facewise-at-most-one / canonical-choice branch on the same fixed embedding, not only as a coarse failed pair of projected-source routes. The new theorem wheelWithInnerTriangleGraph_explicitTait_graphLevelSynthesis_with_fixedEmbedding_without_replacementPositiveProjectedGeometry_or_previousBoundaryWitness extends that same graph-vs-fixed-embedding separation to the newer repaired positive route as well: the graph still admits full Theorem 4.9 synthesis on some embedding, but on the honest source-preserving wheel embedding the same explicit Tait coloring, honest source, and surviving purified carrier still fail both direct replacement-positive packages and the repaired previous-boundary witness geometry. So current graph-level positivity on this benchmark remains strictly upstream of every same-embedding positive route surface now formalized around the live annulus manuscript lane;

- this source-side annulus lane now reaches the Step 2 annihilator endpoint directly as well. In PlanarBoundaryClosedWalkSourceRoute.lean, zero_on_allEdges_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_direct_of_annihilatesKempeClosureGeneratorFamily and its successor-cycle analogue zero_on_allEdges_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_direct_of_annihilatesKempeClosureGeneratorFamily show that honest source data plus same-embedding weak collar geometry, boundary vanishing, and the Kempe-generator annihilation hypothesis already force global vanishing, without routing through the present InteriorDualBoundaryRootAdjDistancePeelData package. The same file now also proves exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_direct and exists_theorem49BoundaryRootSynthesis_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_annulusCollarGeometry_direct, so once honest source data is paired with same-embedding weak collar geometry, the full current theorem-4.9 synthesis endpoint is automatic;

- under the current weak API, even pure annulus decomposition is no longer an open burden on the honest source lane. In Theorem49PlanarBoundaryAnnulusDecomposition.lean, planarBoundaryAnnulusDecomposition_of_boundaryData builds a trivial one-collar decomposition directly from the annulus boundary split, and PlanarBoundaryClosedWalkSource.lean packages that as admitsPlanarBoundaryAnnulusDecomposition_of_admitsPlanarBoundaryClosedWalkAnnulusBoundarySource. So the source shell already reaches pure decomposition; the real remaining burden is witness-edge ownership and stronger collar geometry, not the naked existence of collars;

- Theorem49PlanarBoundaryAnnulusWitnessRegression.lean now shows that this is a real extra burden, not just bookkeeping. The generic theorem not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_exists_two_distinct_interior_edges_on_faceBoundary_of_canonicalDecomposition in Theorem49PlanarBoundaryAnnulusWitness.lean shows that any canonical one-collar decomposition with a face carrying two distinct interior edges has no witness assignment at all, and the concrete annulus boundary model in

Theorem49PlanarBoundaryAnnulusWitnessRegression.lean is an explicit instance. So trivializing pure decomposition from boundary data does not trivialize Step 2 witness ownership;

- that same obstruction now lives on the honest source surface too. The new theorem not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusBoundarySource pushes the canonical one-collar blocker through an honest PlanarBoundaryClosedWalkAnnulusBoundarySource, and the new file Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean realizes it on a finite source model where two outer faces share the consecutive interior edges sip01 and sip12 while a disjoint inner triangle supplies the second boundary component. That model carries the full honest source package but its canonical decomposition still has no witness assignment, so the repair "honest source ⇒ boundary data ⇒ canonical decomposition" is blocked on a genuine same-embedding source witness;

- the newer explicit canonical-witness-choice repair is blocked by the same geometry. In Theorem49PlanarBoundaryAnnulusWitness.lean, not_nonempty_planarBoundaryCanonicalWitnessChoice_of_exists_two_distinct_interior_edges_on_faceBoundary shows that a face carrying two distinct interior edges cannot support the canonical facewise witness choice at all, because such a choice would upgrade to the forbidden canonical one-collar witness assignment. The abstract boundary-data failed converse is not_forall_planarBoundaryAnnulusBoundaryData_implies_nonempty_canonicalWitnessChoice, and the honest source failed converse is not_forall_exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_sharedInteriorPair. Thus the positive theorem admitsPlanarBoundaryAnnulusCollarGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct has a Lean-calibrated extra geometric hypothesis, not a hidden consequence of the honest source shell. That positive hypothesis now constructs actual one-collar geometry on the same embedding: PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry turns any canonical witness choice on any annulus boundary split into a PlanarBoundaryAnnulusCollarGeometry, while PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry_numCollars and PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusCollarGeometry_boundaryData prove that the geometry has exactly one collar and preserves the original boundary split. The honest-source theorem exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice is the broader class version: every honest closed-walk source whose boundary split carries a canonical facewise witness choice has same-embedding one-collar collar geometry with that source boundary split. Lean now also constructs the canonical witness choice from a still more local face condition: PlanarBoundaryCanonicalWitnessChoice.of_atMostOneInteriorEdgePerFace uses the unique interior edge on a face when one exists and a supplied fallback boundary edge otherwise, assuming every face has at most one interior boundary edge and every non-interior face-boundary edge lies on the outer or inner ambient annulus boundary. The latter assumption is now discharged by annulus boundary data itself: PlanarBoundaryAnnulusBoundaryData.mem_outerAmbientBoundary_union_ `innerAmbientBoundary_of_mem_faceBoundary_of_not_mem_interiorEdgeSupport` classifies any non-interior face-boundary edge as selected boundary, hence as outer or inner ambient boundary. Lean now also gets the fallback edge directly from the honest facial closed walks: `PlaneEmbeddingWithBoundary.FaceBoundaryClosedWalk.faceBoundary_nonempty` turns each nonempty face-boundary walk into a nonempty face-boundary edge set, and `PlanarBoundaryClosedWalkAnnulusBoundarySource.fallbackEdge` chooses a boundary edge on every ambient face. Thus `exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_facewiseUniqueInteriorEdge` needs only the equality-style facewise uniqueness of interior boundary edges, and the route-side cardinal theorem `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` needs only the finite at-most-one interior-edge bound. The older explicit source wrapper `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace` is now complemented by the exact converse criterion in `PlanarBoundaryClosedWalkSourceCanonicalWitness.lean`: `nonempty_planarBoundaryCanonicalWitnessChoice_iff_facewiseAtMostOneInteriorEdge_of_closedWalkAnnulusBoundarySource` identifies canonical witness existence on an honest closed-walk source with the facewise at-most-one bound, and `exists_closedWalkAnnulusBoundarySource_and_planarBoundaryCanonicalWitnessChoice_iff_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` records the same equivalence at the graph-level source-search surface. The canonical-witness source attack is therefore exactly the at-most-one face obligation already isolated by the obstruction files. `PlanarBoundaryClosedWalkSourceRoute.lean` now also lowers this lane to the exact split-annulus witness-ownership surface itself: `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct`, `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct`, `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_direct`, and `admitsPlanarBoundaryAnnulusWitnessAssignment_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_direct`. So on the canonical-witness lane, no annulus-decomposition or witness-assignment packaging burden remains below the facewise at-most-one condition itself. `Theorem49AtMostOneNonemptyCarrierImpossibility.lean` now instantiates that exact live successor-cycle facewise-at-most-one shell on the explicit diamond benchmark via `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge_diamondWithTriangleGraph`, `diamondWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_successorCycle_facewiseAtMostOneInteriorEdge`, and `exists_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_facewiseAtMostOneInteriorEdge`. So the live successor-cycle at-most-one branch is genuinely inhabited and already reaches witness assignment and theorem-4.9 synthesis on a concrete graph; the only attempted downstream upgrade on that branch is the nonempty-carrier endpoint, not basic route reachability. The same file now calibrates that constructive branch at the exact current-boundary layer as well: `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_annulusWitnessAssignment_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph` shows that the same live successor-cycle shell already carries source-preserving one-collar current-boundary data while still supporting canonical witness assignment and explicit theorem-4.9 synthesis on the fixed diamond embedding. But the same file now also proves `not_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_hasUnblockedInteriorEndpoint` and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_hasUnblockedInteriorEndpoint`. So on this positive branch the exact current-boundary shell is not the missing burden either: the endpoint-bearing/nonempty-carrier upgrade itself is impossible on the current honest closed-walk and successor-cycle one-collar at-most-one interfaces. The same file now also proves the direct projected-endpoint collapses `not_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge`, `not_theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_facewiseAtMostOneInteriorEdge`, `not_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_theorem49BoundaryRootNonemptyProjectedSynthesis`, and `not_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_theorem49BoundaryRootNonemptyProjectedSynthesis`. So even the actual current projected theorem-4.9 endpoint is structurally blocked on that shell, not just the intermediate endpoint witness. The benchmark packages `exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_annulusWitnessAssignment_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangleGraph` and `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_facewiseAtMostOneInteriorEdge_and_annulusWitnessAssignment_and_taitEdgeColoring_and_theorem49BoundaryRootSynthesis_without_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangleGraph` record the same honest-source and successor-cycle exact-shell diagnosis on the explicit diamond embedding, and `diamondWithTriangleGraph_explicitTait_fixedEmbedding_oneCollarCurrentBoundary_facewiseAtMostOne_annulusWitnessAssignment_and_theorem49BoundaryRootSynthesis_without_theorem49BoundaryRootNonemptyProjectedSynthesis` together with `diamondWithTriangleGraph_explicitTait_fixedEmbedding_closedWalkOneCollarCurrentBoundary_facewiseAtMostOne_annulusWitnessAssignment_and_theorem49BoundaryRootSynthesis_without_theorem49BoundaryRootNonemptyProjectedSynthesis` packages that same conclusion as fixed-embedding explicit-Tait graph-level theorems on the successor-cycle and honest-source shells respectively. `Theorem49AtMostOneNonemptyCarrierImpossibility.lean` now also packages the stronger fallback/count/boundary-rest surface on that benchmark via `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_diamondWithTriangleGraph`, and then instantiates the first corrected projected Definition 4.8 endpoints as `exists_projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` and `exists_span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace`. So this same live successor-cycle at-most-one shell already reaches the corrected projected spanning conclusions on an explicit graph as well; projected spanning is not the blocker on that branch either, because the current exact-shell endpoint-bearing upgrade is already ruled out above. The same benchmark file now pushes one algebraic layer deeper still: `exists_theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` and `exists_theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` show that the explicit live shell also reaches the first chain-dot duality endpoints on the natural purified carrier vertex set. So the remaining burden on that constructive branch is not algebraic projection/duality reachability either; the current exact-shell endpoint-bearing upgrade is already ruled out above. The same file now also reaches the raw finite corrected Definition 4.8 representation there: `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` packages finite Kempe-closure generator sums realizing every class in the corresponding boundary-zero Kirchhoff subspace on that same explicit live shell. So the remaining burden on this constructive branch is not raw Definition 4.8 realization either; the current exact-shell endpoint-bearing upgrade is already ruled out above. The same benchmark file now also reaches the projected-generator detector there: `exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_diamondWithTriangleGraph_for_explicitTaitColoring_via_successorCycle_atMostOneInteriorEdgePerFace` shows that every nonzero class in that same boundary-zero Kirchhoff subspace is detected by some projected Kempe generator on the explicit live shell. So the remaining burden on this constructive branch is not nontrivial projected-generator detection either; the current exact-shell endpoint-bearing upgrade is already ruled out above. The same benchmark file now mirrors that projected Definition 4.8 package on the honest closed-walk shell too. The witness `exists_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_diamondWithTriangleGraph` packages the fallback/count/boundary-rest surface directly for the cyclic source, and the honest-source instantiations `exists_projectedKempeClosureGeneratorSubspace_` eq_planarBoundaryZeroSubmodule_diamondWithTriangleGraph_for_explicitTaitColoring_via_closedWalkAtMostOneInteriorEdgePerFace, exists_span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_diamondWithTriangleGraph_for_explicitTaitColoring_via_closedWalkAtMostOneInteriorEdgePerFace, exists_theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_diamondWithTriangleGraph_for_explicitTaitColoring_via_closedWalkAtMostOneInteriorEdgePerFace, exists_theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_diamondWithTriangleGraph_for_explicitTaitColoring_via_closedWalkAtMostOneInteriorEdgePerFace, exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_diamondWithTriangleGraph_for_explicitTaitColoring_via_closedWalkAtMostOneInteriorEdgePerFace, and exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_diamondWithTriangleGraph_for_explicitTaitColoring_via_closedWalkAtMostOneInteriorEdgePerFace show that the same explicit Tait-colored diamond already reaches the corrected projected subspace, spanning, duality, finite realization, and nonzero- detection endpoints without passing through successor-cycle packaging. So the honest closed-walk side is not blocked below the current projected nonempty endpoint either; it already reaches the same algebraic floor while the exact one-collar projected nonempty upgrade remains refuted above. PlanarBoundaryClosedWalkSourceProjection.lean now removes most of that remaining benchmark-locality too. The new fixed-embedding source-to-projection bridges projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct, span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct, theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct, theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct, exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct, and exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct show that once an honest cyclic source already carries canonical witness choice, the corrected projected subspace, spanning, duality, detector, and finite raw- generator endpoints all follow on that very same embedding. The same file also upgrades the direct local at-most-one shell itself to same-embedding theorems: projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_direct, span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_direct, theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_ orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_direct, theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_direct, exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_direct, and exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_direct. So on the honest at-most-one branch the current formal burden is no longer any corrected Definition 4.8 algebraic step at all, even at fixed-embedding granularity; the surviving obstruction is the projected nonempty carrier endpoint itself. The same file now upgrades the honest same-boundary one-collar geometry shell itself to same-embedding theorems too: projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, and exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct. So once an honest cyclic source already carries same-boundary one-collar geometry with numCollars=1, the corrected projected subspace, spanning, duality, detector, and finite raw-generator endpoints already follow on that very same embedding. The honest one-collar geometry branch is therefore not blocked below corrected Definition 4.8 either; the surviving burden remains the projected nonempty carrier endpoint and the exact-shell upgrades above it. The same file now upgrades the route-facing successor-cycle shell with same-boundary one-collar geometry to matching fixed-embedding direct theorems: projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_ bot_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct, and exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_direct. So even on the explicit boundary-order presentation, once the shell already carries same-boundary one-collar geometry with numCollars=1, the corrected projected subspace, spanning, duality, detector, and finite raw-generator endpoints follow on that very same embedding. The route-facing one-collar geometry branch is therefore not blocked below corrected Definition 4.8 either; it shares the same surviving projected nonempty carrier burden as the honest-source branch. The same file now also upgrades the route-facing successor-cycle canonical witness-choice shell from existential packaging to matching fixed-embedding direct theorems: theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct, theorem49BoundaryRawQuotientErrorPackage_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertices_direct, projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_direct, span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_direct, theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_direct, theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_direct, exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_canonicalWitnessChoice_direct, and exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_ `selectedBoundaryArc_and_canonicalWitnessChoice_direct`. So once the explicit boundary-order shell already carries canonical witness choice, the corrected projected subspace, spanning, duality, detector, and finite raw-generator endpoints follow on that very same embedding; if it also has nonempty purified carrier, then the positive theorem-4.9 endpoint and raw quotient/error package do too. The route-facing canonical-choice branch is therefore no longer blocked at existential repackaging either; it shares the same surviving carrier and exact-shell burdens as the corresponding honest-source branch. The same file now also upgrades the route-facing successor-cycle local at-most-one shell to matching fixed-embedding direct theorems: `projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_direct`, `span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_direct`, `theorem49BoundaryZeroKirchhoffSubspace_inf_chainDot_orthogonal_projectedKempeClosureGeneratorSubspace_eq_bot_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_direct`, `theorem49BoundaryZeroKirchhoffSubspace_chainDot_projectedKempeClosureGeneratorSubspace_eq_zero_iff_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_direct`, `exists_kempeClosureGeneratorFamily_finset_sum_boundaryZeroProjection_of_mem_theorem49BoundaryZeroKirchhoffSubspace_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_direct`, and `exists_projectedKempeClosureGeneratorSubspace_chainDot_ne_zero_of_ne_zero_mem_theorem49BoundaryZeroKirchhoffSubspace_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_direct`. So on the explicit boundary-order presentation, once the local at-most-one hypotheses are already available, the corrected projected subspace, spanning, duality, detector, and finite raw-generator endpoints follow on that very same embedding too. The route-facing at-most-one branch is therefore no longer blocked below corrected Definition 4.8 either; it now matches the honest-source at-most-one branch in leaving the projected nonempty carrier endpoint as the surviving burden. `Theorem49AtMostOneNonemptyCarrierImpossibility.lean` now also lifts that impossibility diagnosis from the facewise shell to the stronger local-cardinality package itself: `not_hasUnblockedInteriorEndpoint_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace`, `not_hasUnblockedInteriorEndpoint_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace`, `not_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace`, `not_theorem49BoundaryRootNonemptyProjectedSynthesis_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFa` and the exact one-collar existential wrappers on both shells. So the extra fallback-edge and boundary-rest data do not rescue this branch either: they only lower it to the corrected Definition 4.8 algebraic floor, while the current endpoint witness and projected nonempty theorem-4.9 endpoint remain structurally false on that stronger shell. still records the fully expanded fallback/count/boundary-rest package. The same cardinal hypothesis now also reaches fixed-embedding collar geometry: `exists_oneCollarAnnulusCollarGeometry_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` constructs one-collar `PlanarBoundaryAnnulusCollarGeometry` preserving the source boundary split from an honest source and facewise at-most-one alone. Thus the minimal extra geometric hypothesis at this layer is precisely the finite at-most-one condition; fallback and non-interior boundary-rest are source consequences. The graph-level endpoint `exists_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdgeCardinality` now packages this as an existential source theorem: an honest closed-walk annulus source plus the facewise cardinal bound directly yields same-embedding one-collar collar geometry preserving the source boundary split. The route file now also repairs this branch to the previous-boundary witness surface: `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_oneCollarAnnulusCollarGeometry` uses the fact that a one-collar geometry has no positive collar index, and `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` applies that repair to the honest-source cardinal criterion without fallback or boundary-rest assumptions. Thus the local at-most-one package reaches the corrected previous-boundary, height-ordered, and collar-layer witness surfaces without another geometric placement hypothesis. The same repaired landing is now available directly from the canonical witness choice surface: `PlanarBoundaryCanonicalWitnessChoice.toPlanarBoundaryAnnulusPreviousBoundaryWitness` builds `PlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry` from the facewise canonical choice, and `exists_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice` packages the honest-source existential form. The result is specific to the canonical one-collar geometry induced by the source boundary split; arbitrary weak collar geometries still have to carry `WitnessOnCurrentBoundary` explicitly. The route theorem `admitsPlanarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_exists_closedWalkAnnulusBour` `and_canonicalWitnessChoice_direct` now lands through this repaired geometry theorem directly, so the canonical source route no longer needs an intermediate one-collar existence detour at this boundary. The route file now also pushes the cardinal form of this local criterion to the theorem-4.9 synthesis endpoint: `exists_planarBoundaryCanonicalWitnessChoice_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge` constructs canonical witness choice from the cardinal at-most-one hypothesis alone, and `exists_theorem49BoundaryRoc` `of_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` then produces `Theorem49BoundaryRootSynthesis` for any Tait coloring. The same source package now has an explicit nonempty-carrier endpoint in the caller-clean cardinal form: `theorem49BoundaryRootNonemptySynthesis_of_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_direct` adds only `(selectedBoundaryInteriorEdgeEndpointVertices` `.Nonempty`, and `exists_theorem49BoundaryRootNonemptySynthesis_of_exists_closedWalkAnnulusBou` `and_facewiseAtMostOneInteriorEdge_with_nonempty_selectedBoundaryInteriorEdgeEndpointVertic` `direct` packages the graph-level form. The positive nonempty burden is therefore a Tait-colorable honest source satisfying facewise cardinal at-most-one while retaining a nonempty purified carrier, with no caller-supplied fallback or boundary-rest proof. The same source-side package now reaches the canonical Definition 4.8 generator spanning bridge. The direct wrappers `exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBounda` `and_atMostOneInteriorEdgePerFace_heightOrdered_direct` and `exists_theorem49_kempeGeneratorSu` `spanning_of_exists_closedWalkAnnulusBoundarySource_and_atMostOneInteriorEdgePerFace_`

collarLayer_direct compose the repaired one-collar source route with the canonical Theorem49KempeGener
lean wrappers, so the only remaining spanning inputs are the target subspace, containment of the raw Kempe-generator span, and boundary-zero interpretation. The stronger one-collar geometry surface also now extracts this hypothesis directly: PlanarBoundaryAnnulusCollarGeometry.toPlanarBoundaryCanonicalWitnessChoice_of_numCollars_eq_one turns any one-collar collar geometry with the fixed boundary split into the canonical facewise witness choice, and exists_planarBoundaryCanonicalWitnessChoice_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData packages the honest-source form where the geometry preserves the source's extracted boundary split. The route file now pushes both positive surfaces to the current theorem-4.9 endpoint: exists_theorem49BoundaryRootSynthes
of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_direct uses the canonical witness choice to feed the split annulus witness-assignment synthesis layer, while exists_theorem49BoundaryRootSynthesis_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData first extracts that canonical witness choice from same-boundary one-collar geometry. The source-facing construction burden may therefore be carried as either the explicit facewise canonical witness choice or the stronger one-collar geometry preserving the source boundary split. The hypothesis is now connected to the canonical Definition 4.8 spanning lane too: exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_heightOrdered_direct and exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_collarLayer_direct start from canonical witness choice itself, while exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_heightOrdered_direct and exists_theorem49_kempeGeneratorSubspace_spanning_of_exists_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_with_sourceBoundaryData_collarLayer_direct start from same-boundary one-collar geometry. Each wrapper factors through the repaired previous-boundary surface before invoking the height-ordered or collar-layer canonical-generator span bridge, so no separate Theorem49SubmoduleDualit
construction remains on these source-side one-collar routes. The corrected boundary-zero target is now available directly as well: PlanarBoundaryClosedWalkSourceProjection.lean routes canonical witness choice, the local at-most-one source package, and same-boundary one-collar geometry to projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule and to the corresponding span_projectedKempeClosureGeneratorFamily_eq_planarBoundaryZeroSubmodule form. This uses the projected Definition 4.8 generator family, which is the generator source that actually lies inside the selected-boundary-zero submodule, so the source-side statement no longer needs an arbitrary target subspace or raw-generator containment premise. The same file now lifts this endpoint to the route-facing successor-cycle source shell: successor-cycle boundary data plus selected-arc contiguity lowers to the honest closed-walk source package while preserving canonical witness choice, the local at-most-one package, or same-boundary one-collar geometry. The resulting projected-span wrappers include exists_projectedKempeClosureGeneratorSubspace_eq_planarBoundaryZeroSubmodule_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEm
and_selectedBoundaryArc_and_canonicalWitnessChoice_heightOrdered_direct and the corresponding finite-collar and explicit-family variants for canonical witness choice, at-most-one, and one-collar geometry. PlanarBoundaryClosedWalkSourceRoute.lean now also has the unprojected raw Definition 4.8 submodule-spanning wrappers on the same successor-cycle source surfaces. The canonical-choice, at-most-one, and same-boundary one-collar successor-cycle packages each feed the height-ordered and finite collar-layer kempeClosureGeneratorSubspace spanning bridge, with the target subspace, raw-generator containment, and boundary-zero interpretation

still explicit. `PlanarBoundaryClosedWalkSourceRoute.lean` now pushes the same route-facing source shell through the full synthesis package. The successor- cycle canonical-choice and facewise cardinal at-most-one surfaces land directly on `Theorem49BoundaryRootSynthesis`; either surface plus nonempty `selectedBoundaryInteriorEdgeEndpointVertices` lands on `Theorem49BoundaryRootNoner` and the same-boundary one-collar source geometry now has both direct and nonempty synthesis wrappers. The facewise at-most-one successor-cycle endpoints are `exists_theorem49BoundaryRootSynthes` `of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundar` `and_facewiseAtMostOneInteriorEdge_direct` and `exists_theorem49BoundaryRootNonemptySynthesis_` `of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundar` `and_facewiseAtMostOneInteriorEdge_with_nonempty_selectedBoundaryInteriorEdgeEndpointVerti` `direct`. These theorems close the packaging gap between the boundary-order source data used upstream and the current theorem-4.9 synthesis API, while leaving the genuine geometric burden exactly at canonical choice, facewise at-most-one, or same-boundary one-collar geometry. The hypothesis is also positively inhabited on the honest diamond-with-triangle source model: `diamondWithTriangl` constructs the canonical facewise witness choice for the boundary split extracted from `diamondWithTriangleC` `exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_diamondWithTri` packages the corresponding source-level existence, and `diamondWithTriangleGraph_admitsPlanarBoundary` `via_closedWalkCanonicalWitnessChoice` lowers it to the canonical one-collar witness-assignment endpoint. This model now also has a concrete coloring witness: `diamondWithTriangleTaitEdgeColoring` is proved by `diamondWithTriangleTaitEdgeColoring_isTait` to be a proper nonzero edge coloring, and `diamondWithTriangleEmbedding_theorem49BoundaryRootSynthesis_for_explicitTaitColor` therefore reaches the theorem-4.9 synthesis surface for an actual named Tait coloring. The same explicit Tait benchmark now also reaches synthesis via the generic at-most-one route: `exists_closedWalkSou` `with_atMostOneInteriorEdgePerFace_diamondWithTriangleGraph` packages the local source hypotheses, `diamondWithTriangleClosedWalkAnnulusBoundarySource_hasCanonicalWitnessChoice_` `via_atMostOneInteriorEdgePerFace` reconstructs canonical witness choice from them, and `diamondWithTri` `theorem49BoundaryRootSynthesis_for_explicitTaitColoring_via_atMostOneInteriorEdgePerFace` proves the fixed-embedding endpoint through that local criterion. The new end-to-end package `diamondWithTriangle_closedWalkSource_explicitTait_atMostOne_to_theorem49RawQuotientErrorPa` then extracts the raw Definition 4.8 representative and selected-boundary-kernel error decomposition for every Kirchhoff chain on that same embedding; `diamondWithTriangle_explicitTait_` `atMostOne_synthesis_has_empty_selectedBoundaryInteriorCarrier` records that this at-most-one derivation still has empty purified selected-boundary interior carrier on the diamond benchmark;

- the same explicit Tait-positive diamond benchmark is now calibrated against the stronger nonempty-carrier target. Lean packages the whole current positive stack as `diamondWithTriangle_explicitTait_currentSufficientSameEmbeddingGeometry`: the honest source, canonical witness choice, explicit Tait coloring, one-collar geometry, collar-layer/height/well-founded peel data, attached rooted certificate, and current boundary-root synthesis all hold on the same embedding. But `diamondWithTriangle_explicitTait_` `synthesis_has_empty_selectedBoundaryInteriorCarrier` also proves that the purified selected-boundary interior carrier is empty and that this same explicit coloring does not inhabit `Theorem49BoundaryRootNonemptySynthesis`. So the remaining positive burden is not another current-synthesis witness; it is a Tait-colorable honest source whose purified selected-boundary carrier is nonempty. This is now recorded at the projected endpoint as well: `not_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangle` rules out `Theorem49BoundaryRootNonemptyProjectedSynthesis` on the diamond-with-triangle embedding for every coloring, and `diamondWithTriangle_explicitTait_synthesis_blocks_nonemptyProjectedSynthesis` packages the explicit Tait-colored ordinary synthesis endpoint with that projected-endpoint obstruction. The stronger theorems `exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_without_theorem49BoundaryRootNonemptyProjectedSynthesis_diamondWithTriangleGraph` and `not_forall_theorem49BoundaryRootNonemptyProjectedSynthesis_of_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_and_taitEdgeColoring_and_currentSufficientSameEmbeddingGeometry_and_theorem49BoundaryRootSynthesis_diamondWithTriangleGraph` show that this is not just a fixed-embedding coincidence: the full current honest-source geometry stack together with explicit-Tait Theorem 4.9 synthesis already fails to force the projected nonempty endpoint generically on that graph;

- a separate carrier-calibration benchmark now shows that the Tait and nonempty purified-carrier conditions can be made concrete without using the shared-interior-pair obstruction model as the only positive witness. `Theorem49PlanarBoundaryAnnulusWheelWitnessRegression.lean` defines `wheelWithInnerTriangleGraph`, whose three interior spokes meet a central vertex while a disjoint inner triangle supplies selected-boundary support. The explicit coloring `wheelWithInnerTriangleTaitEdgeColoring` is proved Tait by `wheelWithInnerTriangleTaitEdgeColoring_isTait`, and `selectedBoundaryInteriorEdgeEndpointVertices_nonempty_wheelWithInnerTriangle` proves that the purified selected-boundary interior carrier is nonempty. The bundle `wheelWithInnerTriangle_tait_and_nonempty_selectedBoundaryInteriorCarrier` is intentionally only a Tait-plus-carrier witness, not a positive canonical witness-choice or same-embedding collar-geometry theorem. Lean now proves the negative calibration directly: `exists_two_distinct_interior_edges_on_wheelWithInnerTriangle_boundary` exhibits a wheel face with two distinct interior spokes, `not_nonempty_planarBoundaryCanonicalWitnessChoice_wheelWithInnerTriangle` rules out canonical witness choice for every boundary split on the embedding, and `not_exists_oneCollar_planarBoundaryAnnulusCollarGeometry_wheelWithInnerTriangle` rules out one-collar weak annulus geometry. The bundled theorem `wheelWithInnerTriangle_tait_and_nonempty_carrier_without_canonical_or_oneCollar` therefore records exactly that this model separates the coloring/carrier obligations from the stronger witness-ownership burden;

- the same wheel benchmark has now been calibrated against the face-local at-most-one sufficient condition. Lean proves `wheelWithInnerTriangle_face0_interiorEdgeFilter_card`: the interior-edge filter on face 0 has cardinality 2. Therefore `not_wheelWithInnerTriangle_atMostOneInteriorEdgePerFace` blocks the local premise used by `PlanarBoundaryCanonicalWitnessChoice.of_atMostOneInteriorEdgePerFace`, and `not_exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_wheelWithInnerTriangleEmbedding` rules out the full fixed-embedding fallback/at-most-one/boundary-rest package for any honest closed-walk source on that embedding. The sharper theorem `wheelWithInnerTriangle_tait_and_nonempty_carrier_without_atMostOne_or_canonical_or_oneCollar` records the exact status: Tait colorability and nonempty purified carrier hold, while the at-most-one premise, canonical witness choice, and one-collar geometry all fail;

- the wheel obstruction now reaches the raw replacement witness-cover interfaces. Lean proves `not_nonempty_planarBoundaryHeightOrderedFacePeelWitnessData_wheelWithInnerTriangle`: once a peeled wheel face chooses one central spoke, the other central spoke on that face must be witnessed at strictly larger height; following the three triangular wheel faces forces a strict height cycle through `wit01`, `wit02`, and `wit03`. The same cycle now also blocks the acyclic parent-peeling surface: `Theorem49PlanarBoundaryPeeling.lean` proves `PlanarBoundaryWellFoundedFacePeelWitnessData.toPlanarBoundaryHeightOrderedFacePeelWitnessData` using the canonical parent-height rank, and the wheel regression applies this as `not_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_wheelWithInnerTriangle`. The finite collar-layer interface is blocked by `not_nonempty_planarBoundaryCollarLayerFacePeelWitnessData_wheelWithInnerTriangle`, because collar-layer data canonically lowers to height-ordered witness-cover data. The same fixed embedding now also blocks weak annulus-collar geometry itself: `not_nonempty_planarBoundaryAnnulusCollarGeometry_wheelWithInnerTriangle` composes the collar-to-height lowering with the strict height-cycle contradiction, and `not_forall_some_acyclicWitnessCover_or_annulusCollarGeometry_of_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_wheelWithInnerTriangle` records the corresponding failed universal from Tait colorability plus nonempty purified carrier. This identifies a real construction-side requirement: the positive route must provide a well-founded orientation or selected-boundary break in the interior-face witness dependency relation before even the direct well-founded/height/collar witness surfaces, and hence weak annulus-collar geometry, can be expected;

- the same wheel model now also instantiates the abstract forcing-edge obstruction. In `Theorem49ForcingInteriorEdgeObstructionRegression.lean`, `wheelWithInnerTriangleForcingInteriorEdgeWitness` chooses the central spoke `wit01`; the two incident wheel faces contain selected-boundary support `wit12` and `wit31`. Therefore Lean derives `not_nonempty_planarBoundaryAnnulusRootAdjDistancePeelData_wheelWithInnerTriangle_via_forcingInteriorEdgeWitness` and `not_nonempty_planarBoundaryAnnulusConstructionFaceLayerData_wheelWithInnerTriangle_via_forcingInteriorEdgeWitness`. The bundle `wheelWithInnerTriangle_tait_nonempty_carrier_and_forcing_obstruction` shows that this Tait-colorable, nonempty-carrier benchmark also carries the same obstruction that blocks the current annulus-root and face-layer construction routes on the fixed embedding;

- the cyclic-source at-most-one derivation attempt has also been converted from commentary into a concrete failed universal. The shared-interior-pair honest source now proves `sharedInteriorPair_sipFace0_interiorEdgeFilter_card`: on `sipFace0`, the interior-edge filter consists of the two shared interior edges `sip01` and `sip12`. Therefore `not_sharedInteriorPair_atMostOneInteriorEdgePerFace` blocks the face-local premise, while `route_impossibility_at_most_one_not_derivable_from_cyclic_source` and `not_forall_closedWalkSource_implies_atMostOneInteriorEdgePerFace` in `Theorem49CyclicSourceAtMostOneDerivation.lean` show that even an honest closed-walk annulus source does not imply at-most-one. This records the at-most-one condition as an independent sufficient hypothesis rather than a consequence of cyclic boundary-order data;

- the strengthened-condition derivation note is now theorem-level as well. `Theorem49StrengthenedConditionAtMostOneDerivation.lean` packages the same shared-interior-pair witness with annulus boundary reachability, dart-successor facial geometry, and selected-boundary arcs in `exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_without_atMostOneInteriorEdgePerFace_sharedInteriorPair`. The failed universals `not_forall_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_implies_atMostOneInteriorEdgePerFace_sharedInteriorPair` and `not_forall_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_boundaryComponentConstant_implies_atMostOneInteriorEdgePerFace_sharedInteriorPair` show that even adding the boundary-component constancy derived from the honest source fields does not force at-most-one;

- `Theorem49DerivationImpossibilitySummary.lean` now packages the separate route obstructions as `theorem49_route_obstruction_matrix`. The theorem records, in one checked bundle, the cyclic-source failure to imply at-most-one, the chained-diamonds failure of source plus at-most-one to imply nonempty purified carrier, the strengthened successor-cycle failed universal with boundary-component constancy, and the wheel-with-inner-triangle Tait/nonempty-carrier forcing-edge obstruction to the current annulus-root and face-layer routes;

- the positive canonical-witness burden is now inhabited by a second concrete honest source model, not merely by another general constructor. `Theorem49PlanarBoundaryAnnulusHonestGeometryRegression.lean` defines `chainedDiamondsWithTriangleGraph`: two diamond cells chained along one outer boundary component, plus a disjoint triangular inner boundary component. The honest source `chainedDiamondsClosedWalkAnnulusBoundarySource` carries closed facial walks and selected-boundary arcs for all five faces. The direct finite witness `chainedDiamondsClosedWalkCanonicalWitnessChoice` chooses the two interior shared edges `cdt12` and `cdt45` on the two diamond pairs, and one boundary edge on the inner triangle. Lean now proves `chainedDiamondsWithTriangle_interiorEdgeSupport_eq`: the interior edge support is exactly $\{\texttt{cdt12}, \texttt{cdt45}\}$, and `chainedDiamondsWithTriangle_atMostOneInteriorEdgePerFace`: each face has at most one interior edge on its boundary. So this model, despite having two distinct interior witness edges across five faces, still satisfies the local face-wise at-most-one condition enabling the canonical-choice constructor. The remaining generic hypotheses are now explicit: `chainedDiamondsCanonicalWitnessEdge_mem` supplies the fallback edge on each face, `chainedDiamonds_nonInteriorOnAmbient` proves that every non-interior face-boundary edge lies on the extracted ambient boundary, and `exists_closedWalkSource_with_atMostOneInteriorEdgePerFace_chainedDiamondsWithTriangleGraph` packages the full local premise. Lean then reconstructs canonical witness choice via `chainedDiamondsClosedWalkAnnulusBoundarySource_hasCanonicalWitnessChoice_via_atMostOneInteriorEdgePerFace` and derives `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_atMostOneInteriorEdgePerFace` and `chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusCollarGeometry_via_atMostOneInteriorEdgePerFace`. The route-level endpoint theorem now applies to the same package: `chainedDiamondsWithTriangleEmbedding_theorem49BoundaryRootSynthesis_via_atMostOneInteriorEdgePerFace` and `exists_theorem49BoundaryRootSynthesis_chainedDiamondsWithTriangleGraph_` via_atMostOneInteriorEdgePerFace show that the local criterion reaches the current theorem-4.9 synthesis surface, conditional on a Tait coloring, without appealing to the hand-built one-collar object. The same model has now been calibrated against the stronger nonempty-carrier target: selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_chainedDiamondsWithTriangle and not_selectedBoundaryInteriorEdgeEndpointVertices_nonempty_chainedDiamondsWithTriangle prove that selected-boundary purification deletes all endpoints of cdt12 and cdt45. Combined with not_exists_taitEdgeColoring_chainedDiamondsWithTriangleGraph, the status theorem chainedDiamonds_atMostOne_source_has_empty_carrier_and_no_tait records that chainedDiamonds contributes at-most-one source geometry, but not the nonempty-carrier or Tait-coloring hypotheses needed by the new endpoint. This obstruction is also recorded as an explicit failed universal in Theorem49AtMostOneNonemptyCarrierImpossibility: Lean proves route_impossibility_at_most_one_does_not_imply_nonempty_carrier, not_forall_atMostOneInteriorEdgePerFace_implies_nonempty_carrier, and not_forall_closedWalkSource_and_atMostOneInteriorEdgePerFace_implies_nonempty_carrier from the same chainedDiamonds honest source, so carrier nonemptiness remains an independent hypothesis for the non-vacuous endpoint. The route-facing successor-cycle source shell is now sharper than an independence result. The diamond-with-triangle model carries diamondWithTriangleDartSuccessorCycleGeometry with selected-arc contiguity proved by diamondWithTriangleDartSuccessorCycleGeometry_selectedBoundaryArcOnFace, satisfies facewise at-most-one, and still has empty purified carrier. Lean packages this as successor_cycle_at_most_one_and_empty_carrier_independent and the failed universal not_forall_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_atMostOneInteriorEdgePerFace_implies_nonempty_carrier. More importantly, Lean now proves this obstruction at the honest closed-walk level: selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_of_closedWalkEmbeddingData_and_facewiseAtMostOneInteriorEdge says that facial closed-walk data plus facewise cardinal at-most-one forces the selected-boundary-purified interior-edge endpoint carrier to be empty on any embedding. Its contradiction forms not_exists_closedWalkEmbeddingData_and_facewiseAtMostOneInteriorEdge_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices and not_exists_closedWalkAnnulusBoundarySource_and_facewiseAtMostOneInteriorEdge_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices therefore rule out the current closed-walk at-most-one non-vacuous endpoint outright. Lean also records the constructive contrapositive as exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkEmbeddingData_and_nonempty_selectedBoundaryInteriorEdgeEndpointVertices and the route-facing exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusBoundarySource_and_hasUnblockedInteriorEndpoint: any honest closed-walk source carrying the live unblocked endpoint witness must have a face boundary with two distinct interior edges. The route-facing positive annulus-collar package now carries the same necessary condition: exists_two_distinct_interior_edges_on_faceBoundary_of_closedWalkAnnulusCollarPositiveProjectedGeometryOn consumes Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn, separating that positive replacement surface from the facewise at-most-one branch before the synthesis endpoint is used. The same route file now also proves exists_two_distinct_interior_edges_on_faceBoundary_of_successorCycleAnnulusCollarPositiveProjectedGeometryOn, so the successor-cycle annulus-collar shell carries that same local two-interior-edge obstruction directly. The package-level negations not_theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn_of_facewiseAtMostOneInteriorEdge and not_theorem49SuccessorCycleAnnulusCollarPositiveProjected of_facewiseAtMostOneInteriorEdge then make the consequence explicit: neither route-facing annulus-collar positive package can inhabit the canonical facewise-at-most-one branch. The stronger selected-boundary-contact construction shell is now blocked there as well: not_theorem49ClosedWalkAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn_of_facewiseAtMostOneInteriorEdge and not_theorem49SuccessorCycleAnnulusConstructionBoundary of_facewiseAtMostOneInteriorEdge show that even that stronger source-facing package cannot inhabit the same branch, while exists_two_distinct_interior_edges_on_faceBoundary_of_successorCycleAnnulusConstructionBoundarySupportPositiveProjectedGeometryOn exposes the same local two-interior-edge obstruction directly on the successor-cycle selected-boundary-contact shell. The older successor-cycle theorem selectedBoundaryInteriorEdgeEndpointVertices_eq_empty_of_dartSuccessorCycleEmbeddingData_and_facewiseAtMostOneInteriorEdge is therefore no longer the deepest route-facing obstruction. The older direct source existence theorem exists_embedding_closedWalkAnnulusBoundarySource_and_canonicalWitnessChoice_chainedDiamondsWithTriangleGraph and the lowered endpoint chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusWitnessAssignment_via_closedWalkCanonicalWitnessChoice show that canonical witness choice exists on a larger honest model with two separate interior witness edges; the new at-most-one route shows the generic local criterion itself is inhabited by that same larger model. The same finite witness now builds the named one-collar geometry chainedDiamondsOneCollarGeometry; Lean proves chainedDiamondsOneCollarGeometry_numCollars and chainedDiamondsOneCollarGeometry_boundaryData, packages the source-level one-collar existence theorem exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCollarGeometry_chainedDiamondsWithTriangleGraph, and records the graph-level admissibility endpoint chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAnnulusCollarGeometry_via_closedWalkCanonicalWitnessChoice. The same concrete source now reaches the downstream certificate branch: chainedDiamondsOneCollarWitnessOnCurrentBoundary and chainedDiamondsOneCollarPreviousBoundaryWitnessGeometry record the repaired placement surface, and Lean constructs chainedDiamondsOneCollarCollarLayerFacePeelWitnessData, chainedDiamondsOneCollarHeightOrderedFacePeelWitnessData, chainedDiamondsOneCollarWellFoundedFacePeelWitnessData, and chainedDiamondsOneCollarAttachedRootedFacePeelCertificate. The graph-level endpoints chainedDiamondsWithTriangleGraph_admitsPlanarBoundaryAttachedRootedFacePeelCertificate_via_oneCollarGeometry and exists_theorem49BoundaryRootSynthesis_chainedDiamondsWithTriangleGraph_via_oneCollarGeometry show that this larger honest-source witness already lands on the current certificate and theorem-4.9 boundary-root synthesis surfaces. The coloring side is explicitly calibrated by not_exists_taitEdgeColoring_chainedDiamondsWithTriangleGraph: the shared vertex 3 has four incident edges, so a Tait coloring would require four pairwise distinct nonzero colors although the color space supplies only three. Hence the chained model is a larger positive source/collar/certificate witness, while its synthesis theorem is vacuous as a Tait-coloring instance;

- that blocker now extends to the whole one-collar weak-geometry branch. The new theorem `not_nonempty_planarBoundaryAnnulusWitnessAssignment_of_numCollars_eq_one_and_exists_two_distinct_interior_edges_on_faceBoundary` in `Theorem49PlanarBoundaryAnnulusWitness.lean` shows that any one-collar witness assignment already fails when one face carries two distinct interior edges. Then `Theorem49PlanarBoundaryAnnulusGeometry.lean` lifts this to `not_exists_planarBoundaryAnnulusCollarGeometry_of_numCollars_eq_one_and_exists_two_distinct_interior_edges_on_faceBoundary`, and `Theorem49PlanarBoundaryAnnulusHonestWitnessRegression.lean` instantiates it on the same honest source model as `not_exists_oneCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair`. So the live weak-collar route from honest source data is now blocked on the entire one-collar branch, not only on the canonical decomposition recovered from the source;

- the same witness first blocked the fixed-source-split two- and three-collar repairs through the current-boundary bridge `not_exists_planarBoundaryAnnulusCollarGeometry_of_not_exists_planarBoundaryAnnulusCurrentBoundaryData_of_numCollars_eq_and_boundaryData`. But Lean now shows that this does *not* make bare current-boundary data a live obstruction surface. In `Theorem49PlanarBoundaryAnnulusGeometry.lean`, `exists_oneCollarAnnulusCurrentBoundaryData_of_closedWalkAnnulusBoundarySource` and the graph-level wrappers `exists_oneCollarAnnulusCurrentBoundaryData_with_sourceBoundaryData_of_exists_closedWalkAnnulusBoundarySource` and `exists_oneCollarAnnulusCurrentBoundaryData_with_sourceBoundaryData_of_exists_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc` show that any honest annulus source already carries same-embedding source-preserving one-collar `PlanarBoundaryAnnulusCurrentBoundaryData`. The shared-interior-pair benchmark positively realizes that weak shell via `closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_with_sharedInteriorPairBoundaryData`, `exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_with_sharedInteriorPairBoundaryData`, and `exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_with_sharedInteriorPairBoundaryData`. What still fails there are the stronger exact-count repairs `not_forall_exists_planarBoundaryAnnulusCurrentBoundaryData_with_sourceBoundaryData_and_numCollars_eq_two_sharedInteriorPair`, `not_forall_exists_planarBoundaryAnnulusCurrentBoundaryData_with_sourceBoundaryData_and_numCollars_eq_three_sharedInteriorPair`, and their successor-cycle analogues. So the current-boundary bridge now marks the first *positive* weak shell above pure annulus boundary data, while the actual obstruction threshold starts only at exact multi-collar or stronger same-embedding geometry;

- the stronger regression no longer fixes the boundary split. The generic terminal-layer theorem `false_of_mem_terminal_collarFaces_and_exists_two_distinct_interior_edges_on_faceBoundary` shows that a terminal collar face cannot carry two distinct interior edges. On `sharedInteriorPairAnnulusEmbedding`, both outer faces carry the two interior edges `sip01` and `sip12`, so `terminal_collarFaces_eq_singleton_sipFace2` forces every terminal collar to be exactly the inner triangular face. Then false_of_intermediateBoundary_before_singleton_sipFace2 rules out any predecessor stage whose remainder is this singleton, yielding not_exists_twoCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair and not_exists_threeCollar_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair without requiring the geometry's boundary split to equal sharedInteriorPairBoundaryData;

- the remaining "start at four collars" escape hatch is still closed by the finite-face bound PlanarBoundaryAnnulusCollarGeometry.numCollars_le_card_faces. The honest source file specializes it as numCollars_le_three_of_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair, since sharedInteriorPairAnnulusEmbedding has exactly three ambient faces. Combining that bound with the one-, two-, and three-collar blockers gives not_nonempty_planarBoundaryAnnulusCollarGeometry_sharedInteriorPair. Thus closedWalkAnnulusBoundarySource_without_annulusCollarGeometry_sharedInteriorPair and exists_embedding_closedWalkAnnulusBoundarySource_without_annulusCollarGeometry_sharedInteriorPair now witness an honest closed-walk annulus source with no weak annulus collar geometry on the same embedding at all, regardless of how a later repair tries to choose the annulus boundary split. The failed universal same-embedding converse is not_forall_exists_planarBoundaryAnnulusCollarGeometry_of_closedWalkAnnulusBoundarySource_sharedInteriorPair. The same absence of weak collar geometry now directly refutes the source-alone previous-boundary route: not_nonempty_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_sharedInteriorPair projects a hypothetical repaired previous-boundary witness back to collar geometry, while closedWalkAnnulusBoundarySource_without_previousBoundaryWitness_sharedInteriorPair and not_forall_exists_planarBoundaryAnnulusPreviousBoundaryWitnessGeometry_of_closedWalkAnnulusBoundarySource_sharedInteriorPair package the concrete failed implication. This does not settle the stronger question with weak collar geometry already present; it records that honest closed-walk source data alone is insufficient. The corresponding repaired source-plus-collar investigation now has no placeholder theorem: Lean proves honest_source_plus_collar_geometry_extending_previous_boundary_witness_iff_witnessOnCurrentBoundary, which identifies the exact remaining fixed-collar obligation as WitnessOnCurrentBoundary. That obligation is now known not to follow from an arbitrary weak collar on an honest source embedding. The connected chained-diamonds source admits the deliberately mismatched two-collar geometry chainedDiamondsBadTwoCollarGeometry: the terminal triangular face chooses cdt97 while its previous intermediate boundary is the singleton cdt45. Lean proves not_witnessOnCurrentBoundary_chainedDiamondsBadTwoCollarGeometry, packages the source-plus-bad-collar witness as exists_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_without_witnessOnCurrentBoundary_chainedDiamonds, and refutes the universal implication as not_forall_honest_source_plus_collar_geometry_forces_previous_boundary_witness. It also proves not_nonempty_previousBoundaryWitnessGeometry_extending_chainedDiamondsBadTwoCollarGeometry, so this exact weak collar cannot be repaired without changing the collar geometry. Thus the repaired lane must use a collar geometry tied to the source boundary split, carry WitnessOnCurrentBoundary explicitly, or derive the collar from stronger annular geometry. The same module now proves the sharper source-side calibration exists_atMostOne_closedWalkAnnulusBoundarySource_and_annulusCollarGeometry_without_witnessOnCurrentBoundary_chainedDiamonds, not_forall_atMostOne_source_plus_collar_geometry_forces_previous_boundary_ witness, and not_forall_canonicalWitnessChoice_source_plus_collar_geometry_forces_previous_boundary_witness: the local at-most-one criterion and source-level canonical witness choice are not enough when the weak collar is supplied independently rather than built from that source data;

- the obstruction now reaches the weaker witness-cover surfaces that no longer require annulus collar geometry. In the same regression file, not_nonempty_planarBoundaryHeightOrderedFacePeelWitnessData_sharedInteriorPair shows that the shared interior pair creates a strict height cycle: a peeled face choosing sip01 needs a strictly later peeled face choosing sip12, and that face needs a strictly later peeled face choosing sip01, but only sipFace0 and sipFace1 can choose either shared edge. The finite collar-layer witness-cover form is blocked by not_nonempty_planarBoundaryCollarLayerFacePeelWitnessData_sharedInteriorPair, because it canonically lowers to the height-ordered surface. The same prefix cycle has now been abstracted as not_exists_mem_of_twoWitnessPrefixCycle, which allows repeated faces in the peel list and still rules out a finite mutual dependency. The support-level form not_mem_facePeelWitnessSupport_pair_of_twoWitnessPrefixCycle now records that both mutually dependent witness edges are absent from the resulting witness support. Instantiating it gives not_nonempty_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair for the raw attached-face BoundaryRootedFacePeelCertificate, and not_nonempty_planarBoundaryWellFoundedFacePeelWitnessData_sharedInteriorPair follows because well-founded-parent witness data lowers to that certificate. Thus closedWalkAnnulusBoundarySource_without_heightOrderedFacePeelWitnessData_sharedInteriorPair and closedWalkAnnulusBoundarySource_without_collarLayerFacePeelWitnessData_sharedInteriorPair now sit beside closedWalkAnnulusBoundarySource_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair and closedWalkAnnulusBoundarySource_without_wellFoundedFacePeelWitnessData_sharedInteriorPair, recording that honest closed-walk annulus source data alone does not force even the raw same-embedding attached certificate accepted by the current Theorem 4.9 synthesis package. The same obstruction model is now Tait-colorable: sharedInteriorPairTaitEdgeColoring is proved by sharedInteriorPairTaitEdgeColoring_isTait, and closedWalkAnnulusBoundarySource_and_taitEdgeColoring_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair packages the honest source, an actual Tait coloring, and failure of the same-embedding attached certificate together;
- this attached-certificate obstruction now also hits the route-facing successor-dart source shell. The same finite model defines sharedInteriorPairDartSuccessorCycleGeometry and proves the induced selected-boundary arc condition sharedInteriorPairDartSuccessorCycleGeometry_selectedBoundaryArcOnFace. The witness theorem exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair packages boundary reachability, successor-cycle facial data, selected-boundary arc contiguity, and failure of the raw attached certificate on one embedding. The failed converse not_forall_exists_attachedBoundaryRootedFacePeelCertificate_

of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_sharedInteriorPair now has a Tait-calibrated companion: exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_without_attachedBoundaryRootedFacePeelCertificate_sharedInteriorPair and not_forall_exists_attachedBoundaryRootedFacePeelCertificate_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_sharedInteriorPair show that even adding an actual Tait edge coloring to the route-facing successor-dart source shell does not force the same-embedding attached certificate. The bundled theorem exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair packages simultaneous failure of height-ordered witness data, collar-layer witness data, the raw attached certificate, well-founded witness data, and annulus collar geometry on that same Tait-colorable source. The companion not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_sharedInteriorPair records that the successor-cycle source shell plus Tait coloring is still not enough to force even one of the currently sufficient same-embedding geometry endpoints. This obstruction is now non-vacuous on the purified carrier: vertex_one_mem_selectedBoundaryInteriorEdgeEndpointVertices_sharedInteriorPair and selectedBoundaryInteriorEdgeEndpointVertices_nonempty_sharedInteriorPair show that vertex 1 survives selected-boundary purification, while exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair and not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_sharedInteriorPair show that adding this nonempty carrier hypothesis still does not force any of the current sufficient same-embedding geometry endpoints. The same source package now still fails even at the honest closed-walk layer after adding exact one-collar current-boundary data preserving the same annulus boundary split: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair and not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_sharedInteriorPair show that the exact one-collar current-boundary shell still adds no forcing power even before passing to the successor-cycle boundary-order interface. The same exact one-collar obstruction survives when the raw purified-carrier nonemptiness is replaced by the named local endpoint witness HasUnblockedInteriorEndpoint: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_without_ currentSufficientSameEmbeddingGeometry_sharedInteriorPair, not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair, and not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair show that even exact one-collar current-boundary data together with a real Tait coloring and an explicit unblocked endpoint still adds no forcing power at either the honest-source or boundary-order source shell. The same strengthening now reaches the direct replacement-positive layer too: exists_embedding_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_without_any_replacementPositiveProjectedGeometryOn_sharedInteriorPair, not_forall_any_replacementPositiveProjectedGeometryOn_of_closedWalkAnnulusBoundarySource_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair, exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_without_any_replacementPositiveProjectedGeometryOn_sharedInteriorPair, and not_forall_any_replacementPositiveProjectedGeometryOn_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_hasUnblockedInteriorEndpoint_sharedInteriorPair show that even this exact one-collar endpoint-witness shell still does not force either direct replacement package or the route-facing annulus-collar positive packages. The same failure persists on that route-facing shell: exists_embedding_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_without_currentSufficientSameEmbeddingGeometry_sharedInteriorPair and not_forall_exists_some_currentSufficientSameEmbeddingGeometry_of_boundaryReachabilityData_and_dartSuccessorCycleEmbeddingData_and_selectedBoundaryArc_and_oneCollarAnnulusCurrentBoundaryData_and_taitEdgeColoring_and_nonempty_selectedBoundaryInteriorCarrier_sharedInteriorPair show that the exact one-collar current-boundary shell itself adds no forcing power at this boundary-order endpoint either. The same source package now has matching failed converses for PlanarBoundaryAnnulusCollarGeometry, PlanarBoundaryHeightOrderedFacePeelWitnessData, PlanarBoundaryCollarLayerFacePeelWitnessData, and PlanarBoundaryWellFoundedFacePeelWitnessData, so this stronger route-facing source shell does not force any of the currently sufficient same-embedding geometric endpoints on the shared-interior-pair witness. Theorem49PositiveGeometricSourceReplacementRegression.lean now reflects that obstruction at the newer positive package layer as well: the same witness refutes both `Theorem49ClosedWalkAnnulusCollarPositiveProjectedGeometryOn` and `Theorem49SuccessorCycleAnnulusCollarPositiveProjectedGeometryOn`. The honest closed-walk and successor-source failed converses are now factored through reusable generic source principles in `PlanarBoundaryClosedWalkSource.lean` and target-specific wrappers in `Theorem49PlanarBoundaryAnnulusGeometry.lean`, with the shared-interior-pair model supplying the concrete missing-target witnesses;

- the honest long-term frontier is still to derive this ordered or cyclic data from real embedding-side boundary walks rather than postulating it.

So the present formalization is no longer hiding the gap. The gap is explicit, localized, and regression-tested.

4 Relevant Lean files

- `Mettapedia/GraphTheory/OrderedPlanarEmbeddingWithBoundary.lean`
- `Mettapedia/GraphTheory/OrderedPlanarEmbeddingWithBoundaryRegression.lean`
- `Mettapedia/GraphTheory/WalkPlanarEmbeddingWithBoundary.lean`
- `Mettapedia/GraphTheory/WalkPlanarEmbeddingWithBoundaryRegression.lean`
- `Mettapedia/GraphTheory/FourColor/PlanarBoundaryOrderedEmbedding.lean`
- `Mettapedia/GraphTheory/FourColor/PlanarBoundaryFaceBoundaryRunRegression.lean`
- `Mettapedia/GraphTheory/FourColor/Theorem49ClosedWalkOuterRootObstruction.lean`